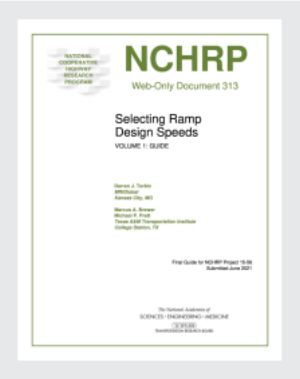
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NCHRP

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Selecting Ramp Design Speeds

VOLUME 1: GUIDE

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Final Guide for NCHRP Project 15-56
Submitted June 2021

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Summary

In the American Association of State Highway and Transportation Officials' (AASHTO) *A Policy on Geometric Design of Highways and Streets*, commonly referred to as the *Green Book*, design speed is defined as a selected speed used to determine the various geometric design features of the roadway (AASHTO, 2018). Design speed is used to determine all pertinent highway features to obtain a balanced design. Over an extended length of highway, it is desirable to select a uniform design speed. Where high-speed facilities meet low-speed facilities, drivers are expected to reduce their speeds suitable to the roadway environment they are entering. This change in speed should be achieved over an extended distance; but in the case of an interchange ramp, this is not possible. Application of the design speed concept and selection of an appropriate ramp design speed as presented in the AASHTO *Green Book* is overly simplistic and does not consider many factors.

This document provides guidelines for selecting appropriate ramp design speeds for ramps. The guidelines facilitate the selection of an appropriate ramp design speed based on a combination of contextual considerations and quantitative information. The guidelines define ramp design speed and all terms and elements related to its application. The guidelines address the following topics related to selecting an appropriate ramp design speed:

- System and service interchanges.
- Ramp configuration.
- Entrance and exit ramps.
- Freeway mainline ramp terminal configuration (i.e., taper vs. parallel).
- Type of crossroad ramp terminal (i.e., stop control, signal control, free-flow).
- Ramp grade.
- Type of horizontal curvature (i.e., simple, spiral, compound, and reverse curves).
- Number of lanes (i.e., single vs. multilane ramps).
- Crossover crown line.
- Superelevation and superelevation transition.
- Ramp metering.
- Collector-distributor (C-D) and frontage roads.
- Physical site constraints, environmental and social impacts, and right-of-way costs.

Ramp design speed is defined as the selected speed used to determine the various geometric design features of a ramp. The selected ramp design speed should be a logical one with respect to the type of intersecting highways, area type (i.e., urban or rural), ramp configuration, and site constraints (including physical, environmental, and social). Guide values for ramp design speeds are presented for system and service interchanges. The guide values are presented as a range of values given the context, ramp configuration, and type of interchange. The guide values were developed to be relatively consistent with the guide values presented in the 2018 edition of the *Green Book*.

Where the horizontal alignment of the ramp is curvilinear, the ramp design speed applies to the controlling curve on the ramp proper. For an entrance ramp, the last curve encountered along the ramp proper that significantly affects vehicle speed is considered the controlling curve. For an exit ramp, the controlling curve is the first curve encountered along the ramp proper that

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significantly affects vehicle speed. For direct connection and outer connection ramps, the sharpest curve on the ramp proper is considered the controlling curve. The freeway mainline ramp terminal, the crossroad ramp terminal, and the adjoining sections of the ramp proper that connect to the controlling curve should be designed to allow for appropriate speed transitions, consistent with driver behavior and expectations and vehicle performance capabilities, for the selected ramp design speed.

Where the horizontal alignment of the ramp is relatively straight and has little or no influence on vehicle speeds, the ramp design speed is based on the operational characteristics of the freeway mainline ramp terminal for an entrance ramp and the operational characteristics of the crossroad ramp terminal for an exit ramp and applies to the tangential section of the ramp proper.

Each individual section of the ramp proper (i.e., tangent and curve) has its own design speed. The design speed of the controlling feature of the ramp (i.e., the ramp design speed) is selected first, and then the design speeds of the other elements or segments of the ramp proper should be determined accordingly. It is desirable (but not essential) that the design speeds of the contiguous segments along the ramp proper increase or decrease in a stepwise manner. The change in design speed between adjoining sections should be limited to no more than 10 to 15 mph. For entrance ramps, this means that the design speeds of the contiguous segments along the ramp proper where practical should increase in a stepwise manner; and for exit ramps the design speeds of the contiguous segments along the ramp proper should decrease in a stepwise manner. The design speed of each individual curve and tangent along the ramp proper should be consistent with the speeds that drivers are likely to be traveling along the respective segment.

The freeway mainline ramp terminal portion of a ramp does not have an associated design speed. However, the operational and design conditions of the freeway mainline ramp terminal are related to and should be coordinated with the operational and design conditions of the adjoining sections of the ramp proper and freeway.

Each component of the ramp should be designed such that a vehicle has sufficient distance to accelerate or decelerate within the performance capabilities of the vehicle and driver expectations given the design speeds and features of the individual components of the ramp. Several design controls and criteria can be used to design ramps so that the individual components and features of the ramps are designed in a consistent manner, accounting for sequential speed transitions from one component or section to the next.

A spreadsheet-based tool was developed to estimate vehicle speeds along interchange ramps to help designers and traffic engineers conduct operational analyses and make design decisions. The spreadsheet tool is applicable to entrance and exit ramps at service interchanges including diagonal ramps, loop ramps, and outer connection ramps. The spreadsheet-based tool incorporates several speed prediction models based on vehicle speeds observed on ramps. With the capability to adjust several input parameters, the spreadsheet-based tool can be used to predict vehicle speeds based upon anticipated driver behaviors and design conditions. Output from the tool can be interpreted to help design an interchange ramp in a consistent manner that meets driver expectations.

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Section 1. Introduction

1.1 Background

In the AASHTO *A Policy on Geometric Design of Highways and Streets*, commonly referred to as the *Green Book*, design speed is defined as a selected speed used to determine the various geometric design features of the roadway (AASHTO, 2018). In principle, design speed is used to determine all pertinent highway features to obtain a balanced design. The design speed concept has been the centerpiece of the geometric design process since introduced in 1936 (Barnett, 1936).

Over an extended length of highway, it is desirable to select a uniform design speed. Selection of a design speed should be based upon the anticipated operating speed, topography, adjacent land use, modal mix, and functional classification of the roadway (AASHTO, 2018). Where high-speed facilities meet low-speed facilities, drivers are expected to reduce their speeds suitable to the roadway environment they are entering. This change in speed should be achieved over an extended distance; but in the case of an interchange ramp, this is not possible. Application of the design speed concept and selection of an appropriate ramp design speed as presented in the AASHTO *Green Book* may be overly simplistic and does not consider many factors.

The design speed of a ramp is related to the speeds of the intersecting highways that the ramp connects. According to AASHTO policy, the ramp design speed should desirably approximate the low-volume running speed on the intersecting highways; however, this is often not practical. *Green Book* Table 10-1 (Table 1) provides guidance for ramp design speeds based upon a range of percentages (i.e., upper, middle, and lower) of the higher design speed for the two intersecting highways. Further guidance is provided in the *Green Book* for selecting a ramp design speed based on various conditions and ramp types. For example, ramp design speeds between the middle and upper ranges of the highway speeds should be used for direct and semidirect connections, and lower-range values of ramp design speeds are generally used for loop ramps.

Table 1. Guide Values for Ramp Design Speed as Related to Highway Design Speed (AASHTO, 2018)

U.S. Customary											
Highway design speed (mph)	30	35	40	45	50	55	60	65	70	75	80
Ramp design speed (mph)											
Upper range (85%)	25	30	35	40	45	50	50	55	60	65	70
Middle range (70%)	20	25	30	30	35	40	45	45	50	55	60
Lower range (50%)	15	20	20	25	25	30	30	30	35	40	45
Corresponding minimum radius (ft)	See (*Green Book*) Table 3-7										

The guide values in *Green Book* Table 10-1 (Table 1) are applicable to the sharpest, or controlling, curve on the ramp, usually on the ramp proper, and do not pertain to the ramp terminals. The ramp design speed affects many aspects and elements of a ramp, including right-of-way requirements, horizontal and vertical alignments, speed-change lane lengths, ramp type, superelevation, and stopping sight distance, in addition to project costs and potential environmental and social impacts. Directly or indirectly, the selection of ramp design speed should consider physical and economic constraints and anticipated operating speeds on the ramp.

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The primary components of a ramp at a service interchange include the freeway mainline ramp terminal (i.e., an acceleration or deceleration lane), the ramp proper, and the crossroad ramp terminal; and they must be considered together with the two intersecting roadways during the design of a ramp (see Figure 1). At a system interchange, the primary components of a ramp include two freeway mainline ramp terminals (i.e., one exit and one entrance) and the ramp proper (see Figure 2).

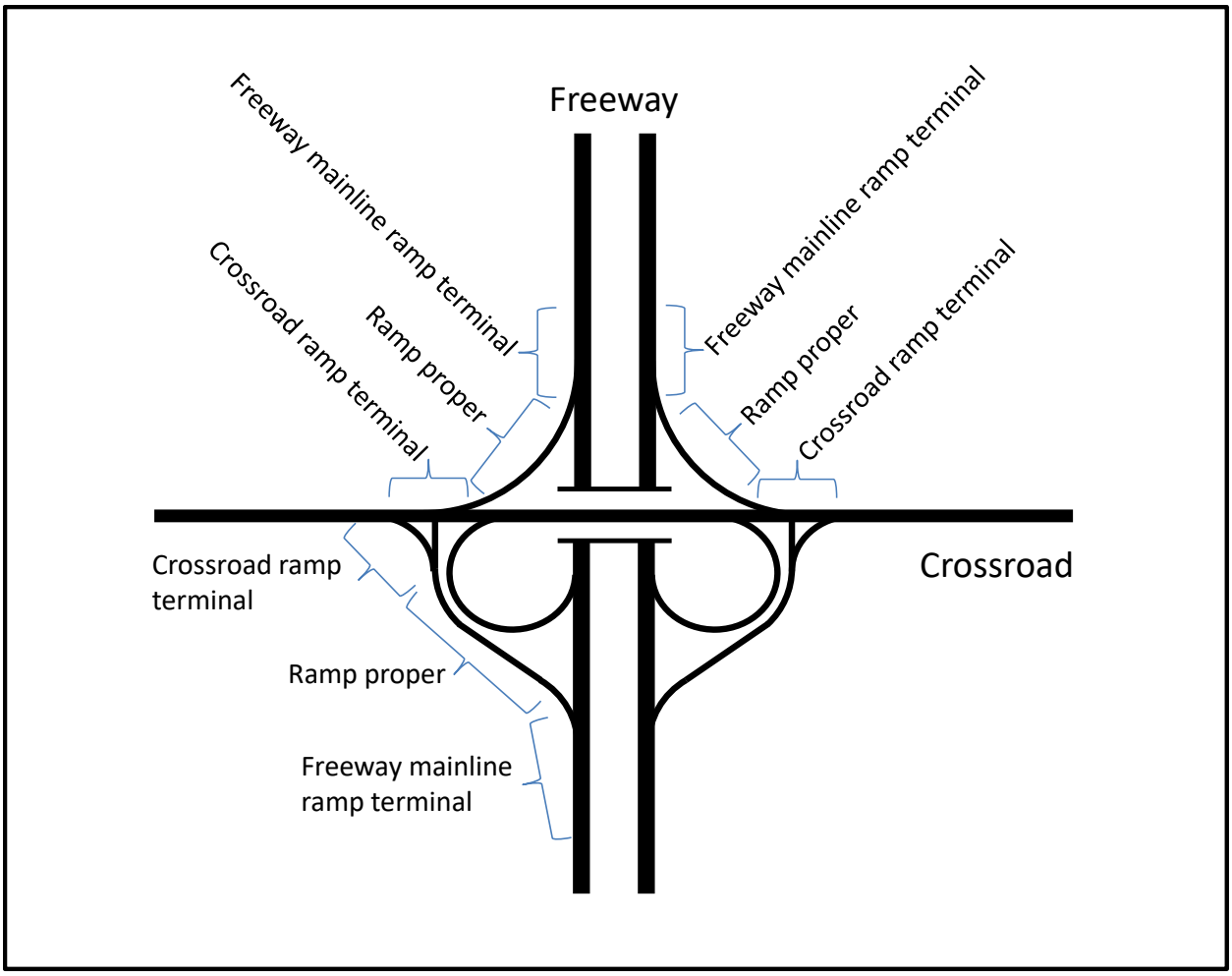


Figure 1. Primary Components of a Ramp at a Service Interchange

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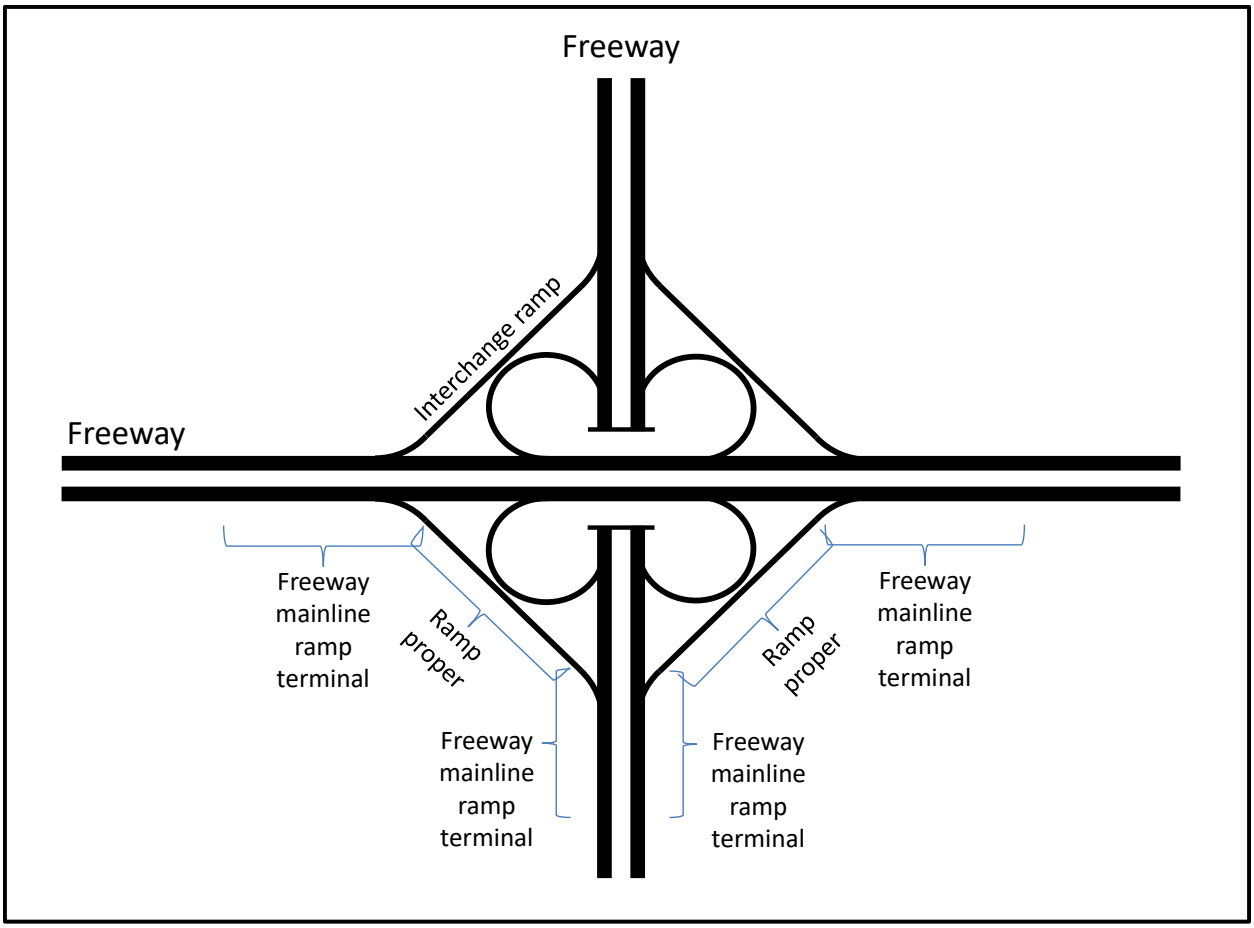


Figure 2. Primary Components of a Ramp at a System Interchange

The design guidelines presented herein provide further detail for selecting an appropriate ramp design speed than presented in the 2018 *Green Book*, to address several overarching challenges that may lead to confusion or inconsistent interpretation of existing AASHTO guidance for selecting an appropriate ramp design speed. In particular, the design guidelines presented herein addresses the following issues:

- **The *Green Book* does not provide a specific definition for ramp design speed**: The *Green Book* provides general guidance on selecting appropriate values for ramp design speed, describes the portion of the ramp to which the ramp design speed applies, and indicates which aspects and elements of the ramp that may be affected by the ramp design speed; however, the *Green Book* does not provide a specific definition for ramp design speed.
- ***Green Book* Table 10-1 does not provide sufficient detail for selecting ramp design speeds for specific ramp types and configurations:** *Green Book* Table 10-1 is the centerpiece of AASHTO's guidance on selecting an appropriate ramp design speed for the design of all ramp types (i.e., entrance, exit, and freeway-to-freeway ramps) and ramp configurations (e.g., diagonal, loop, semidirect, direct). Rather than *Green Book* Table 10-1 addressing entrance and exit ramps separately and different ramp

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configurations separately, *Green Book* Table 10-1 provides upper-, middle-, and lower-range values for ramp design speeds. AASHTO policy then provides further guidance regarding desired, practical, and/or minimum ramp design speeds corresponding to the upper-, middle-, and lower-speed ranges for different ramp configurations. However, it would be desirable to revise *Green Book* Table 10-1 to address different ramp types and configurations separately.

- **AASHTO policy provides little guidance on how the selected ramp design speed is to be used to develop a balanced design among all components of a ramp:** AASHTO policy states that the ramp terminals are to be properly transitioned and provided with speed-change facilities adequate for the speed of the highway being considered. For freeway mainline ramp terminals, *Green Book* Tables 10-4 and 10-6 provide the minimum acceleration and deceleration lengths for entrance and exit ramps, respectively. The minimum acceleration and deceleration lengths are determined based on the highway design speed and initial speeds at the beginning of the acceleration length and final speeds at the end of the deceleration length. *Green Book* Tables 10-4 and 10-6 do not specify a corresponding design speed for acceleration or deceleration lengths, nor is there guidance tying the conditions specified in *Green Book* Tables 10-4 and 10-6 for acceleration and deceleration lengths to the ramp design speed values in *Green Book* Table 10-1 for the controlling curve. Beyond AASHTO policy stating that the ramp terminals are to be properly transitioned and provided with speed-change facilities adequate for the speed of the highway being considered, there is no guidance on how this is to be achieved.

  The other type of ramp terminal is the crossroad ramp terminal. For exit ramps, the *Green Book* provides some guidance regarding how the functional area of an intersection relates to speed transitions approaching a crossroad ramp terminal. For entrance ramps, there is no corresponding guidance in the *Green Book*.

  Guidance in *Green Book* Chapter 10 implies the use of the vehicle performance curves for acceleration and deceleration distances for passenger cars for coordinating the design of the ramp terminals with the controlling curve on the ramp proper. However, use of the vehicle performance curves provided in Chapter 2 of the *Green Book* is not explicitly stated or recommended in Chapter 10 as it relates to the use of *Green Book* Tables 10-1, 10-4, and 10-6. Also, associated with the need to provide transitions between the controlling curve and the ramp terminals, there may be connecting portions of the ramp that are tangent or have a radius larger or smaller than the controlling curve that have to be addressed in the design of the entire ramp. Little guidance is provided within the *Green Book* on how the ramp terminals and adjoining sections are to be properly transitioned to connect with the controlling curve over the entire length of the ramp for a balanced ramp design that meets driver expectations.

- **There is no consensus on the portion of the ramp to which ramp design speed applies:** Even though the *Green Book* specifically states that the guide values for ramp design speed in *Green Book* Table 10-1 apply to the sharpest, or controlling, ramp curve, usually on the ramp proper, and the ramp design speeds do not pertain to the ramp terminals, there is no consensus amongst practitioners as to what portion of the ramp the ramp design speed is applicable. Some practitioners believe that ramp design speed

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should be applicable to other ramp components, and not simply the controlling curve on the ramp proper.

- **There is no consensus on whether the ramp design speed should be a single value or should vary along the ramp:** Again, the *Green Book* states that the guide values in *Green Book* Table 10-1 are applicable to the sharpest, or controlling, curve on the ramp, usually on the ramp proper, and do not pertain to the ramp terminals; but as speeds may vary along a ramp, it is not directly evident whether a single ramp design speed should be used or the ramp design speed should vary along the ramp. As a result, there is confusion or inconsistent interpretation by practitioners of whether the ramp design speed should be a single value or vary along the ramp.

1.2 Objective and Scope of Guidelines

This document provides guidelines for selecting appropriate design speeds for ramps. The term "ramp" includes all types, arrangements, and sizes of turning roadways that connect two or more legs at an interchange. The guidelines presented herein facilitate the selection of an appropriate ramp design speed based on a combination of contextual considerations and quantitative information. The guidelines define ramp design speed and all terms and elements related to its application. The guidelines are based on known safety and operational relationships of ramps over a range of interchange forms, ramp types, and environments (i.e., rural and urban) and are expected to result in ramp designs consistent with driver expectations and behaviors over a range of traffic conditions and the functional classification of the two intersecting roads.

The guidelines address the following topics related to selecting an appropriate ramp design speed:

- System and service interchanges.
- Ramp configuration.
- Entrance and exit ramps.
- Freeway mainline ramp terminal configuration (i.e., taper vs. parallel).
- Type of crossroad ramp terminal (i.e., stop control, signal control, free-flow).
- Ramp grade.
- Type of horizontal curvature (i.e., simple, spiral, compound, and reverse curves).
- Number of lanes (i.e., single vs. multilane ramps).
- Crossover crown line.
- Superelevation and superelevation transition.
- Ramp metering.
- Collector-distributor (C-D) and frontage roads.
- Physical site constraints, environmental and social impacts, and right-of-way costs.

The focus of this document is on providing assistance on selecting appropriate ramp design speeds for ramps at service interchanges that connect freeways to lesser facilities and ramps at system interchanges. In other words, this guide addresses selecting ramp design speeds for intersecting facilities, where at least one of the intersecting facilities is a freeway. The design speeds applied in the design of early freeways in urban areas were generally 50 to 60 mph, but today 60 mph is usually the minimum design speed for urban freeways and freeways in mountainous areas; while design speeds of freeways in rural areas are usually 70 to 80 mph or

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higher (Leisch, 2005). Thus, this guide addresses selecting ramp design speeds for ramps where at least one of the interconnecting facilities has a design speed of at least 50 mph or higher. Guide values for ramp design speeds that connect nonfreeway, arterial roadways to lesser facilities are provided in this document, but are merely an extrapolation of the guide values for interchange ramps where at least one of the intersecting facilities is a freeway.

1.3 Concept and Definition of Ramp Design Speed

The definition of ramp design speed is derived from the definition for design speed, from which the term originates. Recall that design speed is defined as a selected speed used to determine the various geometric design features of the roadway, and the selected design speed should be logical with respect to the anticipated operating speed, topography, adjacent land use, modal mix, and functional classification of the roadway (AASHTO, 2018). Similarly, ramp design speed is defined herein as a selected speed used to determine the various geometric design features of a ramp. Where the horizontal alignment of the ramp is curvilinear, the ramp design speed applies to the controlling curve on the ramp proper. For an entrance ramp, the last curve encountered along the ramp proper that significantly affects vehicle speed is considered the controlling curve. For an exit ramp, the controlling curve is the first curve encountered along the ramp proper that significantly affects vehicle speed. For direct connection and outer connection ramps, the sharpest curve on the ramp proper is considered the controlling curve.

The freeway mainline ramp terminal, the crossroad ramp terminal, and the adjoining sections of the ramp proper (i.e., adjoining tangents and horizontal curves in the direction of travel) that connect to the controlling curve should be designed to allow for appropriate speed transitions, consistent with driver behavior and expectations and vehicle performance capabilities, for the selected ramp design speed. If the horizontal alignment of the ramp is relatively straight and has little or no influence on vehicle speeds, as with some configurations of diagonal ramps, the ramp design speed is based on the operational characteristics of the freeway mainline ramp terminal for an entrance ramp and the operational characteristics of the crossroad ramp terminal for an exit ramp and applies to the tangential section of the ramp proper. The ramp design speed should be a logical one with respect to the type of intersecting highways, area type (i.e., urban or rural), ramp configuration, and site constraints (including physical, environmental, and social). The amount of speed transition to be accommodated along a ramp is dictated by the design speeds of the intersecting freeways at a system interchange and the design speed of the freeway and operations near the crossroad ramp terminal at a service interchange.

This definition of ramp design speed includes several key concepts:

1. Ramp design speed is used to determine the various geometric features of the ramp, which means that various geometric features of the ramp are related to the ramp design speed, but are not necessarily designed with the same design speed. For example, if the ramp proper consists of a compound curve with two curves with separate design speeds (e.g., 25 mph and 35 mph), only one of the design speeds will be designated as the ramp design speed for the entire ramp. As another example, if the freeway mainline ramp terminal of an exit ramp (i.e., a deceleration lane) connects to the controlling curve on the ramp proper, the freeway mainline ramp terminal should be designed such that operating

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speeds of vehicles exiting the freeway mainline ramp terminal will be consistent with the anticipated operating speeds entering the controlling curve.

2. The freeway mainline ramp terminal, the crossroad ramp terminal, and the adjoining sections of the ramp proper should be designed to allow for appropriate speed transitions, consistent with driver behaviors and expectations and vehicle performance capabilities, for the selected ramp design speed. For practical purposes, the ramp proper should be divided into individual tangents and horizontal curves. Figures 3 and 4 illustrate this principle for entrance and exit ramps, respectively. In Figure 3, in the direction of travel, the first section of the ramp proper is a tangent (designated RP-Tangent 1) followed by a curve (designated RP-Curve 1). Design criteria for the freeway mainline ramp terminal component of the ramp are slightly different from the design controls for the ramp proper, so the freeway mainline ramp terminal portion of the ramp is not divided into tangents and curves. In this example, RP-Curve 1 would be considered the controlling curve for the ramp; as the horizontal alignment of the ramp is curvilinear, and RP-Curve 1 is the last and only curve encountered along the ramp proper that significantly affects vehicle speed. Thus, the design speed for RP-Curve 1 would be designated as the ramp design speed for the entire ramp.

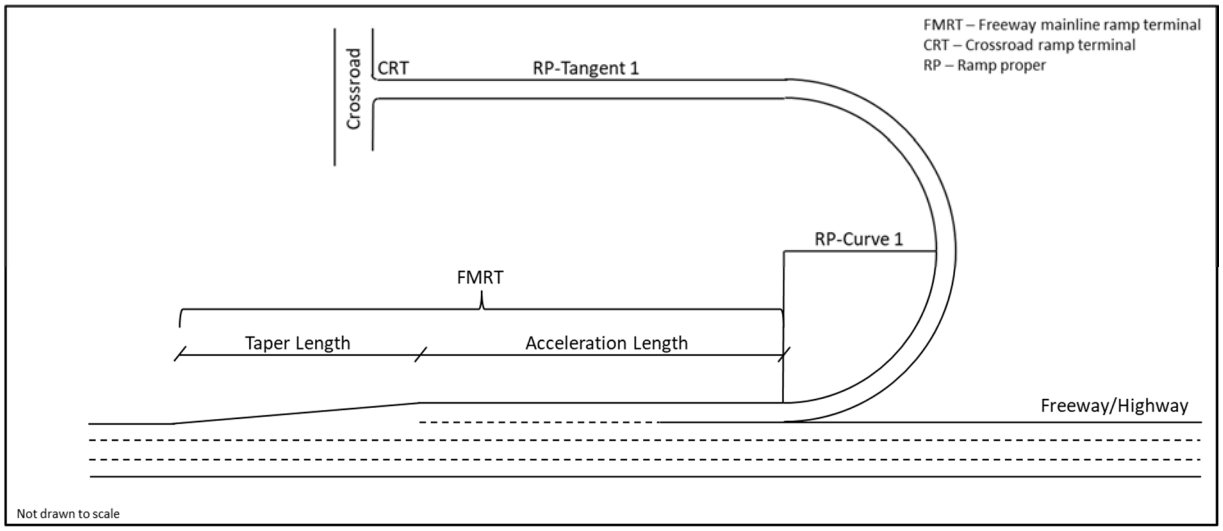


Figure 3. Illustration of Dividing the Ramp Proper of an Entrance Ramp into Individual Tangents and Curves

In Figure 4, the first curve encountered on the ramp proper is RP-Curve 1, followed by a second curve (RP-Curve 2), and then a tangent (RP-Tangent 1). The first curve encountered along the ramp proper (RP-Curve 1) would likely significantly impact vehicle speed so it would be designated as the controlling curve for the ramp. As such, the design speed for RP-Curve 1 would be designated as the ramp design speed for the entire ramp. All other sections of the ramp would be designed consistent with the ramp design speed.

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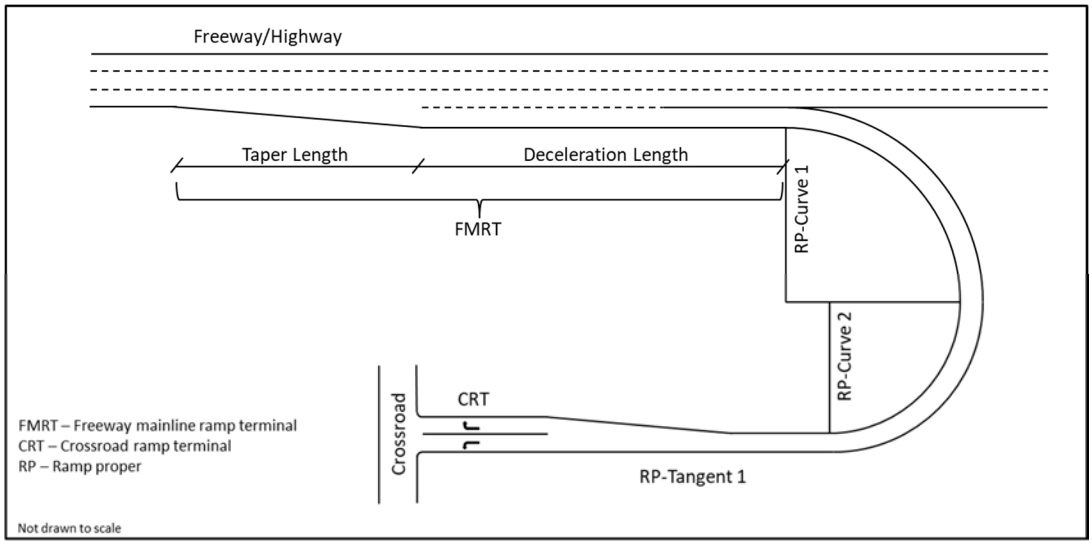


Figure 4. Illustration of Dividing the Ramp Proper of an Exit Ramp into Individual Tangents and Curves

3. At a service interchange, a driver is expected to either (a) increase speed sequentially transitioning from one adjoining section of the ramp to the next, in the case of an entrance ramp, or (b) decrease speed sequentially transitioning from one adjoining section of the ramp to the next, in the case of an exit ramp. From a practical standpoint, this may not be achieved in the design of every ramp depending upon the alignment. For example, depending upon the entry speed to a curve and the sharpness of the curve, it may be expected that vehicle speeds will decrease slightly or remain constant along a curve or tangent rather than accelerate along the section.

   Each individual section of the ramp proper (i.e., tangent and curve) has its own design speed. Depending upon the length and the number of individual sections, the change in design speed between adjoining sections should be limited to no more than 10 to 15 mph. Also, the assumed exiting speed of one section should be the assumed entering speed for the adjoining section downstream. Design speed has a direct impact on the minimum radius for a horizontal curve and may affect the design of other geometric elements on horizontal curves, as well. On tangent sections, the design speed may have only a limited effect on the roadway design, unless a critical sight distance limitation is present.

4. The concept of a design speed for an individual section of a ramp is slightly different from the traditional concept of design speed. With the traditional concept of design speed, an individual roadway segment is designed assuming that a vehicle travels at a constant speed; however, with interchange ramps, vehicles are expected to accelerate or decelerate over the length of an individual section. Theoretically, when designing a horizontal curve based on a point mass and the basic curve equation (see Equation 1), it is assumed that a vehicle travels at a constant speed. However, for an entrance ramp, it is assumed that vehicle speed will increase over the length of the curve. Similarly, for an exit ramp, it is assumed that vehicle speeds will decrease over the length of the curve. Thus, for the design of a horizontal curve on an interchange ramp, rather than designing a curve based on a constant speed, the anticipated speed at the midpoint of the curve may

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be used to design the curve rather than the anticipated speeds at the beginning or end of the curve.

The basic curve equation is:

$$R_{min} = \frac{V_{DS\ Curve}^2}{15(0.01e_{max} + f_{max})} \quad \textbf{(1)}$$

where:

R_{min} = minimum radius of curvature, ft.
$V_{DS\ Curve}$ = design speed, mph.
e_{max} = maximum superelevation rate, percent.
f_{max} = maximum side friction factor.

For the design of a tangent section on an interchange ramp, the speed at the downstream end is considered the design speed for the tangent.

5. If the horizontal alignment of the ramp is relatively straight and has little to no influence on vehicle speeds, as with some configurations of diagonal ramps, the ramp design speed is based on the operational characteristics of the freeway mainline ramp terminal for an entrance ramp and the operational characteristics of the crossroad ramp terminal for an exit ramp and applies to the tangent section of the ramp proper. For an entrance ramp, the design speed of the tangent section of the ramp proper (i.e., the ramp design speed) should be consistent with the assumed initial speed entering the freeway mainline ramp terminal. For an exit ramp, the design speed of the tangent section of the ramp proper (i.e., the ramp design speed) should be consistent with the operational conditions at the crossroad ramp terminal.
6. The ramp design speed should be a logical one with respect to the type of intersecting highways, area type (i.e., urban or rural), ramp configuration, and site constraints (including physical, environmental, and social). This indicates that a combination of contextual issues should be considered when selecting a ramp design speed, and the selected design speed should result in the design of a ramp consistent with driver behaviors and expectations (e.g., anticipated acceleration or deceleration along the ramp), vehicle performance capabilities, and the context of the local environment. For instance, for the design of an individual tangent or curve, the assumed acceleration or deceleration expected to take place over the length of a section should not exceed driver comfort levels nor the performance capabilities of vehicles. In addition, the selected ramp design speed should be consistent within the context of the environment. For example, at a system interchange that connects two rural freeways, drivers expect to maintain a high speed through a connecting ramp so a higher ramp design speed for the ramp would be a logical choice. However, at a service interchange in a downtown urban area, drivers would likely expect moderate to significant speed adjustment along a ramp, and with limited right-of-way, a lower ramp design speed would be logical.

1.4 Related Terms and Definitions

This section provides definitions and descriptions of key terms related to ramp design and selection of an appropriate ramp design speed. For some terms, existing definitions from the *Green Book* or other resources are provided; however, for many terms, the 2018 *Green Book*

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does not provide an explicit definition, but rather the definition or meaning of the term is understood. For these terms an explicit definition of the term is provided herein. Several figures are presented as well to provide additional clarity. The terms are organized according to the following topics:

- Interchange type
- Primary components of interchange ramps
- Speed-related terms
- Ramp configurations
- Characteristics of ramps
- Horizontal alignment
- Vertical alignment
- Sight distance

Interchange Type:

Service interchange — A service interchange connects a freeway to a lesser facility.

System interchange — A system interchange connects two intersecting freeway facilities.

Primary Components of Interchange Ramps:

Freeway mainline ramp terminal (Figure 5) — The freeway mainline ramp terminal is that portion of the ramp adjacent to the through traveled way of the primary highway or freeway and can extend beyond the length of the speed-change lane as necessary. At freeway mainline ramp terminals, ramp traffic merges with or diverges from high-speed through traffic at flat angles. Freeway mainline ramp terminals are further classified as either single lane or multilane, according to the number of lanes on the ramp at the terminal, and as either a taper or parallel type, according to the configuration of the speed-change lane.

The boundary between the freeway mainline ramp terminal and ramp proper can differ based on ramp configuration and horizontal and vertical alignment. Where the speed on the ramp proper is limited by horizontal curvature, the beginning or end of the horizontal curvature generally serves as the boundary between the freeway mainline ramp terminal and the ramp proper. For other situations, the boundaries between the freeway mainline ramp terminal and ramp proper are based on engineering judgement.

The freeway mainline ramp terminal does not have an associated design speed.

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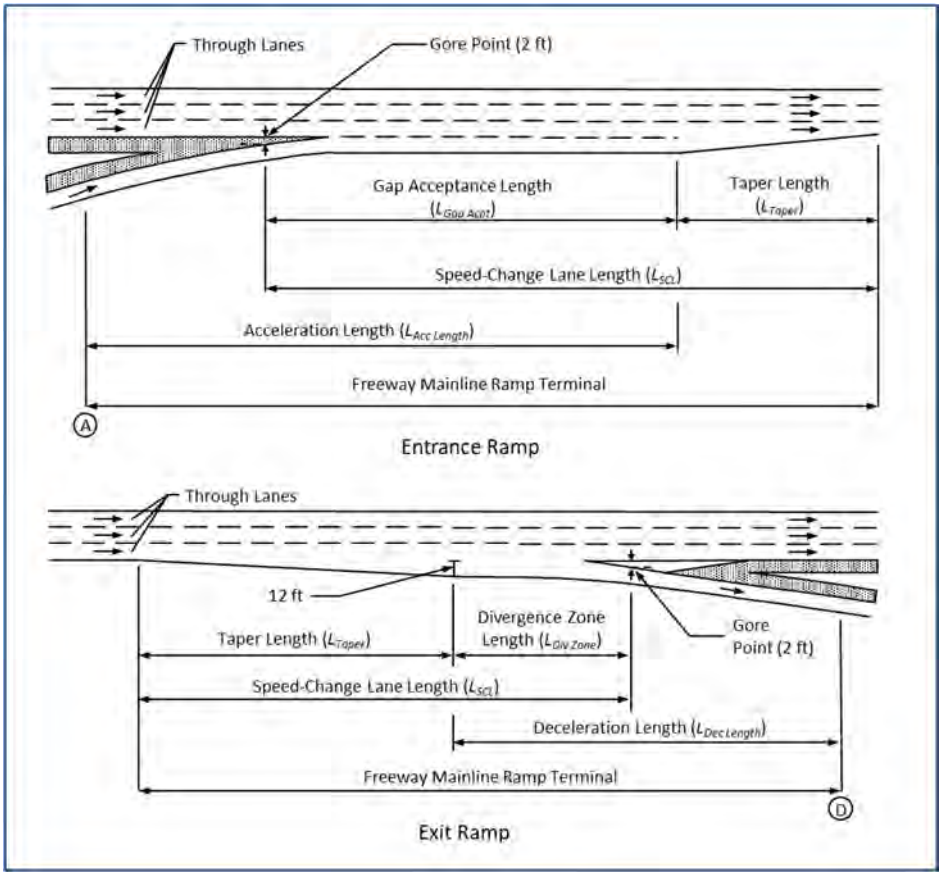


Figure 5. Freeway Mainline Ramp Terminals (Entrance and Exit Ramps)

Ramp proper (Figures 1 and 2)

The ramp proper is the roadway that connects the two terminals of the ramp. The geometry of the connecting road usually involves some curvature and grade.

Crossroad ramp terminal (Figure 1)

The crossroad ramp terminal is that portion of the ramp that connects the ramp proper to the crossroad. The crossroad ramp terminal may be the at-grade type with interruption of traffic destined to or from the ramp, as at the crossroad terminal of diamond or partial cloverleaf interchanges, or the free-flow type where ramp traffic merges with or diverges from high-speed through traffic at flat angles, or a combination of the two.

The crossroad ramp terminal does not have an associated design speed.

Speed-Related Terms:

Design speed

Design speed is a selected speed used to determine the various geometric design features of the roadway. The selected design speed should be a logical one with respect to the anticipated operating speed, topography, the adjacent land use, modal mix, and the functional classification of the highway (AASHTO, 2018).

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Ramp design speed

Ramp design speed is a selected speed used to determine the various geometric design features of the ramp. Where the horizontal alignment of the ramp is curvilinear, the ramp design speed applies to the controlling curve on the ramp proper. For an entrance ramp, the last curve encountered along the ramp proper that significantly affects vehicle speed is considered the controlling curve. For an exit ramp, the controlling curve is the first curve encountered along the ramp proper that significantly affects vehicle speed. For direct connection and outer connection ramps, the sharpest curve on the ramp proper is considered the controlling curve. The freeway mainline ramp terminal, the crossroad ramp terminal, and the adjoining sections of the ramp proper should be designed to allow for appropriate speed transitions, consistent with driver behavior and expectations and vehicle performance capabilities, for the selected ramp design speed. If the horizontal alignment of the ramp is relatively straight and has little or no influence on vehicle speeds, as with some configurations of diagonal ramps, the ramp design speed is based on the operational characteristics of the freeway mainline ramp terminal for an entrance ramp and the operational characteristics of the crossroad ramp terminal for an exit ramp and applies to the tangential section of the ramp proper. The ramp design speed should be a logical one with respect to the type of intersecting highways, area type (i.e., urban or rural), ramp configuration, and site constraints (including physical, environmental, and social). The amount of speed transition to be accommodated along a ramp is dictated by the design speeds of the intersecting freeways at a system interchange and the design speed of the freeway and operations near the crossroad ramp terminal at a service interchange.

Operating speed

Operating speed is the speed at which drivers are observed operating their vehicles during free-flow conditions. The 85th percentile of the distribution of observed speeds is the most frequently used measure of the operating speed associated with a particular location or geometric feature (AASHTO, 2018).

Posted speed

Posted speed is the highest speed that can legally be traveled on a section of roadway. Posted speeds are usually set to approximate the 85th percentile speed of traffic as determined by measuring the speeds of a sizable sample of vehicles.

Running speed

Running speed is the speed at which an individual vehicle travels over a section of highway. The running speed is the length of the highway section divided by the time for a typical vehicle to travel through the section. Peak and off-peak running speeds are used in design and operation, while average running speeds for an entire day are used in economic analyses (AASHTO, 2018).

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Ramps Configurations:

Diagonal ramp (Figure 6)

A diagonal ramp is a turning roadway that connects two intersecting roadways at an interchange. Diagonal ramps are almost always one-way but usually have both a left- and right-turning movement at the terminal on the minor intersecting road. A diagonal ramp may be largely tangent or wishbone in shape with a reverse curve.

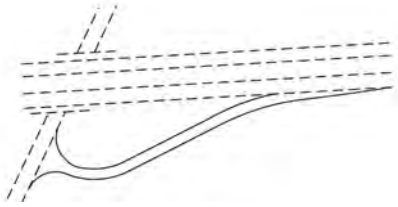


Figure 6. Diagonal Ramp (AASHTO, 2018)

Loop ramp (Figure 7)

A loop ramp is a turning roadway that connects two intersecting roadways at an interchange. Its shape is circular in nature. A loop ramp may have single turning movements (left or right) or double turning movements (left and right) at either or both ends. Loop ramps usually involve more indirect travel distance than other types of ramps.

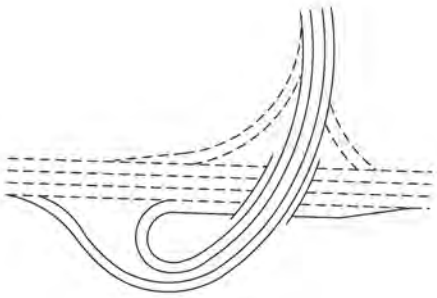


Figure 7. Loop and Semidirect Connection Ramps (AASHTO, 2018)

Semidirect connection (Figure 7)

A semidirect connection is a ramp where the driver exits to the right first, heading away from the intended direction, gradually reversing, and passing around other interchange ramps before entering the intersecting road with a free-flow movement. A semidirect connection may also be used for right turns, but there is little reason for its use if a conventional diagonal ramp can be provided. A descriptive term frequently associated with this type of ramp is “jug-handle,” the obvious plan shape. Travel distance on this ramp is typically less than for a comparable loop and more than for a direct connection.

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Direct connection (Figure 8)

A direct connection is defined as a ramp that does not deviate greatly from the intended direction of travel. Direct connections are designed to encourage high-speed merging with freeway traffic and are generally designed with higher design speeds than semidirect connections. Direct connections typically are designed with parallel-type terminals at both ends.

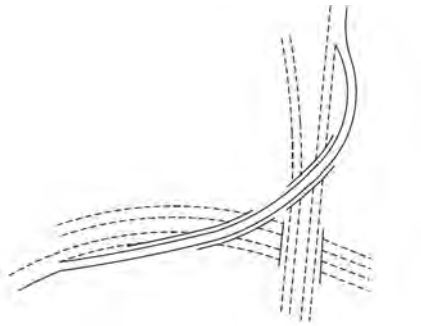


Figure 8. Direct Connection Ramp (AASHTO, 2018)

Outer connection (Figure 9)

An outer connection is a turning roadway where the driver exits in the intended direction and merges with the other leg of the interchange and another ramp (normally a loop ramp) is located between the outer connection and center of the interchange. Outer connections are designed to encourage high-speed merging with traffic.

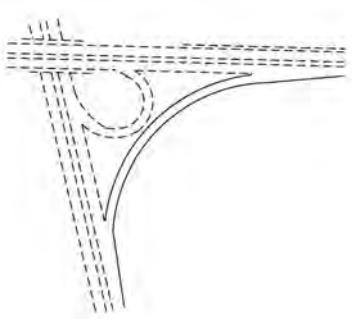


Figure 9. Outer Connection Ramp (AASHTO, 2018)

Characteristics of Ramps:

Auxiliary lane

An auxiliary lane is defined as the portion of the roadway adjoining the through lanes for speed change, turning, storage for turning, weaving, truck climbing, and other purposes that supplement through-traffic movement (AASHTO, 2018).

Speed-change lane (Figure 5)

A speed-change lane is an auxiliary lane, including tapered areas, primarily for acceleration or deceleration of vehicles entering or leaving through traffic. The term speed-change lane applies broadly to the added lane that joins the traveled way of the highway to the turning roadway and does not necessarily imply a definite lane of uniform width. For an entrance ramp the speed-change lane length is measured from the gore point to the end of the taper, and for an exit ramp the speed-change length is measured from the beginning of the taper to the gore point. A speed-change lane should have sufficient length to enable a driver to make the appropriate change in speed between the intersecting highway and the turning roadway.

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Acceleration length ($L_{Acc\ Length}$) (Figure 5)

The acceleration length is associated with the freeway mainline ramp terminal of an entrance ramp and merging maneuvers of a system interchange and is the distance needed for acceleration upon exiting the controlling feature of the ramp proper or final section of the ramp proper upstream of the gore point to the point of convergence with the primary highway or freeway. This distance for acceleration is governed by the speed differential between the operating speed on the entrance curve or other controlling feature of the ramp and the operating speed of the primary highway. The acceleration length begins at Point A and ends where the gap acceptance length ends and the taper begins. Point A represents the location where the ramp proper ends and the freeway mainline ramp terminal begins. Point A may be located upstream of the gore point, but it cannot be located downstream of the gore point. The acceleration length should not start on a curve of the ramp unless the radius equals 1,000 ft or more and a driver on the ramp has a clear view of freeway right-lane traffic.

Gap acceptance length ($L_{Gap\ Acpt}$) (Figure 5)

The gap acceptance length is associated with the freeway mainline ramp terminal of an entrance ramp and merging maneuvers of a system interchange and is the distance necessary to allow motorists to evaluate gaps in the freeway traffic and position their vehicles to use the gap. The gap acceptance length begins at the gore point and ends at the end of the acceleration length where the width of the auxiliary lane reduces to less than 12 ft. The gap acceptance length is always less than or equal to the acceleration length.

Deceleration length ($L_{Dec\ Length}$) (Figure 5)

The deceleration length is associated with the freeway mainline ramp terminal of an exit ramp and diverging maneuvers of a system interchange and is the distance needed for deceleration after clearing the through-traffic lane of the freeway and before reaching the first location that significantly affects vehicle speed on the ramp proper. The deceleration length is governed by the speed of traffic on the through lane of the freeway and the speed to be attained on the ramp. Deceleration may end in a complete stop, as at a crossroad terminal for a diamond ramp, or may be governed by the operating speed of the curvature of the ramp proper. The deceleration length begins where the width of the auxiliary lane increases to 12 ft or greater (i.e., the end of taper) and ends at Point D. Point D represents the location where the freeway mainline ramp terminal ends and the ramp proper begins. Point D cannot be located upstream of the gore point but may be located downstream of the gore point.

Divergence zone length ($L_{Div\ Zone}$) (Figure 5)

The divergence zone length is associated with the freeway mainline ramp terminal of an exit ramp and diverging maneuvers of a system interchange and allows for deceleration after clearing the through-traffic lane of the freeway up to the gore point. The divergence zone length begins where the width of the auxiliary lane increases to 12 ft or greater (i.e., the end of taper) and ends at the gore point. The divergence zone length is always less than or equal to the deceleration length.

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Taper length (L_{Taper}) (Figure 5)

The taper length is a transition length of roadway where the lane width gradually increases to add a lane or gradually narrows to remove a lane. The taper length typically begins or ends where the auxiliary lane transitions to a width less than 12 ft.

Taper-type entrance and exit (Figure 10)

A taper-type entrance and exit is a general form of a speed-change lane that provides direct entry or exit at a flat angle. For exit ramps, the divergence angle is usually between 2 and 5 degrees.

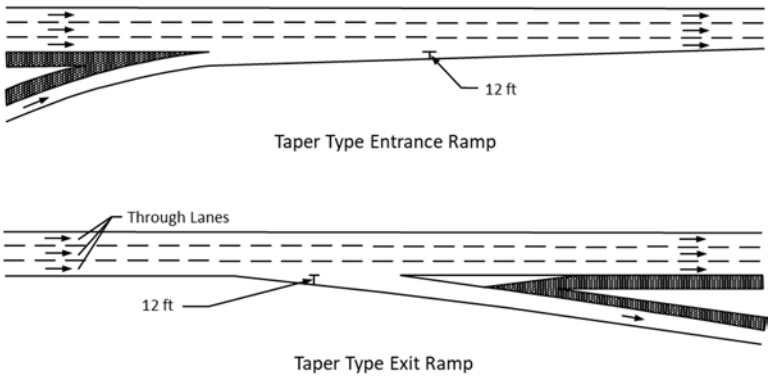


Figure 10. Taper-Type Entrance and Exit Ramps

Parallel-type entrance and exit (Figure 11)

A parallel-type entrance and exit is a general form of a speed-change lane that has an added lane for changing speed adjacent to the freeway mainline.

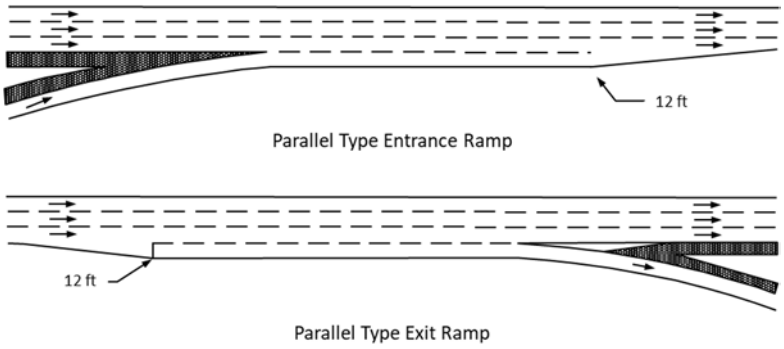


Figure 11. Parallel-Type Entrance and Exit Ramps

Acceleration lane

An acceleration lane is a speed-change lane of sufficient length to enable a driver to increase speed from the speed of the turning roadway to the higher speed of operation of the highway or street prior to merging. Moreover, there should be additional length to permit adjustments in speeds of both through and entering vehicles so that the entering driver can position the vehicle opposite a gap in the through-traffic stream and maneuver into the stream before the acceleration lane ends. The terms “speed-change lane” and “acceleration lane” can be used synonymously for an entrance ramp.

Deceleration lane

A deceleration lane is a speed-change lane of sufficient length to enable a driver to reduce speed from the speed of operation on the highway or freeway to the lower speed on the turning roadway. The terms “speed-change lane” and “deceleration lane” can be used synonymously for an exit ramp.

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Painted nose (Figure 12) The painted nose is a point, having no dimensional width, where the pair of solid white pavement edge markings from the ramp and intersecting roadway meet.

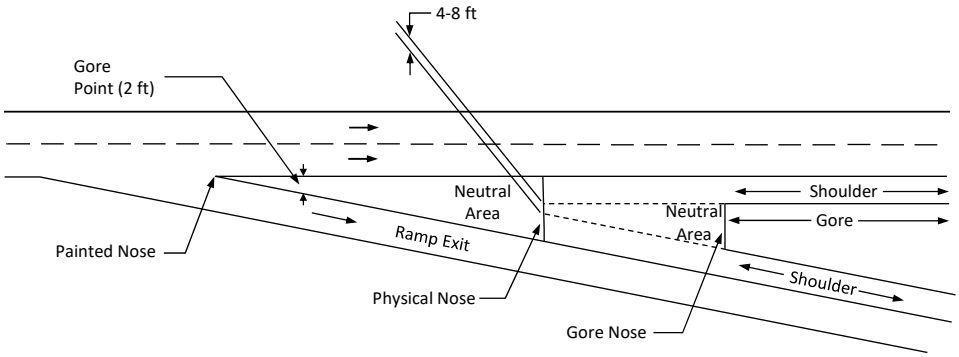


Figure 12. Typical Exit Gore Area Characteristics (adapted from AASHTO, 2018)

Gore point (Figure 12) The gore point is located where the pair of solid white pavement edge markings that separate the ramp from the intersecting roadway are 2 ft apart. If the markings do not extend to a point where they are 2.0 ft apart, then the gore point is found by extrapolating both markings until the extrapolated portion is 2.0 ft apart. The gore point is often used to define the beginning or ending milepost of the ramp. The gore point is also used to define the beginning of the gap acceptance length and speed-change lane length for an entrance ramp and the end of the divergence zone length and speed-change lane length for an exit ramp.

Physical nose (Figure 12) The physical nose is a point upstream from the gore, having some dimensional width (typically 4 to 8 ft) that separates the ramp from the intersecting roadway.

Neutral area (Figure 12) The neutral area refers to the triangular area between the painted nose and the gore nose and incorporates the gore point and physical nose.

Gore nose (Figure 12) The gore nose is a point delineating the downstream end of the neutral area at the shoulder intersection points. The gore nose generally is the location where the neutral area pavement ends.

Gore (Figure 12) The term “gore” indicates an area downstream from the shoulder intersection points.

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Crossover crown line	At the boundary between the right lane of the freeway and the adjacent auxiliary lane of a freeway mainline ramp terminal, the cross slope rates of the freeway lane and auxiliary lane may differ. This boundary location is referred to as the crossover crown line. The crossover crown line (not to be confused with the crown line normally provided at the centerline of a roadway) is measured as the algebraic difference in cross slope rates of the two adjacent lanes.
Traveled way	The portion of the roadway for the movement of vehicles, exclusive of shoulders and bicycle lanes (AASHTO, 2018).
Ramp meter	A traffic signal installed on an entrance ramp to control the number of vehicles entering the freeway.
Diverge	A movement in which a single stream of traffic separates into two streams without the aid of traffic control devices (TRB, 2010).
Merge	A movement in which two separate lanes of traffic combine to form a single lane without the aid of traffic signals or other right-of-way controls (TRB, 2010).

Horizontal Alignment:

Simple curve	A circular arc with a single, constant radius.
Compound curve	A compound curve is where two or more circular arcs of different radii are joined in succession, turning in the same direction, without any tangent between adjoining arcs.
Spiral curve	The Euler spiral, which is also known as the clothoid, is used in the design of spiral transition curves. The radius varies from infinity at the tangent end of the spiral to the radius of the circular arc at the end that adjoins that circular arc. By definition, the radius of curvature at any point on an Euler spiral varies inversely with the distance measured along the spiral. In the case of a spiral transition that connects two circular curves having different radii, there is an initial radius rather than an infinite value (AASHTO, 2018).
Controlling curve	For an entrance ramp, the last curve encountered along the ramp proper that significantly affects vehicle speed is considered the controlling curve. For an exit ramp, the controlling curve is the first curve encountered along the ramp proper that significantly affects vehicle speed. For direct connection and outer connection ramps, the sharpest curve on the ramp proper that significantly affects vehicle speed is considered the controlling curve.

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Superelevation	Superelevation is the rotation of the pavement on the approach to and through a horizontal curve. Superelevation is intended to assist the driver by counteracting the lateral acceleration produced by tracking the curve (TRB, 2010).
Superelevation transition section	The superelevation transition section consists of the superelevation runoff and tangent runout sections. The superelevation runoff section consists of the length of roadway needed to accomplish a change in outside-lane cross slope from zero (flat) to full superelevation, or vice versa. The tangent runout section consists of the length of roadway needed to accomplish a change in outside-lane cross slope from the normal cross slope rate to zero (flat), or vice versa (AASHTO, 2018).
Curve radius	The radius of the circular arc used to specify the horizontal alignment of the roadway. For ramps, the minimum radii used for design should preferably be measured from the inner edge of the traveled way rather than the middle of the vehicle path or the centerline of the traveled way.

Vertical Alignment:

Grade and profile	The profile of a typical ramp usually consists of a central portion on an appreciable grade, coupled with terminal vertical curves and connections to the profiles of the intersection legs (AASHTO, 2018).
Vertical curve	Vertical curves effect gradual changes between tangent grades. For simplicity, a parabolic curve with an equivalent vertical axis centered on the Vertical Point of Intersection (VPI) is usually used in roadway profile design.

Sight Distance:

Stopping sight distance (SSD)	Stopping sight distance is the sum of two distances: (1) the distance traversed by the vehicle from the instant the driver sights an object necessitating a stop to the instant the brakes are applied, and (2) the distance needed to stop the vehicle from the instant brake application begins. These are referred to as brake reaction distance and braking distance, respectively (AASHTO, 2018).
Decision sight distance (DSD)	Decision sight distance is the distance needed for a driver to detect an unexpected or otherwise difficult to-perceive information source or condition in a roadway environment that may be visually cluttered, recognize the condition or its potential threat, select an appropriate speed and path, and initiate and complete complex maneuvers. Because decision sight distance offers drivers additional margin for error and affords them sufficient length to maneuver their vehicles at the same or reduced speed, rather than to just stop, its values are substantially greater than stopping sight distance (AASHTO, 2018).

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Horizontal sight distance	Sight distance is the length of the roadway ahead visible to the driver. On horizontal curves, the obstruction that limits the driver's sight distance may be some physical feature outside of the traveled way, and on a crest vertical curve the obstruction may be the road surface at some point.
Vertical sight distance	Sight distance is the length of the roadway ahead visible to the driver. On a tangent roadway, the obstruction that limits the driver's sight distance may be the road surface at some point on a crest vertical curve.

1.5 Outline of Document

The remainder of this document is organized as follows:

Section 2—Ramp Elements Related to Ramp Design Speed
Section 3—Guide for Designing Ramps in a Consistent Manner Based on the Selected Ramp Design Speed
Section 4—Design Tool to Evaluate Ramp Designs for Consistency with the Selected Ramp Design Speed
Section 5—Case Studies
Section 6—Summary of Design Guide and Future Research Needs
Section 7—References
Section 8—Abbreviations, Acronyms, Initialisms, and Symbols

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Section 2.
Ramp Elements Related to Ramp Design Speed

The definition of ramp design speed indicates a combination of contextual issues such as the type of intersecting highways, area type, ramp configuration, and site constraints that should be considered when selecting a ramp design speed. This section of the design guidelines discusses in greater detail the most relevant elements that affect selecting an appropriate ramp design speed. For completeness, additional elements are discussed that do not impact the selection of an appropriate ramp design speed, but are key elements of ramp design. The elements addressed in this section include:

- System and service interchanges.
- Ramp configuration.
- Entrance and exit ramps.
- Freeway mainline ramp terminal configuration (i.e., taper vs. parallel).
- Type of crossroad ramp terminal (i.e., stop control, signal control, free-flow).
- Ramp grade.
- Type of horizontal curvature (i.e., simple, spiral, compound, and reverse curves).
- Number of lanes (i.e., single vs. multilane ramps).
- Crossover crown line.
- Superelevation and superelevation transition.
- Ramp metering.
- Collector-distributor (C-D) and frontage roads.
- Physical site constraints, environmental and social impacts, and right-of-way costs.

For some of the ramp elements, guide values for ramp design speeds are provided. These should be considered general guide values for ramp design speeds related to the individual ramp elements. However, no single element is the primary determinant in selecting an appropriate ramp design speed. Rather, a combination of contextual considerations should be made in selecting an appropriate ramp design speed. The elements presented in this section, and associated recommended ramp design speeds, should be considered collectively to select an appropriate ramp design speed for a given ramp. In Section 2.14, a table of guide values for ramp design speeds for system and service interchanges is provided.

2.1 System and Service Interchanges

A system interchange, which connects two or more freeways, and a service interchange, which connects a freeway to lesser facilities (i.e., non-controlled access facilities), differ from one another in how they relate to ramp design speed. With a system interchange, vehicles exit a high-speed facility and enter another high-speed facility, so it is desirable to provide a ramp between the two facilities that enables vehicles to maintain a speed as close as possible to freeway speeds, minimizing the need for deceleration or acceleration along the ramp. In rural or unconstrained areas, drivers expect minimal acceleration or deceleration along a ramp of a system interchange; while in urban and constrained areas, drivers expect moderate acceleration or deceleration along the ramp of a system interchange.

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For situations where the design speeds of both intersecting freeways of a ramp are equivalent, the highest possible ramp design speed for the connecting ramp would be the design speed of the intersecting freeways. However, due to physical constraints and related right-of-way costs, in most cases it is not practical for the ramp design speed to be equivalent to the design speed of the two intersecting freeways. Therefore, the ramp design speed may be less than the design speed of the two intersecting freeways, but as close as practical. For the situation where two intersecting freeways do not have equivalent design speeds, the ramp design speed may be a speed between the two highway design speeds of the intersecting freeways; but again, for practical purposes this may not be feasible so the ramp design speed may be less than the lower design speed of the two intersecting freeways.

With a service interchange, vehicles either accelerate from a near-stop condition at the crossroad ramp terminal to a higher speed on the freeway (e.g., in the case of entrance ramps) or decelerate from a higher speed on the freeway to a stop or near-stop condition at the crossroad ramp terminal (e.g., in the case of exit ramps). In either case, the ramp needs to be designed to accommodate the necessary acceleration or deceleration. In rural or unconstrained areas where the intersecting crossroad is a primary highway or major street, minimal to moderate acceleration or deceleration along the ramp is expected by the driver; while if the intersecting crossroad is a minor arterial or collector, moderate acceleration or deceleration is expected by the driver along the ramp. In urban or constrained areas where the intersecting facility is a primary highway or major street, moderate to significant acceleration or deceleration along the ramp is expected by the driver; while if the intersecting road is a minor arterial or collector, significant acceleration or deceleration is expected by the driver along the ramp.

Table 2 summarizes the anticipated acceleration/deceleration adjustment (i.e., either speed reduction or increase) by drivers in rural/unconstrained and urban/constrained areas by intersecting facility type and interchange type and corresponding ramp design speed.

Table 2. Anticipated Driver Behavior and Ramp Design Speeds by System and Service Interchanges (adapted from Leisch, 2005)

	Intersecting Facility	Rural / Unconstrained	Urban / Constrained
Service Interchange	Primary highway or major arterial	Minimal to moderate acceleration/deceleration adjustment	Moderate to substantial acceleration/deceleration adjustment
		High to low ramp design speed	Intermediate to low ramp design speed
	Minor arterial or collector	Moderate acceleration/deceleration adjustment	Substantial acceleration/deceleration adjustment
		Intermediate to low ramp design speed	Low ramp design speed
System Interchange	Freeway	Minimal acceleration/deceleration adjustment	Moderate acceleration/deceleration adjustment
		High ramp design speed	Intermediate ramp design speed

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As a guide, high ramp design speeds generally correspond to values of 50 mph and above; intermediate ramp design speeds are in the range of 35 to 45 mph; and low ramp design speeds correspond to values of 30 mph or less.

2.2 Ramp Configuration

When selecting the appropriate ramp design speed for a particular ramp configuration, the primary considerations are the intended purpose of the ramp and the controlling feature of the ramp. For example, is the ramp intended to facilitate movement between two high-speed facilities in the case of a system interchange or is the ramp intended to facilitate larger speed transitions between high- and low-speed facilities? Some ramp configurations, such as direct and semidirect ramps, are primarily used at system interchanges, while diagonal ramps are primarily used at service interchanges. Loop ramps are commonly used at both system and service interchanges. Another consideration is the controlling feature of the ramp. Is the horizontal alignment the primary feature of the ramp that will control speeds, or will speed primarily be controlled by the operations of the terminals? Several considerations are discussed below regarding the selection of an appropriate ramp design speed for the following ramp configurations and are relatively consistent with guidance provided in the *Green Book* (AASHTO, 2018):

- Direct connection ramps
- Semidirect connection ramps
- Loop ramps
- Diagonal ramps

Direct connection ramps (see Figure 8) do not deviate greatly from the intended direction of travel. They can be designed as right exit/right entrance ramps or as left exit/left entrance ramps that utilize the shortest and most direct path with curvilinear, diagonal, or reverse curve alignment. Direct connection ramps are commonly used at system interchanges. As such, they are intended to encourage high-speed, free-flow merging with freeway traffic and may be designed with ramp design speeds ranging from 35 to 80 mph. Direct connection ramps are also used at service interchanges, and in these situations the ramp design speeds may range from 30 to 65 mph. For both system and service interchanges, the horizontal alignment is commonly the controlling feature of direct connection ramps. For direct connection ramps, the sharpest curve on the ramp proper that significantly affects vehicle speed is considered the controlling curve.

Semidirect connection ramps (see Figure 7) provide a less direct path than direct connection ramps and typically include smaller curve radii than direct connection ramps, and thus may have associated lower ramp design speeds compared to direct connection ramps. Semidirect connection ramps are commonly used at system interchanges and are intended to encourage high-speed, free-flow merging with freeway traffic. As such, high ramp design speeds are preferred but may not be practical due to the curvilinear alignment. For semidirect connection ramps, ramp design speeds may range from 30 to 65 mph. For a semidirect connection ramp, the horizontal alignment is commonly the controlling feature of the ramp. For semidirect connection ramps, the sharpest curve on the ramp proper that significantly affects vehicle speed is considered the controlling curve.

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Loop ramps (see Figure 7) may be used in both system and service interchanges. The design speed of a loop ramp is limited by the radius of the controlling curve. The radii of controlling curves on loop ramps generally range from 150 to 250 ft, with corresponding design speeds of about 25 to 30 mph. Loop ramps with design speeds above 30 mph involve large land areas rarely available in urban areas. Loop ramps with ramp design speeds above 30 mph are costly and require drivers to travel a considerable extra distance, but if less restrictive conditions exist, a ramp design speed above 30 mph may be used and the corresponding radius may be increased. For both system and service interchanges, the horizontal alignment is the controlling feature of loop ramps.

Diagonal ramps are used for service interchanges and typically have both a left- and right-turn movement at the crossroad ramp terminal. Diagonal ramps consist of a high-speed freeway mainline ramp terminal, a tangent or curved alignment on the ramp proper, and stop or yield conditions at the crossroad ramp terminal. Depending on the configuration of the crossroad, a free-merge right-turn maneuver may be provided.

The “shape” of a diagonal ramp largely depends on the pattern of turning traffic and the right-of-way available. Diagonal ramps may be designed with (1) a tangent throughout most, or all, of the ramp; (2) a continuous curve to the right, with a spur to the left for left turns; (3) or a combination of tangent and reverse curvature. For diagonal ramps where the horizontal alignment is curvilinear, the horizontal alignment will be the controlling feature of the ramp. For diagonal ramps where the horizontal alignment is relatively flat (i.e., where the horizontal alignment has little or no impact on the speeds of vehicles), the ramp design speed is based on the operational characteristics of the freeway mainline ramp terminal for an entrance ramp and the operational characteristics of the crossroad ramp terminal for an exit ramp. For an entrance ramp, the design speed of the tangent section of the ramp proper (i.e., the ramp design speed) should be consistent with the speed entering the freeway mainline ramp terminal. For an exit ramp, the design speed of the tangent section of the ramp proper (i.e., the ramp design speed) should be consistent with the operating conditions at the crossroad ramp terminal. Because diagonal ramps are used for service interchanges and the controlling feature of a diagonal ramp may be either the horizontal alignment or the operational characteristics of one of the terminals, diagonal ramps may be designed with ramp design speeds as low 15 to 20 mph and potentially ranging as high as 45 to 70 mph. Diagonal ramps are unique compared to other ramp configurations in that the controlling feature of a diagonal ramp may be either the horizontal alignment or the operational characteristics of one of the terminals. With the other ramp configurations, the horizontal alignment of the ramp will likely be the controlling feature.

2.3 Entrance and Exit Ramps

Entrance and exit ramps are more associated with service interchanges than with system interchanges. The primary components of entrance and exit ramps include the freeway mainline ramp terminal, the ramp proper, and the crossroad ramp terminal. With system interchanges, ramps function more as a freeway mainline ramp terminal, a ramp proper, and another freeway mainline ramp terminal. There is no crossroad ramp terminal component to a system interchange ramp, which makes it difficult to classify ramps at system interchanges as entrance or exit ramps. Therefore, the discussion below applies primarily to ramps at service interchanges.

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2.3.1 Entrance Ramps

On entrance ramps, vehicles travel from the crossroad ramp terminal to the ramp proper to the freeway mainline ramp terminal. At the crossroad ramp terminal vehicles speeds are low and increase along the ramp proper. On entrance ramps, the controlling feature is either a curve on the ramp proper immediately upstream of the freeway mainline ramp terminal or vehicle operations near the freeway mainline ramp terminal. Where the horizontal alignment of the ramp is curvilinear, the last curve encountered along the ramp proper immediately upstream of the freeway mainline ramp terminal that significantly affects vehicle speed is considered the controlling curve (i.e., it will control the speeds of vehicles transitioning from the ramp proper to the freeway mainline ramp terminal). For design purposes, it may be assumed that horizontal curves with radii less than 1,000 ft will significantly control vehicle speeds. For entrance ramps where the horizontal alignment is relatively flat (i.e., diagonal ramps), the ramp design speed is based on the operational characteristics of the freeway mainline ramp terminal so the design speed of the tangent section of the ramp proper (i.e., the ramp design speed) should be consistent with the speed entering the freeway mainline ramp terminal.

The boundary between the ramp proper and the freeway mainline ramp terminal is dependent upon the line of sight of the driver and the horizontal alignment of the ramp. Typically, the gore point serves as the boundary between the ramp proper and freeway mainline ramp terminal. Research shows that drivers tend to begin the merge, acceleration process only after gaining a clear view of the freeway right-traffic lane. Thus, if the driver's view from the ramp is obstructed, the acceleration length will not begin until near the gore point where the view of the freeway becomes unobstructed (Hunter and Machemehl, 1997). However, if the driver's view from the ramp is unobstructed, the acceleration length may extend upstream from the gore point where the horizontal curvature has a radius less than 1,000 ft, thus extending the freeway mainline ramp terminal upstream of the gore point.

Entrance ramps can be designed with taper- or parallel-type freeway mainline ramp terminal configurations. A taper-type freeway mainline ramp terminal is a general form of a speed-change lane that provides direct entry at a flat angle (see Figure 10). Vehicles merge onto the freeway from a long, uniform taper between the outer edge of the acceleration lane and the edge of the through-traffic lane. A parallel-type freeway mainline ramp terminal is a general form of a speed-change lane that has an added lane for acceleration adjacent to the freeway mainline (see Figure 11).

The type of freeway mainline ramp terminal configuration, taper or parallel, has no direct impact on the selection of an appropriate ramp design speed for an entrance ramp. Entrance ramps with either type of freeway mainline ramp terminal configuration are designed in the same manner for a given ramp design speed. For the same assumed entrance speed at the beginning of the freeway mainline ramp terminal onto the acceleration lane and the same assumed merge speed at the end of the freeway mainline ramp terminal, the minimum acceleration length would be exactly the same for a taper-type entrance ramp and a parallel-type entrance ramp. In a recent study (Torbic et al., 2012), it was found that vehicles are more likely to use the full length of a tapered speed-change lane to accelerate to near-freeway speeds before merging; in contrast to a parallel speed-change lane, where vehicles may merge earlier along the speed-change lane and at lower speeds. However, for design purposes, vehicles are assumed to accelerate at the same rates on both taper- and parallel-type entrance ramps and use the full acceleration length prior to merging. Therefore,

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acceleration lengths are equivalent for both taper- and parallel-type entrance ramps given the same initial speed ($V_{Acc\ Length\ (i)}$) and speed reached ($V_{Acc\ Length\ (f)}$), so the type of freeway mainline ramp terminal configuration (i.e., taper or parallel) does not affect the selection of the ramp design speed.

Differences in the general alignment of ramps between taper- and parallel-type entrances, however, could impact whether the selection of the ramp design speed is based on the last curve that significantly influences vehicle speeds on the ramp proper prior to the freeway mainline ramp terminal or the operational characteristics of the freeway mainline ramp terminal. With a taper-type entrance providing direct entry at a flat angle, there is a greater likelihood that the controlling feature of the ramp will be the operational characteristics of the freeway mainline ramp terminal. With a parallel-type entrance, there is a greater likelihood that the horizontal alignment immediately upstream of the acceleration length on the ramp proper will be curvilinear, affect the speed of vehicles, and be the controlling feature of the ramp. For both taper- and parallel-type entry ramps, acceleration lengths should include only the portions of the ramp proper from which drivers can clearly view the freeway right-lane traffic.

Because the controlling feature of an entrance ramp will always be associated with speeds or operational characteristics towards the end of the ramp near the freeway mainline ramp terminal, ramp design speeds are generally at least 35 mph or greater for entrance ramps.

2.3.2 Exit Ramps

Exit ramps should be designed to facilitate deceleration from the mainline freeway speed to a speed slightly below the ramp design speed, and then potentially to a stop condition (at service interchanges). Exit ramps may be designed as taper-type exits that provide direct exits at flat angles and fit the preferred path of most exiting drivers and allow vehicles to leave the through lane of the freeway at relatively high speeds (see Figure 10). Deceleration can take place off of the freeway mainline, as the exiting vehicle moves along the taper onto the ramp proper. Exit ramps may also be designed with parallel-type exits that have an added lane adjacent to the freeway for deceleration (see Figure 11).

Similar to entrance ramps, the type of freeway mainline ramp terminal configuration (i.e., taper or parallel) of an exit ramp has no direct impact on the selection of an appropriate ramp design speed. For the same assumed diverge speed from the mainline freeway onto the deceleration lane at the beginning of the deceleration length and the same assumed speed at the end of the deceleration length, the minimum deceleration length would be exactly the same for a taper-type exit ramp and a parallel-type exit ramp. The key point, though, is that the assumed speed at the end of the freeway mainline ramp terminal and the beginning of the first section of the ramp proper would be the same whether an exit ramp is designed as a taper-type or parallel-type. Therefore, the type of freeway mainline ramp terminal configuration (i.e., taper or parallel) has no direct impact on the selection of the appropriate ramp design speed for an exit ramp.

Differences in the general alignment of ramps between taper- and parallel-type exits, however, may affect whether the horizontal alignment or the operations near the crossroad ramp terminal is the controlling feature of the ramp and will impact the selection of the ramp design speed. With a taper-type exit providing direct exit at a flat angle, there is a greater likelihood that the controlling feature of the ramp will be vehicle operations near the crossroad ramp terminal. With

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a parallel-type exit, there is a greater likelihood that the horizontal alignment immediately downstream of the deceleration lane on the ramp proper will be curvilinear, affect the speed of vehicles, and be the controlling feature of the ramp. If the controlling feature of the ramp is the horizontal alignment immediately downstream of the deceleration lane on the ramp proper, the ramp design speed may be higher than if the operational characteristics of the crossroad ramp terminal are the controlling feature and ramp design speeds will be lower.

In general, the controlling feature of an exit ramp may be the horizontal alignment or operational characteristics of the crossroad ramp terminal. A wide range of ramp design speeds are applicable to exit ramps ranging from 15 mph to 65 mph or higher. In particular, where the horizontal alignment of the ramp is curvilinear and the first curve encountered on the exit ramp is the controlling feature, ramp design speeds will likely be higher. If the horizontal alignment of the ramp is relatively straight and the operational characteristics of the crossroad ramp terminal are the controlling feature, the ramp design speed may be as low as 15 or 20 mph.

2.4 Freeway Mainline Ramp Terminal Configuration

As discussed in Section 2.3, the type of freeway mainline ramp terminal configuration (i.e., taper or parallel), has no direct impact on the selection of an appropriate ramp design speed. For design purposes, vehicles are assumed to accelerate at the same rates on both taper- and parallel-type entrance ramps and use the full acceleration length prior to merging. Similarly, for exit ramps, vehicles are assumed to decelerate at the same rates on both taper- and parallel-type exit ramps. As a result, the manner in which speed transitions occur between adjoining sections of a ramp is the same whether the freeway mainline ramp terminal is designed with a taper- or parallel-type configuration. Thus, the type of freeway mainline ramp terminal configuration does not impact the ramp design speed of a given ramp.

2.5 Type of Crossroad Ramp Terminal

A crossroad ramp terminal, associated with a service interchange, is that portion of the ramp that connects the ramp proper to the crossroad. Crossroad ramp terminals can be classified in several ways based upon distinguishing factors such as number of ramp legs, number of left-turn movements, location of crossroad left-turn storage, and type of traffic control.

In many cases, the controlling feature of an exit ramp may be the horizontal alignment immediately downstream of the freeway mainline ramp terminal. However, where the horizontal alignment of an exit ramp is relatively straight and has little impact on vehicle speeds, which is possible at diagonal ramps, then the operational characteristics of the crossroad ramp terminal will be the controlling feature of the ramp, and the type of traffic control at the crossroad ramp terminal (stop-controlled, signal controlled, or free-flow) will directly affect the selection of the ramp design speed.

The *Highway Safety Manual* (HSM) (AASHTO, 2010; AASHTO, 2014) indicates that the average speed of vehicles at the point where the ramp connects to the crossroad is typically around 15 mph where the crossroad ramp terminal is stop-, yield-, or signal-controlled and around 30 mph for all other ramps at service interchanges, which implies ramps with free-flow operations. However, for stop-, yield-, and signal-controlled terminals, which would be the case for all diagonal exit ramps, the ramps need to be designed to accommodate queued vehicles

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stopped for the traffic control. Thus, rather than designing to accommodate vehicles traveling 15 mph at the intersection of the ramp and crossroad, the ramp design speed needs to be selected based on vehicles stopped at the end of the queue storage leading to the crossroad. Thus, for all diagonal exit ramps where the horizontal alignment is relatively straight and has little impact on vehicle speeds and the crossroad ramp terminal is stop-, yield-, or signal-controlled, for practical purposes the ramp design speed should be 15 to 20 mph, which will still require vehicles to decelerate to a complete stop upon entering the functional area of the crossroad ramp terminal.

As described in Section 2.3.1, the controlling feature of an entrance ramp will always be based upon speeds or operational characteristics towards the end of the ramp near the freeway mainline ramp terminal. The speeds of vehicles entering a ramp will be affected by the type of traffic control at the crossroad ramp terminal; but as the length of the ramp increases, the speeds of vehicles towards the end of the ramp near the freeway mainline ramp terminal become less of a function of the vehicle speeds entering the ramp. Therefore, the type of crossroad ramp terminal of an entrance ramp indirectly impacts the selection of an appropriate ramp design speed. As ramp length increases, the type of traffic control at the crossroad ramp terminal of an entrance ramp has less impact on the selection of an appropriate ramp design speed.

2.6 Ramp Grade

The vertical profile of a ramp usually assumes the shape of the letter “S” with a sag curve at the lower end, a central portion on an appreciable grade, followed by a crest vertical curve at the upper end. Additional vertical curvature may be needed for ramps that cross other roadways. Most passenger vehicles can readily negotiate grades as steep as 4 to 6 percent without any appreciable loss in speed, but the effect of grades on truck speeds is much more pronounced.

Little is known about the relationship between roadway grades and design speeds for conventional roadways (i.e., arterials, collectors, etc.), and the same is true for the relationship between ramp grades and ramp design speeds (AASHTO, 2018). Therefore, ramp grades are not directly related to selecting appropriate ramp design speeds; however, ramps with higher ramp design speeds should generally have flatter gradients than ramps with lower ramp design speeds. The *Green Book* provides the following general criteria for upgrades on ramps:

- Upgrades on ramps with ramp design speeds of 45 to 50 mph should be limited to 3 to 5 percent.
- Upgrades on ramps with ramp design speeds of 35 to 40 mph should be limited to 4 to 6 percent.
- Upgrades on ramps with ramp design speeds of 25 to 30 mph should be limited to 5 to 7 percent.
- Upgrades on ramps with ramp design speeds of 15 to 20 mph should be limited to 6 to 8 percent.

The Institute of Transportation Engineers (ITE) *Freeway and Interchange Handbook* (Leisch, 2005) provides the following guidance for maximum gradients on ramps:

- Normal conditions: 4 to 6 percent.
- Heavy truck traffic: 3 to 4 percent.

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- Rugged terrain: 6 to 8 percent.

The lower values for each range of gradient are applicable to the higher ramp design speeds, particularly for major interchanges.

While the selection of ramp grade is important, the AASHTO *Green Book* states that "adequate sight distance is more important than a specific gradient control and should be favored in design," but acknowledges that these two controls are usually compatible (AASHTO, 2018).

2.7 Type of Horizontal Curvature

The horizontal alignment of an interchange ramp consists of a combination of tangents and curves. When pieced together, the individual tangents and curves form the shape of the ramp.

Four basic types of horizontal curves are used in the design of interchange ramps to achieve the desired shape and alignment:

- Simple curves with constant or uniform radii.
- Spiral curves with varying radii.
- Compound curves which consist of two or more simple curves with different radii, joined together, curving in the same direction.
- Reverse curves which consist of two simple curves joined together, curving in the opposite direction.

The specific shape of a ramp may be influenced by factors such as the traffic pattern, traffic volume, ramp design speed, topography, culture, intersection angle, and type of ramp terminal (AASHTO, 2018). In particular, compound and spiral curves may be used to obtain the desired alignment and fit the natural path of vehicles; but from the perspective of selecting an appropriate ramp design speed, there is no difference in the type of horizontal curves used in the design of a ramp. In other words, for a given ramp design speed and maximum superelevation rate, a simple curve, a spiral curve, and the individual simple curve of a compound or reverse curve are designed with the same curve radius and would be expected to result in the same operational conditions.

With respect to the horizontal curvature and selection of the appropriate ramp design speed, the key is designing the ramp to provide appropriate speed transitions from one adjoining section of the ramp to the next. Each individual horizontal curve and tangent has its own design speed, and the design speeds of adjoining sections should incrementally increase or decrease by no more than 10 to 15 mph to accommodate sequential increasing or decreasing speeds along the ramp.

2.8 Number of Lanes

As traffic volumes continue to increase and right-of-way for infrastructure becomes more limited, a potential treatment to add capacity at interchanges is to construct multilane ramps or convert single-lane ramps to multilane ramps. With higher traffic volumes and increased capacity, multilane ramps may naturally be considered or viewed as higher functional classification facilities compared to single-lane ramps; and it is common for higher functional class facilities to have

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higher design speeds than lower functional class facilities. For example, multilane loop ramps have higher capacities and typically consist of:

- More significant movements.
- Larger curve radii (ranging from 180 to 200 ft).
- Higher ramp design speeds.

Single-lane loop ramps, on the other hand, have lower capacities and typically consist of:

- Less significant movements.
- Smaller curve radii (ranging from 100 to 170 ft).
- Lower ramp design speeds.

However, the higher ramp design speeds found on multilane loop ramps are directly related to accommodating the more significant movements, and not to the number of lanes on the ramp.

Acceleration and deceleration properties of vehicle speeds and capabilities and driver behaviors are expected to be the same whether the ramp is single-lane or multilane ramp. For example, minimum acceleration lengths for entrance terminals and minimum deceleration lengths for exit terminals are assumed to be the same for design purposes of both single-lane and multilane ramps. Furthermore, research on single-lane and multilane loop ramps indicates that vehicle speeds and driver behaviors are comparable in either lane of multilane ramps (Torbic et al., 2016), which supports using the same ramp design speed for single-lane and multilane ramps.

2.9 Crossover Crown Line

At the boundary between the right lane of the freeway and the adjacent auxiliary lane of a freeway mainline ramp terminal, the cross slopes of the right lane of the freeway and auxiliary lane may differ. This boundary location is referred to as the crossover crown line and should not to be confused with the crown line normally provided at the centerline of the roadway or the cross-slope break between the traveled way and the shoulder. The maximum allowable difference in cross slopes at the crossover crown line, measured as is the algebraic difference in cross slopes of the right lane of the freeway and adjacent auxiliary lane, is an important control in developing superelevation along the freeway mainline ramp terminal and should not be designed such that it could cause driver discomfort or impact control of the vehicle.

Chapter 10 of the *Green Book* states that the maximum algebraic differences in the cross-slope rates at the crossover crown line should be based on the values in *Green Book* Table 9-18 which are intended for use with turning roadway terminals at intersections (AASHTO, 2018). *Green Book* Table 9-18, presented here as Table 3, provides suggested maximum algebraic differences in cross-slope rates at the crossover crown line for design speeds of exit and entrance curves for three speed categories: 20 mph and under, 25 and 30 mph, and 35 mph and over. The suggested maximum algebraic differences decrease as the speed categories increase.

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Table 3. Maximum Algebraic Difference in Cross Slope at Turning Roadway Terminals (AASHTO, 2018)

Design Speed of Exit or Entrance Curve (mph)	Maximum Algebraic Difference in Cross Slope at Crossover Crown Line (%)
20 and under	5.0 to 8.0
25 and 30	5.0 to 6.0
35 and over	4.0 to 5.0

From a design perspective, it seems logical to limit the maximum algebraic difference in cross-slope rates at the crossover crown line based on speed. At higher speeds of 35 mph and over, the maximum algebraic difference in cross slope at the crossover crown line is 4.0 to 5.0 percent. As speeds decrease, the maximum algebraic difference in cross slope at the crossover crown line increases to as high as 8.0 percent, and recent research (Torbic et al., 2016) shows that under normal scenarios vehicles will be able to recover when encountering a cross slope break of 8 percent or less. It is only under the most extreme departure scenarios that some types of heavy trucks may not recover (i.e., rollover) when encountering a cross slope break of 8 percent or less.

At the same time, there is no logical reason for the maximum allowable algebraic difference in cross slope at the crossover crown line to directly affect the selection of an appropriate ramp design speed for a given ramp. After the ramp design speed is selected, the ramp should be designed consistent with the current design criteria for the cross slope break at the crossover crown line.

2.10 Superelevation and Superelevation Transition

Superelevation is the rotation of the pavement surface on the approach to and through a horizontal curve. As necessary, horizontal curves are designed with superelevation to offset a portion of the lateral acceleration of the vehicle. The practical limits of superelevation on horizontal curves are controlled by several factors including climate conditions, terrain conditions, adjacent land use, and frequency of slow-moving vehicles (AASHTO, 2018). There is no universally applicable maximum superelevation rate for every region. Transportation agencies typically establish their own policies concerning the maximum superelevation rate that will be used on horizontal curves within their jurisdiction. Most transportation agencies use maximum superelevation rates of either 6 or 8 percent; but in regions where snow and ice are not a factor, a superelevation rate of 12 percent represents a practical maximum limit. One maximum superelevation rate should be used within a region to promote design consistency.

The maximum superelevation rate of a curve does not impact the selection of an appropriate ramp design speed for a given ramp. It will impact the minimum radius that can be designed for a given ramp design speed, but the maximum superelevation rate used by an agency does not factor into the selection of an appropriate ramp design speed for a given ramp.

The critical issue related to superelevation and ramp design speed is that adjoining sections of the ramp need to be of sufficient length to allow for appropriate speed transitions between sections and superelevation transition of the pavement between sections. For longer ramps, providing sufficient lengths to accommodate superelevation transitions should not be an issue; however, in constrained areas where individual components of ramps are designed near their

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minimum length requirements, superelevation transition of the pavement could become the controlling factor in the design of the adjoining sections of ramps.

2.11 Ramp Metering

Ramp metering regulates the flow of vehicles at entrance ramps to reduce turbulence in the merge area to improve the efficiency of freeway operations and reduce crash frequency. Ramp metering consists of traffic signals installed on the ramp proper near the freeway mainline ramp terminal to control the number of vehicles entering the freeway from the ramp. Ramp meters generally only operate during peak hours.

Ramp metering does not directly impact the selection of an appropriate ramp design speed because ramps should be designed for the higher ramp operating speeds that would be present when ramp meters are not operational. Other design considerations such as the number of lanes at the ramp meter stop bar, ramp traffic volume, ramp grade, and freeway design speed factor into the design of the length and geometry of the ramp proper and longer acceleration lengths necessary during metering operations. With the exception of the freeway design speed, these other design considerations do not affect the selection of the ramp design speed.

2.12 Collector-Distributor and Frontage Roads

The purpose of a collector-distributor (C-D) road is to eliminate weaving on the mainline freeway and reduce the number of entrances and exits on the through roadways while satisfying the demand for access to and from the freeway. The design speeds of C-D roads are usually equal to or less than the design speeds of the mainline freeway roadways. Frontage roads serve numerous functions depending on the facility they serve and the character of the surroundings. For example, frontage roads may control access to an arterial, function as a street facility serving adjoining properties, and maintain circulation of traffic on either side of the arterial. Frontage roads separate local traffic from higher speed through traffic and provide access to residences and commercial establishments along the highway.

C-D and frontage roads function as one of the intersecting highways connected by a ramp. A C-D road can serve as the equal or lower speed intersecting facility where connected to a freeway so the ramp design speed for a ramp connecting a freeway to a C-D road would be similar to that of a ramp for a system interchange. A C-D road can also serve as the higher speed facility where the ramp connects the C-D road to a surface street or crossroad. In this situation, the ramp design speed for a ramp connecting a C-D road to a surface street or crossroad would be similar to that of ramp for a service interchange. Similarly, the ramp design speed for a ramp connecting a freeway to a frontage road would be similar to that of a ramp for a service interchange. Otherwise, C-D and frontage roads do not affect the selection of the ramp design speed.

2.13 Physical Site Constraints, Environmental and Social Impacts, and Right-of-Way Costs

A combination of contextual considerations factor into the process for determining interchange type and location, and the length and orientation of each interchange ramp, including:

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- Physical site constraints such as topography, bodies of water (e.g., streams, lakes, and rivers), existing infrastructure, buildings, and right-of-way limits.
- Environmental conditions such as wetlands and endangered species.
- Social impacts related to displacement, barrier effects, accessibility, visual quality, land use, and noise.
- Accommodation of pedestrians and bicyclists.
- Economic factors such as construction, maintenance, vehicle operating, and right-of-way costs.

In general, any combination of these contextual considerations may affect the selection of an appropriate ramp design speed for a given ramp, and any one aspect of these contextual considerations does not necessarily carry more weight in the decision-making process. In other words, physical site constraints may be just as relevant and carry equal weight as environmental and social impacts and economic factors in the decision-making processes for determining interchange type and location and the length and orientation of each interchange ramp. Ultimately, these contextual considerations impact the overall design of each ramp, and since ramp design speed is the selected speed used to determine the various geometric design features of a given ramp, these contextual considerations directly impact the selection of the ramp design speed.

Conceptually, where these contextual considerations factor into the selection process for determining an appropriate ramp design speed for a given ramp, these contextual considerations can be viewed from a qualitative perspective and classified as either creating constrained or unconstrained conditions. For example, where the physical site characteristics for a given ramp may be considered typical or less restrictive than normal for a given rural or urban area, the site should be classified as being unconstrained. On the other hand, where physical site characteristics exist beyond what should be considered as typical or normal for a given rural or urban area, the site should be classified as constrained. Similarly, sites should be classified as unconstrained or constrained as they pertain to environmental and social impacts and economic factors.

As a general rule of thumb, ramp design speeds are higher for unconstrained ramps and lower for constrained ramps, consistent with driver behavior and expectations.

2.14 Ramp Design Speed Guide Values

Sections 2.1 through 2.13 address key elements related to interchange ramp design and discuss how the various elements directly or indirectly impact the selection of an appropriate ramp design speed or do not factor into the selection of ramp design speed. In particular, selection of an appropriate ramp design speed is most influenced by the type of interchange (i.e., system or service) and the type of ramp configuration. However, none of these elements (i.e., type of interchange, ramp configuration, entrance vs. exit ramp, freeway mainline ramp terminal configuration, type of crossroad ramp terminal, ramp grade, type of horizontal curvature, number of lanes, crossover crown line, superelevation, ramp metering, and C-D and frontage roads) is the single primary determinant in selecting an appropriate ramp design speed. In addition to the key ramp elements that directly or indirectly impact the selection of an appropriate ramp design, a combination of contextual considerations such as physical site constraints, environmental and social impacts, and right-of-way costs factor into the processes for determining interchange type

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and location and the length and orientation of each ramp and directly impact the selection of the ramp design speed as part of the geometric design process. Collectively, these key ramp elements and contextual considerations factor into the selection of an appropriate ramp design speed for a given ramp.

Based on the key elements and contextual considerations that factor into the geometric design process for ramps, Table 4 provides a range of guide values for ramp design speeds for system and service interchanges. In general, higher ramp design speeds are associated with ramps:

- For system interchanges compared to service interchanges.
- For ramp configurations that have mild or gentle horizontal curvature compared to sharp curvature.
- In rural areas compared to urban areas.
- At service interchanges that intersect higher functional class roadways (i.e., primary/major arterials) compared to lower functional class roadways (i.e., minor arterials/collectors).
- In unconstrained conditions compared to constrained conditions.

Table 4 provides a range of guide values given the context, ramp configuration, and type of interchange. For the specified design condition, it is recommended that the ramp design speed for a given ramp be within the range of guide values. The lower guide values are generally associated with constrained conditions, and the higher guide values are associated with unconstrained conditions. Typically, the range in guide values for a given context and ramp configuration is between 5 and 15 mph. The maximum range in guide values for a given context and ramp configuration is 30 mph. Where no guide value(s) is provided in the table, it is implied that the type of context and ramp configuration is not applicable. For example, it is uncommon for freeways in urban areas to designed with design speeds of 80 to 85 mph. Similarly, system interchanges always connect two or more freeways. Thus, at a minimum the design speed of the intersecting freeway will be at least 50 mph. Therefore, no guide value or values for ramp design speed are provided for these uncommon design conditions.

Where the horizontal alignment of the ramp is curvilinear, the ramp design speed applies to the controlling curve on the ramp proper. For an entrance ramp, the last curve encountered along the ramp proper that significantly affects vehicle speed is considered the controlling curve (see Figure 13). For an exit ramp, the controlling curve is the first curve encountered along the ramp proper that significantly affects vehicle speed. For design purposes, it can be assumed that horizontal curves with radii less than 1,000 ft will significantly affect vehicle speeds. For direct, semidirect, and outer connection ramps, the sharpest curve on the ramp proper is considered the controlling curve (see Figure 14).

As noted at the bottom of Table 4, where the horizontal alignment of the ramp is relatively straight and has little or no influence on vehicle speeds, the ramp design speed applies to the primary tangential section of the ramp proper. For entrance ramps that are relatively straight, it is recommended that the ramp design speed be within 15 to 20 mph of the highway design speed. For exit ramps that are relatively straight, it is recommended that the ramp design speed be around 15 to 20 mph due to operational characteristics of the crossroad ramp terminals.

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Table 4. Range of Guide Values for Ramp Design Speed as Related to Highway Design Speed, Interchange Type, Ramp Configuration, and Contextual Considerations

Context		Ramp configuration	Highway design speed (mph)											
			30	35	40	45	50	55	60	65	70	75	80	85
			Ramp design speed (mph)											
System Interchanges														
Rural		Direct connections	-	-	-	-	40-45	40-50	40-55	45-60	45-65	50-70	50-75	50-80
		Semidirect connections	-	-	-	-	30-40	30-45	35-45	35-50	40-55	40-60	40-65	45-65
		Loop ramps	-	-	-	-	25-30	25-30	25-35	25-35	30-35	30-40	35-45	35-45
Urban		Direct connections	-	-	-	-	35-40	35-40	35-50	40-50	40-55	40-60	-	-
		Semidirect connections	-	-	-	-	30-35	30-40	30-45	30-45	35-50	35-55	-	-
		Loop ramps	-	-	-	-	20-25	20-30	25-30	25-30	25-35	25-40	-	-
Service Interchanges														
Rural:	Primary or major arterial	Direct connections	-	-	-	35-40	35-45	40-45	40-45	40-50	45-50	45-55	50-60	50-65
		Diagonal ramps[1]	20-25	20-25	25-30	25-35	30-40	30-45	30-50	30-55	35-55	40-60	45-65	45-70
		Loop ramps	15-20	20-25	20-25	20-25	20-25	25-30	25-30	25-30	30-35	30-40	35-45	35-45
	Minor arterial or collector	Diagonal ramps[1]	20	20-25	25	25-30	25-35	30-40	30-45	30-50	35-50	40-55	45-60	45-65
		Loop ramps	15-20	15-20	15-20	15-25	20-25	20-30	20-30	25-30	25-35	30-40	30-45	30-45
Urban:	Primary or major arterial	Direct connections	-	-	-	30-35	35-40	35-40	35-40	40-45	40-50	40-55	-	-
		Diagonal ramps[1]	15-20	20-25	25-30	25-30	30-35	30-40	30-45	30-45	35-50	40-55	-	-
		Loop ramps	15-20	15-20	15-20	20-25	20-25	20-30	25-30	25-30	25-35	25-40	-	-
	Minor arterial or collector	Diagonal ramps[1]	15	20	20-25	20-30	25-35	25-40	25-45	30-45	30-50	35-55	-	-
		Loop ramps	15-20	15-20	15-20	15-25	15-25	20-30	20-30	20-30	20-35	25-40	-	-

[1] Where the horizontal alignment of a diagonal entrance ramp is relatively straight and has little impact on vehicle speeds, the ramp design speed is dependent on the operational characteristics of the freeway mainline ramp terminal. For this situation, it is recommended that the ramp design speed be within 15 to 20 mph of the highway design speed.

Where the horizontal alignment of a diagonal exit ramp is relatively straight and has little impact on vehicle speeds, the ramp design speed is dependent on the operational characteristics of the crossroad ramp terminal. The recommended ramp design speed for this situation is 15 or 20 mph, as it is expected that the crossroad ramp terminal at all diagonal ramps will either be stop-, yield-, or signal-controlled.

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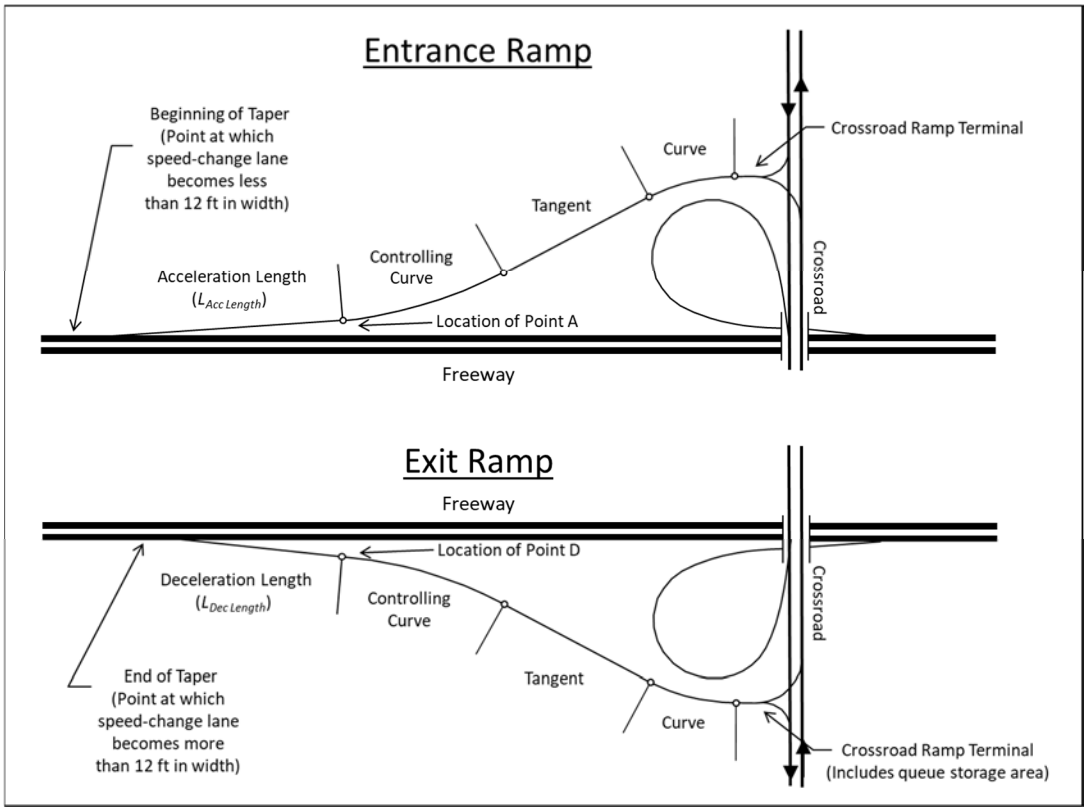


Figure 13. Primary Components and Controlling Curves at Entrance and Exit Ramps at Service Interchanges

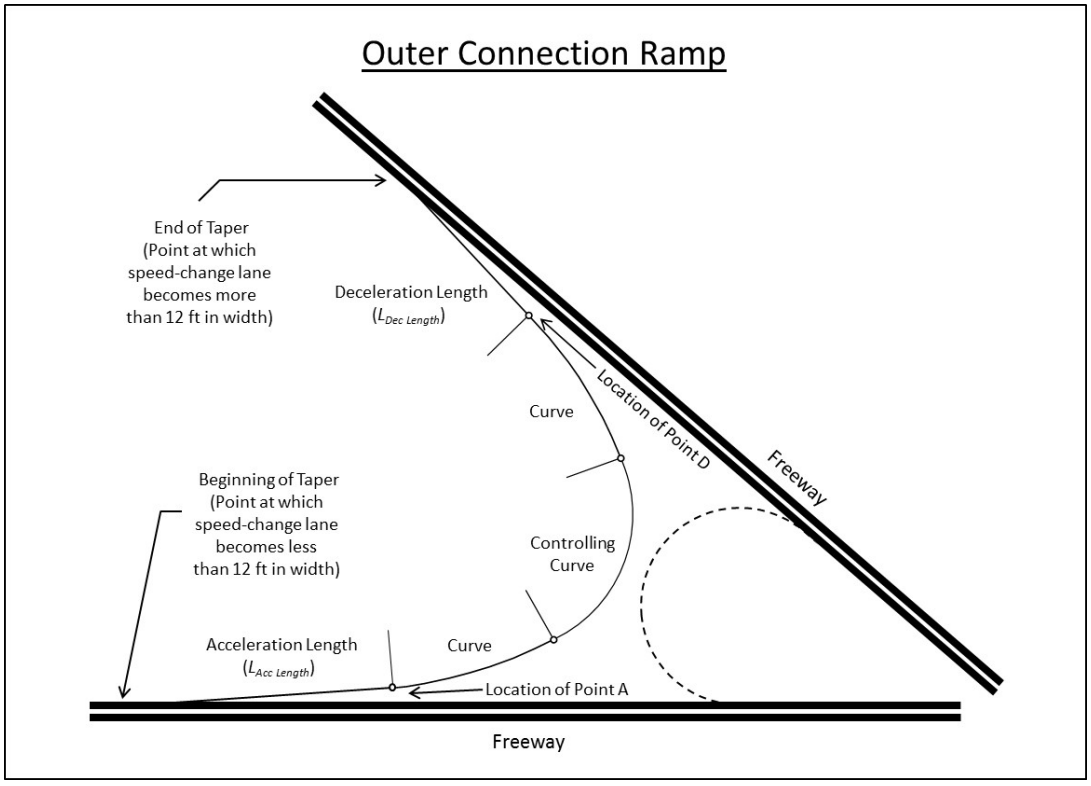


Figure 14. Primary Components and Controlling Curve for Outer Connection Ramp at a System Interchange

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By providing a range of guide values for ramp design speeds, Table 4 provides designers with flexibility in determining the contextual and design considerations that apply for a given ramp (e.g., in determining whether a ramp is located in an unconstrained or constrained area). The range of guide values in Table 4 were developed to be relatively consistent with guidance presented in the *Green Book* (AASHTO, 2018). The *Green Book* states that for direct connections ramp design speeds between the middle and upper ranges shown in *Green Book* Table 10-1 should be used, and 40 mph is preferably the minimum ramp design speed for a direct connection. As presented in Table 4, for a majority of the conditions, the minimum ramp design speed is 40 mph or greater for direct connections, and the range of values includes values in the middle and upper ranges of *Green Book* Table 10-1.

For semidirect connections, again the *Green Book* states ramps design speeds between the middle and upper ranges shown in *Green Book* Table 10-1 should be used, and ramp design speeds for semidirect connections are typically 30 to 40 mph. The *Green Book* continues indicating that ramp design speeds less than 30 mph should not be used for semidirect connections; and for a short single-lane ramp, a ramp design speed of 50 mph is generally not practical. As presented in Table 4, for all of the conditions, the minimum ramp design speed is 30 mph or greater; and for all but one of the conditions, values of 30, 35, and/or 40 mph are included in the range of guide values. It is only for the rural system interchange with a highway design speed of 80 mph where values of 30, 35, and/or 40 mph are not included in the range of guide values. For this condition (i.e., rural system interchange with a highway design speed of 80 mph), the minimum ramp design speed is 45 mph.

For loop ramps, the *Green Book* states that upper-range values of ramp design speeds shown in *Green Book* Table 10-1 are generally not attainable. Ramp design speeds of 30 mph for loops encompass large land areas rarely available in urban areas. Long loop ramps needed for higher design speeds are costly and require drivers to travel a considerable extra distance. In addition, minimum values usually control; but for highway design speeds of 55 mph and greater, the ramp design speed should be no less than 20 mph. Where less restrictive conditions exist, the ramp design speed and consequently the radius of the loop ramp may be increased. As presented in Table 4, for a majority of the conditions for service interchanges, the minimum ramp design speed for a loop ramp is either 15, 20, or 25 mph; and for a majority of the conditions for system interchanges, the range of values includes values in the lower range of *Green Book* Table 10-1.

In some cases, the range of guide values in Table 4 extend slightly beyond the range of values suggested in the *Green Book*. This was done intentionally. A table of guide values should not be too prescriptive that it limits flexibility and inhibits designers from exercising engineering judgement. In other cases, extending beyond the range of values suggested in the *Green Book* was necessary to address the fact that many states are posting freeway speed limits as high as 75, 80, and 85 mph, and assistance for selecting ramp design speeds to address such conditions is necessary.

Selecting the appropriate ramp design speed is only the first step in designing a ramp that meets driver expectations. Another critical step in the design process for designing a ramp that meets driver expectations is designing the other components and sections of the ramp consistent with the overall ramp design speed such that appropriate speed transitions between adjoining components and sections can take place. For this to occur, in most situations, the change in design speed between adjoining sections should be limited to no more than 10 to 15 mph.

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Section 3.
Guidelines for Designing Ramps in a Consistent Manner Based on the Selected Ramp Design Speed

Interchange design and the design and orientation of each individual interchange ramp is an art form that requires consideration of a wide range of factors including:

- Number of intersecting legs.
- Classification of intersecting facilities.
- Volume and pattern of existing and future traffic.
- Environmental requirements.
- Local access and circulation considerations.
- Physical constraints and right-of-way considerations.
- Construction and maintenance costs.
- Road-user costs (i.e., costs related to safety and operations).
- Design controls.
- Signing.

The decision on selecting an appropriate ramp design speed and then designing the horizontal and vertical alignments to allow for appropriate speed transitions along a ramp is still left to the engineering judgement of the designer. The design guidelines presented in this section can be used during the design process to help design a ramp in a consistent manner based on the selected ramp design speed. First, several general design guidelines are presented followed by design controls and criteria for entrance and exit ramps.

3.1 Design Guidelines

Several guidelines for designing an interchange ramp in a consistent manner based on the selected ramp design speed are as follows:

1. The ramp design speed is typically associated with an individual segment of a ramp near the upstream or downstream end of the ramp. For an entrance ramp, the ramp design speed is normally associated with a downstream portion of the ramp; and with an exit ramp, the ramp design speed is usually associated with a segment on the upstream portion of the ramp. It is desirable (but not essential) that the design speeds of the contiguous segments along the ramp proper increase or decrease in a stepwise manner; and unless a segment is rather long, the change in design speed between adjoining sections should be limited to no more than 10 to 15 mph. For entrance ramps, this means that the design speeds of the contiguous segments along the ramp proper should typically increase in a stepwise manner; and for exit ramps the design speeds of the contiguous segments along the ramp proper should decrease in a stepwise manner. Decreasing design speeds sequentially along a ramp in a stepwise manner is more critical for exit ramps than increasing design speeds along a ramp for entrance ramps, as unexpected and abrupt changes in design speeds along an exit ramp are more likely to be associated with potential lane keeping and/or loss of control issues.

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The design speed of the controlling feature of the ramp (i.e., the ramp design speed) is first selected, and then the design speeds of the other elements or segments of the ramp proper should be determined accordingly.

2. Each component of the ramp should be designed such that a vehicle has sufficient distance to accelerate or decelerate within the performance capabilities of the vehicle and driver expectations given the design speeds and features of the individual components of the ramp. Several design controls and criteria can be used in the design of entrance and exit ramps so that the individual components and features of the ramps are designed in a consistent manner, accounting for sequential speed transitions from one component or section to the next.
3. Each ramp component should be designed in a consistent manner based on the design speed of the component and anticipated operating speed along the component. The design speed of each individual curve and tangent along the ramp proper should be consistent with the speeds that drivers are likely to be traveling along the respective curve and tangent. If the design speed of an individual curve or tangent (or multiple curves and tangents) along the ramp proper is not consistent with the speeds that drivers are likely to be traveling along the respective curve and tangent, consider potential options so that the design speed would be more consistent with operating speeds. For example, if the anticipated operating speed exceeds the design speed, consider changing (e.g., raising or potentially decreasing) the design speed for the respective section. If changing the design speed of an individual section means that the design speeds of the contiguous segments along the ramp proper no longer increase or decrease in a stepwise manner, consider changing the ramp design speed. If the design speed or ramp design speed cannot be changed (e.g., due to physical site constraints, environmental or social impacts, economic constraints, etc.), consider implementing speed-control measures so that the design speed or ramp design speed would be more consistent with operating speeds.

3.2 Design Controls and Criteria for Ramps

This section describes several design controls and criteria that could be used to design a ramp in a consistent manner based on the selected ramp design speed. Design controls are attributes, values, or qualities that influence discrete geometric element dimensions or considerations. Design criteria are dimensions and values that meet design control needs, such as curve radius, acceleration lengths, and deceleration lengths. Section 3.2.1 addresses design controls and criteria for entrance ramps, followed by Section 3.2.2 which addresses design controls and criteria for exit ramps. Section 3.2.3 addresses design controls and criterial for freeway-to-freeway ramps.

3.2.1 Guidelines for Entrance Ramps

With entrance ramps, vehicle acceleration is a primary design control. Individual components and features of the ramp should be designed with sufficient length such that vehicles may accelerate in a comfortable and controlled manner from an initial speed to a final speed over the length of the ramp so vehicles can merge onto the through lanes of the primary highway or freeway at a speed approximately equal to the operating speed of the primary highway or freeway. The design controls and criteria discussed below for entrance ramps are presented as

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appropriate in order of the primary components of an entrance ramp based on the direction of travel. As such, applicable design controls and criteria are first discussed for the crossroad ramp terminal, then the ramp proper, and finally the freeway mainline ramp terminal.

The first component of an entrance ramp is the crossroad ramp terminal. The crossroad ramp terminal does not have an associated design speed assigned to it. Therefore, the design of the entrance ramp is based on the operating speed of vehicles entering the ramp proper from the crossroad ramp terminal. As vehicles enter from the crossroad ramp terminal onto the ramp proper, typically they enter the ramp proper at lower speeds depending upon the design and type of traffic control of the crossroad ramp terminal. Where stop-, yield-, or signal control are implemented at a crossroad ramp terminal, vehicles are expected to enter the ramp proper at speeds around 15 mph (AASHTO, 2014). If free-flow operations are provided at the crossroad ramp terminal, vehicle speeds may be higher entering the ramp proper.

For the ramp proper of an entrance ramp, AASHTO policy does not provide specific design criteria for determining the minimum lengths of individual tangent and curve sections to allow for vehicles to accelerate to an assumed or defined speed prior to entering the freeway mainline ramp terminal. However, *Green Book* Figure 2-33 (Acceleration of Passenger Cars, Level Conditions) (see Figure 15) may be used as guide for determining minimum practical lengths of individual tangent and curve sections along the ramp proper (AASHTO, 2018). *Green Book* Figure 2-33 depicts acceleration capabilities of lower performance vehicles which may be used for design. Table 5 presents acceleration distances from *Green Book* Figure 2-33 in tabular form for entering and exiting speeds that would be consistent for the design of entrance ramps.

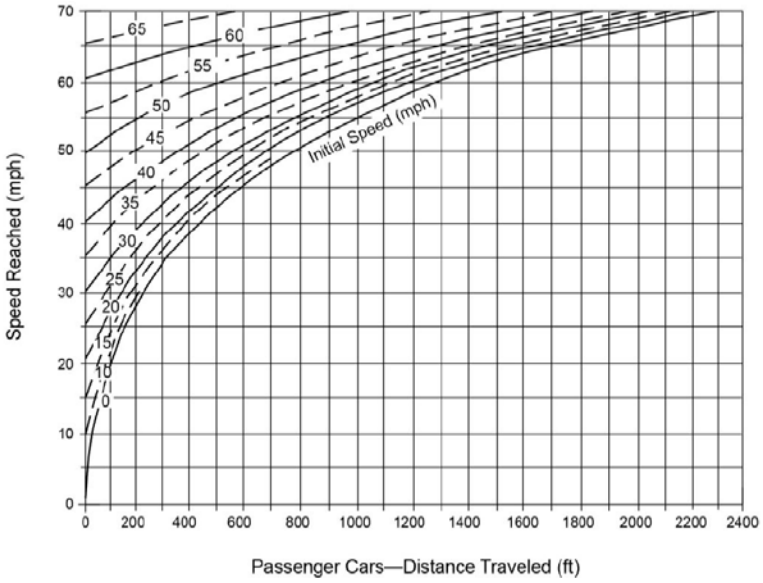


Figure 15. Acceleration of Passenger Cars, Level Condition (AASHTO, 2018)

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Table 5. Minimum Lengths for Tangents and Curves on Ramp Proper of Entrance Ramps

Speed Exiting the Tangent or Curve on the Ramp Proper (Final Speed) (mph)	Recommended Minimum Length of Tangent or Curve on Ramp Proper to Provide for Vehicle Acceleration[1] (ft)								
	Initial Speed Entering Tangent or Curve on the Ramp Proper (Initial Speed) (mph)								
	15	20	25	30	35	40	45	50	55
20	50								
25	110	70							
30	175	140	75						
35	275	245	175	100					
40	380	355	300	230	105				
45	540	500	445	365	265	140			
50	735	685	615	560	445	355	200		
55	945	875	825	775	670	575	415	210	
60	1230	1170	1125	1050	965	860	730	500	285

[1] Lengths based on acceleration capabilities of passenger cars on level conditions as depicted in *Green Book* Figure 2-33.

Green Book Figure 2-33 may be used when referring to both initial and final design speeds and initial and final anticipated operating speeds. The speed distance relationships depicted in *Green Book* Figure 2-33 do not account for the effects of horizontal or vertical alignments. The lengths should be adjusted accordingly to account for the effects of horizontal curvature and/or grade.

The selected design speed of an individual tangent section should be consistent with the speeds that drivers are likely to be traveling at the end of the tangent. With curves on entrance ramps, it may also be assumed that vehicle speeds will increase slightly over the length of the curve depending on the curve radius, so the speeds of vehicles exiting a curve may be slightly higher than the speeds of vehicles entering the curve. Speeds associated with the midpoint of the curve are to be used to design the curve rather than the speeds at the beginning or end of the curve. Again, the design speed of the curve should be consistent with the speeds that drivers are likely to be traveling at the midpoint of the curve.

For consistency, the anticipated operating speed exiting a tangent section or curve should be equivalent to the anticipated operating speed entering the adjoining downstream tangent section or curve. As operating speeds are expected to increase along an entrance ramp from section to section, subsequently the design speeds of the individual sections should also increase. As a general rule, the change in design speed between adjoining sections should be limited to no more than 10 to 15 mph. This general rule is not as critical for longer sections of ramps.

As vehicles transition from the ramp proper to the freeway mainline ramp terminal of an entrance ramp, the primary design criterion provided in AASHTO policy addresses minimum acceleration lengths (see *Green Book* Table 10-4) along the freeway mainline ramp terminal. The freeway mainline ramp terminal portion of a ramp does not have an associated design speed.

Green Book Table 10-4 is presented below as Table 6 with a few adaptions. The headings were adapted to be more consistent with the design guidelines presented herein. In addition, the design values were expanded to include additional design conditions for higher ramp design speeds and higher highway (or freeway) design speeds. The expanded design values were estimated for use with the case studies in Section 5 and recognizing the limitations of Table 10-4 in the 2018

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edition of the *Green Book* to be used in the design of higher speed facilities. The new design values (shown in red text in Table 6) were estimated through numeric extrapolation of the minimum acceleration lengths for the lower speed design conditions and to be consistent with the trends and acceleration rates of the design values in the original table. They were not developed based on field data or vehicle dynamics models that estimate vehicle performance. It is recommended that additional research be conducted to verify and/or modify these design values for higher speed facilities.

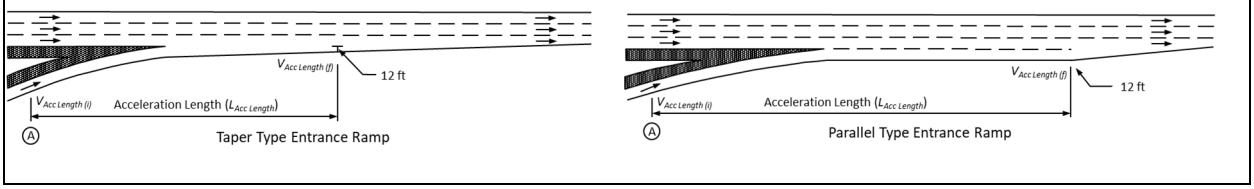
With *Green Book* Table 10-4 and Table 6 presented below, the design speed of the final section of the ramp proper upstream of the gore point and the design speed of the freeway are used to determine the minimum acceleration length for the freeway mainline ramp terminal. Both tables also provide anticipated speeds entering the freeway mainline ramp terminal and merging with freeway traffic for given design speeds. For the freeway mainline ramp terminal to be designed in a consistent manner with the upstream components of the ramp, the anticipated speed of vehicles exiting the ramp proper should be equivalent to the speed of vehicles entering the freeway mainline ramp terminal [i.e., the initial speed of vehicles entering the acceleration length ($V_{Acc\ Length(i)}$)]. Thus, although the freeway mainline ramp terminal does not have an associated design speed, the operational and design conditions of the freeway mainline ramp terminal are related to and coordinated with the operational and design conditions of the individual sections of the upstream components of the ramp.

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Table 6. Minimum Acceleration Lengths for Entrance Terminals with Flat Grades of Less Than 3 Percent (adapted from AASHTO, 2018)

Highway		Acceleration Length, $L_{Acc\ Length}$ (ft), for Design Speed (V_{DS}) of Controlling Feature on the Ramp Proper (or the Final Section of the Ramp Proper Upstream of the Gore Point)											
		Stop Condition	15	20	25	30	35	40	45	50	55[1]	60[1]	65[1]
Design Speed $V_{DS\ Hwy}$ (mph)	Merge Speed $V_{Acc\ Length(f)}$ (mph)	Average Running Speed (i.e., Initial Speed) Exiting the Controlling Feature on the Ramp Proper (or the Final Section of the Ramp Proper Upstream of the Gore Point), $V_{Acc\ Length(i)}$											
		0	14	18	22	26	30	36	40	44	48	52	56
30	23	180	140										
35	27	280	220	160									
40	31	360	300	270	210	120							
45	35	560	490	440	380	280	160						
50	39	720	660	610	550	450	350	130					
55	43	960	900	810	780	670	550	320	150				
60	47	1200	1140	1100	1020	910	800	550	420	180			
65	50	1410	1350	1310	1220	1120	1000	770	600	370	135		
70	53	1620	1560	1520	1420	1350	1230	1000	820	580	350	135	
75	55	1790	1730	1630	1580	1510	1420	1160	1040	780	550	300	
80	57	2000	1900	1800	1750	1680	1600	1340	1240	980	725	500	200
85[1]	59	2230	2090	1985	1930	1865	1790	1525	1455	1250	940	700	500

NOTE: Uniform 50:1 to 70:1 tapers are recommended where acceleration lengths exceed 1,300 ft.



[1] Acceleration lengths estimated based on extrapolation.

The minimum acceleration lengths associated with freeway mainline ramp terminals of entrance ramps are calculated using a simple kinematic equation (see Equation 2) with constant acceleration. The minimum acceleration length is based on:

A. The speeds at which drivers enter the freeway mainline ramp terminal (i.e., the speed at the end of the ramp proper's controlling feature).

B. An assumed acceleration rate.

C. The speeds at which drivers merge with through traffic (i.e., the speed at the end of the acceleration length).

$$L_{Acc\ Length} = \frac{\left(1.47V_{Acc\ Length(f)}\right)^2 - \left(1.47V_{Acc\ Length(i)}\right)^2}{2a} \quad \textbf{(2)}$$

where:

$L_{Acc\ Length}$ = acceleration length, ft.

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$V_{Acc\ Length(f)}$	=	merge speed (i.e., the speed at which a driver merges with through traffic at the end of the acceleration length), mph.
$V_{Acc\ Length(i)}$	=	initial speed entering the freeway mainline ramp terminal of an entrance ramp from the ramp proper (i.e., the speed exiting the controlling feature of the ramp proper), mph.
A	=	acceleration rate, ft/s^2.

Green Book Table 10-4 (and corresponding Table 6) provides minimum acceleration lengths for various combinations of design conditions for both taper- and parallel-type exit terminals on level grades. The minimum acceleration lengths are to be adjusted in accordance to *Green Book* Table 10-5 where grades are present.

Several points are worth noting concerning the basis for determining the minimum acceleration lengths within AASHTO policy:

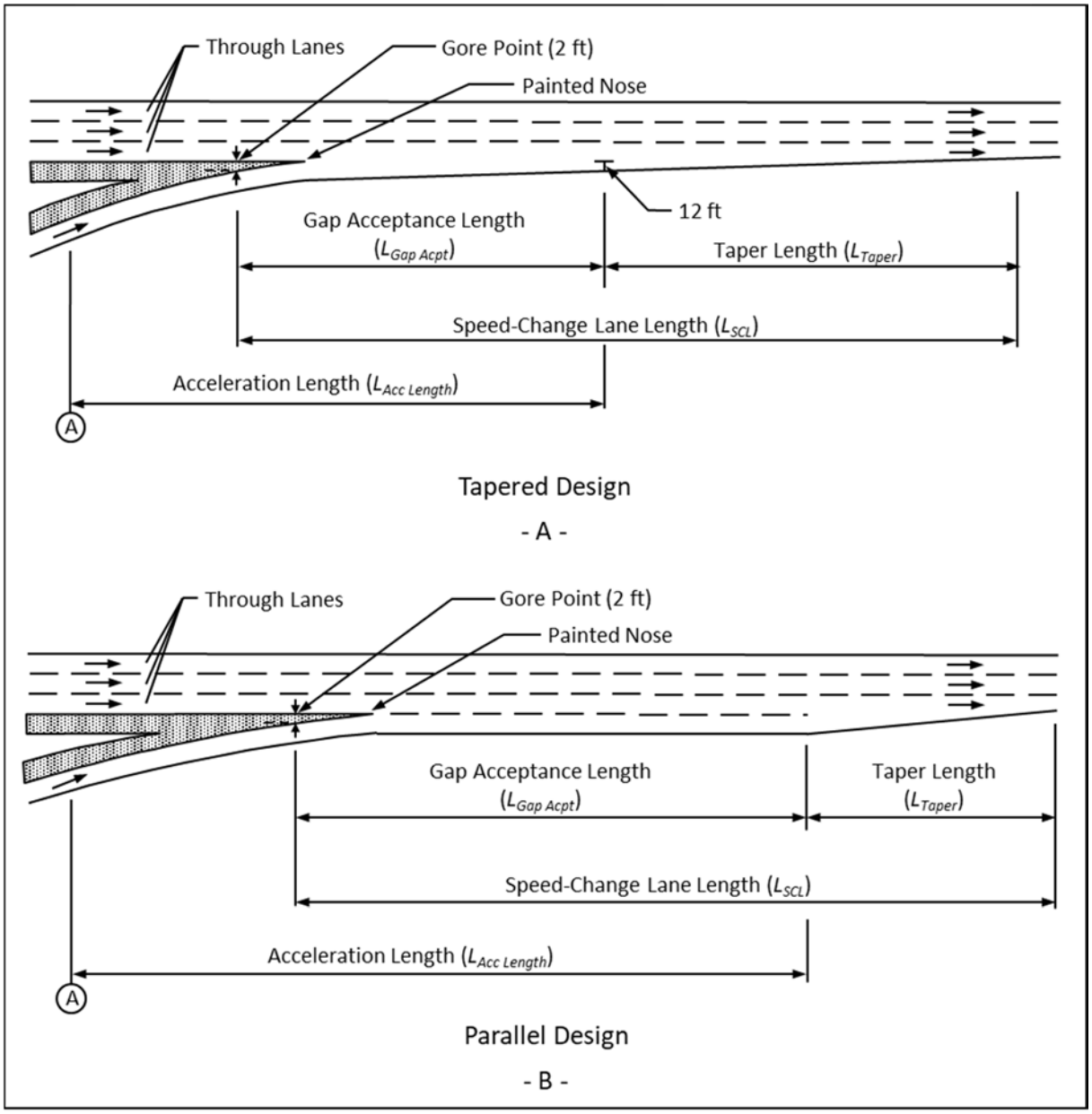
- The acceleration rates used to determine the acceleration distances are based on "normal acceleration" for passenger vehicles on level grade.
- The speed of vehicles entering the primary highway or freeway from the freeway mainline ramp terminal should approximate that of the through traffic on the highway or freeway. In addition, it is satisfactory and does not unduly inconvenience through traffic if vehicles enter the highway at a speed approximately 5 mph less the speed of traffic on the highway.

Figure 16 illustrates the primary components associated with the freeway mainline ramp terminal of an entrance ramp, including but not limited to:

- A **speed-change lane** or auxiliary lane, including tapered area, is provided primarily for acceleration of vehicles entering through traffic. The speed-change lane joins the traveled way of the turning roadway to the highway and does not necessarily imply a definite lane of uniform width. For an entrance ramp, the length of the speed-change lane is measured from the gore point to the end of the taper. The speed-change lane should have sufficient length to enable a driver to make the appropriate change in speed between the ramp proper and intersecting highway.
- The **gap acceptance length** is that portion of the speed-change lane necessary to allow motorists to evaluate gaps in the freeway traffic and position their vehicles to use the gap. The gap acceptance length begins at the gore point and ends where the width of the speed-change lane reduces to less than 12 ft. The gap acceptance length should be a minimum of 300 to 500 ft depending on the nose width. The gap acceptance length is always less than or equal to the acceleration length.
- The **acceleration length** is the distance needed for acceleration upon exiting the controlling feature of the ramp proper or the final section of the ramp proper upstream of the gore point to the point of convergence with the freeway. The acceleration length begins at Point A and ends where the gap acceptance length ends and the taper begins. Point A represents the location where the ramp proper ends and the freeway mainline ramp terminal begins. Conceptually, the acceleration length begins at the end of the controlling curve or feature of the ramp proper; however, if the radius of the final curve of the ramp is greater than 1,000 ft and drivers on the ramp have a clear view of traffic in

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the right lane of the freeway, the beginning of the acceleration length may be located somewhere along the curve. This means the beginning of the acceleration length may be located upstream of the gore point.



Notes:

1. $L_{Acc\ Length}$ is the recommended acceleration length as shown in Table 6 or adjusted due to grade.
2. $L_{Gap\ Acpt}$ is the gap acceptance length and should be a minimum of 300 to 500 ft depending on the nose width.
3. The value of $L_{Acc\ Length}$ or $L_{Gap\ Acpt}$ whichever produces the greater distance downstream from the gore point is suggested for use in the design of the freeway mainline ramp terminal.

Figure 16. Primary Components of Freeway Mainline Ramp Terminal of an Entrance Ramp (adapted from AASHTO, 2018)

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3.2.2 Guidelines for Exit Ramps

With exit ramps, a primary design control is vehicle deceleration. Individual components and features of the ramp should be designed with sufficient length such that vehicles decelerate in a controlled manner from an initial to final speed over the length of the section and do not exceed an assumed maximum deceleration. The design controls and criteria discussed below for exit ramps are presented as appropriate in order of the primary components of an exit ramp based on the direction of travel. As such, applicable design controls and criteria are first discussed for the freeway mainline ramp terminal, then the ramp proper, and finally the crossroad ramp terminal.

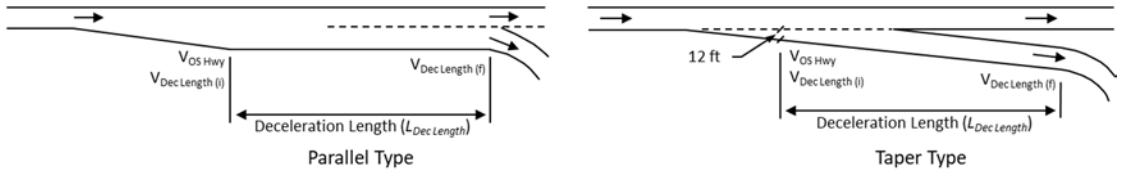
As vehicles transition from the primary highway or freeway to the freeway mainline ramp terminal of an exit ramp and then to the ramp proper, the primary design criterion associated with vehicle deceleration addresses the minimum deceleration lengths along the freeway mainline ramp terminal (see *Green Book* Table 10-6). *Green Book* Table 10-6 is presented below as Table 7 with a few adaptions. The headings were adapted to be more consistent with the design guidelines presented herein. In addition, the design values were expanded to include additional design conditions for higher ramp design speeds and higher highway (or freeway) design speeds. The expanded design values were estimated recognizing the limitations of Table 10-6 in the 2018 edition of the *Green Book* to be used in the design of higher speed facilities. The new design values (shown in red text in Table 7) were estimated through numeric extrapolation of the minimum deceleration lengths for the lower speed design conditions and to be consistent with the trends and acceleration rates of the design values in the original table. They were not developed based on field data or vehicle dynamics models that estimate vehicle performance. It is recommended that additional research be conducted to verify and/or modify these design values for higher speed facilities.

With *Green Book* Table 10-6 (Table 7), the minimum deceleration length for the freeway mainline ramp terminal is determined based on the design speed of the freeway and the design speed of the controlling feature of the ramp proper or the first section of the ramp proper downstream of the gore point. *Green Book* Table 10-6 (Table 7) also provides anticipated speeds of vehicles diverging from the freeway and exiting the freeway mainline ramp terminal. For the freeway mainline ramp terminal to be designed in a consistent manner with the downstream components of the ramp, the anticipated speed of vehicles exiting the freeway mainline ramp terminal [i.e., the final speed of vehicles exiting the deceleration length ($V_{Dec\ Length(f)}$)] should be equivalent to the speed of vehicles entering the ramp proper. In this way, the operational and design conditions of the freeway mainline ramp terminal are related to and coordinated with the operational and design conditions of the individual sections of the ramp proper.

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Table 7. Minimum Deceleration Lengths for Exit Terminals with Flat Grades of Two Percent or Less (adapted from AASHTO, 2018)

Highway Design Speed ($V_{DS\ Hwy}$) (mph)	Diverge Speed ($V_{OS\ Hwy}$) ($V_{Dec\ Lenth\ (i)}$) (mph)	Deceleration Length, $L_{Dec\ Length}$ (ft), for Design Speed (V_{DS}) of Controlling Feature on the Ramp Proper (or the Initial Section of the Ramp Proper Downstream of the Gore Point) (mph)								
		Stop Condition	15	20	25	30	35	40	45	50
		Average Running Speed at the End of the Deceleration Length, $V_{Dec\ Length\ (f)}$ (mph), Entering the Controlling Feature on the Ramp Proper (or the Initial Section of the Ramp Proper Downstream of the Gore Point)								
		0	14	18	22	26	30	36	40	44
30	28	235	200	170	140	-	-	-	-	-
35	32	280	250	210	185	150	-	-	-	-
40	36	320	295	265	235	185	155	-	-	-
45	40	385	350	325	295	250	220	-	-	-
50	44	435	405	385	355	315	285	225	175	-
55	48	480	455	440	410	380	350	285	235	-
60	52	530	500	480	460	430	405	350	300	240
65	55	570	540	520	500	470	440	390	340	280
70	58	615	590	570	550	520	490	440	390	340
75	61	660	635	620	600	575	535	490	440	390
80	64	705	680	665	645	620	580	535	490	440
85[1]	67	750	725	710	690	665	625	580	540	485



[1] Deceleration lengths estimated based on extrapolation.

The basis for the recommended minimum deceleration lengths in the *Green Book* is as follows. It is assumed that a vehicle maintains its speed in the freeway lanes and first decelerates after exiting the freeway mainline and upon entering the deceleration length portion of the speed-change lane (see Figure 17). It is assumed that a vehicle completely diverges from the freeway mainline lanes where the width of the auxiliary lane increases to 12 ft or greater. This point is the end of taper and the beginning of both the divergence zone and the deceleration length. The divergence zone allows for deceleration after clearing the through-traffic lane of the freeway and extends to the gore point. For a taper-type exit ramp, the divergence zone may be relatively short. The deceleration length is the distance needed for deceleration after clearing the through-traffic lane of the freeway and before reaching the first location that significantly limits vehicle speeds on the ramp proper or the first section of the ramp proper downstream of the gore point. As illustrated in Figure 17, the deceleration length ends at Point D, which represents the location where the freeway mainline ramp terminal ends and the ramp proper begins. Point D cannot be located upstream of the gore point on the speed-change lane but may be located downstream of the gore point.

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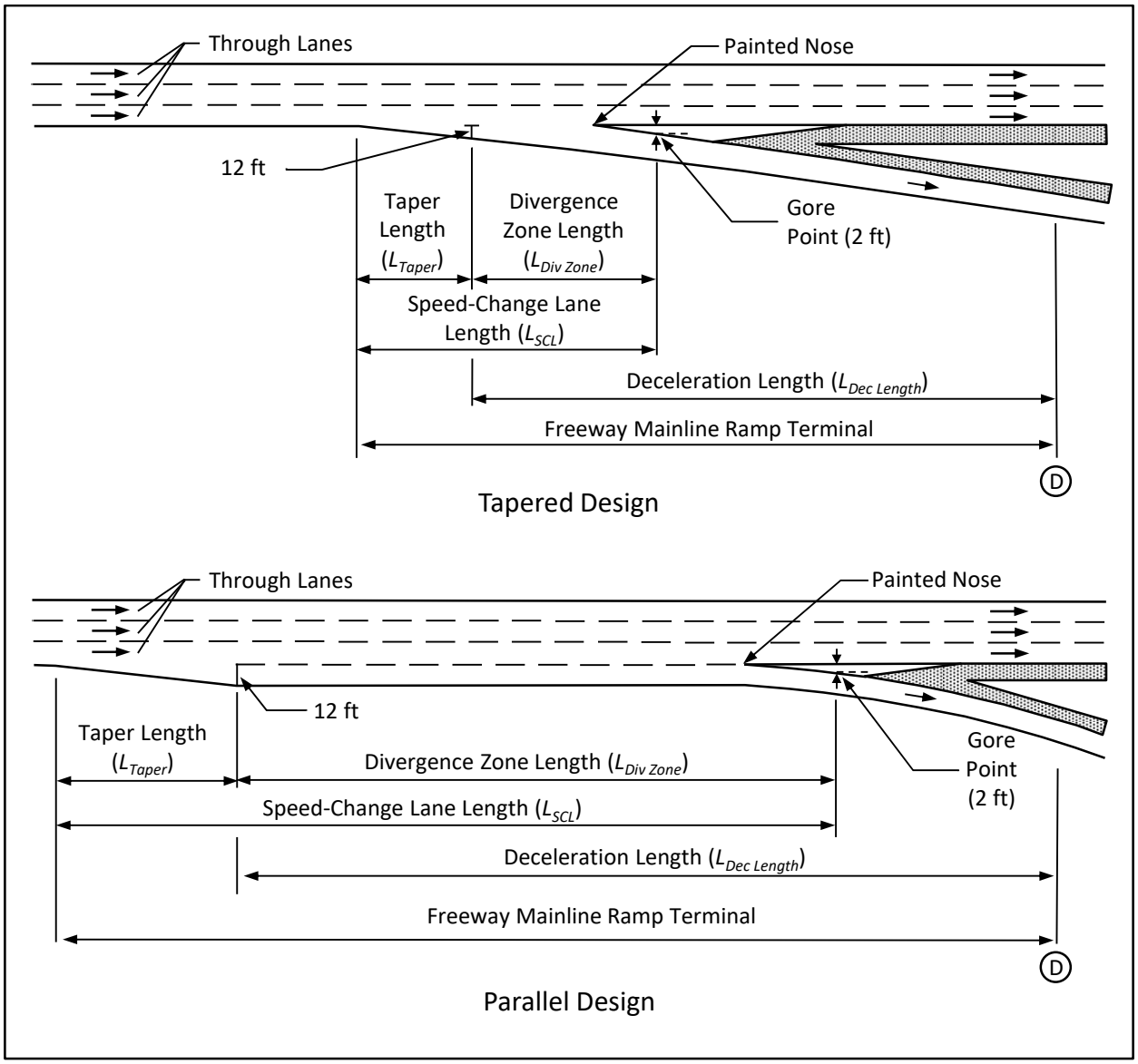


Figure 17. Freeway Mainline Ramp Terminal Elements of an Exit Ramp

AASHTO policy uses a basic two-step process for establishing the minimum deceleration length criteria for exit ramps. Deceleration is accomplished first as the driver removes his or her foot from the accelerator pedal and the vehicle slows in gear for a period of time (assumed to be 3 sec) without the use of brakes, and then as the driver applies the brakes and decelerates at a comfortable rate. Equations 3 and 4 represent this two-step process. The minimum deceleration length is based on the combination of the following inputs:

A. The speed at which drivers maneuver onto the auxiliary lane.
B. The speed at which drivers turn after traversing the deceleration lane.
C. The manner of deceleration.

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$$L_{Dec\ Length} = 1.47V_{Dec\ Length(i)}t_n - 0.5d_n(t_n)^2 + \frac{\left(1.47V_{Dec\ Length\ (f)}\right)^2 - (1.47V_a)^2}{2d_{wb}} \quad \textbf{(3)}$$

$$V_a = \frac{1.47V_{Dec\ Length\ (i)} - d_n t_n}{1.47} \quad \textbf{(4)}$$

where:

$L_{Dec\ Length}$ = Deceleration length, ft.
$V_{Dec\ Length\ (i)}$ = Speed at the beginning of the deceleration length, mph.
V_a = Speed after t_n sec of deceleration without brakes, mph.
$V_{Dec\ Length\ (f)}$ = Speed at the end of the deceleration length, mph.
t_n = Deceleration time without brakes (assumed to be 3 sec), sec.
d_n = Deceleration rate without brakes, ft/s^2.
d_{wb} = Deceleration rate with brakes, ft/s^2.

Green Book Table 10-6 (Table 8) provides recommended minimum deceleration lengths for various combinations of design conditions for both taper- and parallel-type exit terminals. The minimum recommended deceleration lengths are to be adjusted in accordance to *Green Book* Table 10-5 where grades are present. The speed-change lane for an exit ramp should meet the recommended minimum deceleration lengths based on the operating speed of the highway and the anticipated speed entering the controlling feature on the ramp proper or the initial section of the ramp proper downstream of the gore point.

A key factor in designing the freeway mainline ramp terminal of an exit ramp is determining where the deceleration length and the freeway mainline ramp terminal end (i.e., location of Point D). The deceleration length and freeway mainline ramp terminal may end at the gore point; however, the deceleration length and freeway mainline ramp terminal may also extend beyond the gore point.

To determine the boundary between the freeway mainline ramp terminal and the ramp proper, the ramp beyond the gore point should be divided into individual tangent and curve sections. If the horizontal alignment of the ramp is curvilinear, the beginning of the first curve encountered at the gore point or downstream of the gore point that significantly affects vehicle speed is the boundary between the freeway mainline ramp terminal and the ramp proper. It should be assumed that any curve radius less than or equal to 1,000 ft will significantly affect vehicle speed. If the first curve that significantly affects vehicle speed begins at the gore point, then the boundary between the freeway mainline ramp terminal and the ramp proper is at the gore point; however, if the beginning of the first curve that significantly affects vehicle speed is downstream of the gore point, then the boundary between the freeway mainline ramp terminal and the ramp proper may be downstream of the gore point.

Thus, in some situations the portion of the ramp immediately downstream of the gore point is part of the ramp proper, while in other situations the portion of the ramp immediately downstream of the gore point is part of the freeway mainline ramp terminal for some extended length. If the horizontal alignment of the ramp is relatively straight and has little or no influence

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on vehicle speeds, then conceptually the freeway mainline ramp terminal may extend beyond the gore point all the way to the beginning of the functional area of the crossroad ramp terminal.

No design criteria exist for specifying minimum length requirements of the individual tangent and curve sections of the ramp beyond the gore point; but the deceleration distances depicted in *Green Book* Figure 2-34 (Deceleration Distances for Passenger Vehicles Approaching Intersections) (see Figure 18) may be used as guide for determining the minimum practical lengths of the individual sections. These deceleration distances depict performance capabilities of lower performance vehicles decelerating with steady breaking suitable for design application. For each individual section, entering and exiting speeds should be assumed for the design, consistent with the speed exiting the freeway mainline ramp terminal and decreasing sequentially along the adjoining sections of the ramp. The individual component or section of the ramp should at least be as long as the deceleration distances depicted in *Green Book* Figure 2-34, based on the assumed entering and exiting speeds. Table 8 presents the deceleration distances in *Green Book* Figure 2-34 in tabular form.

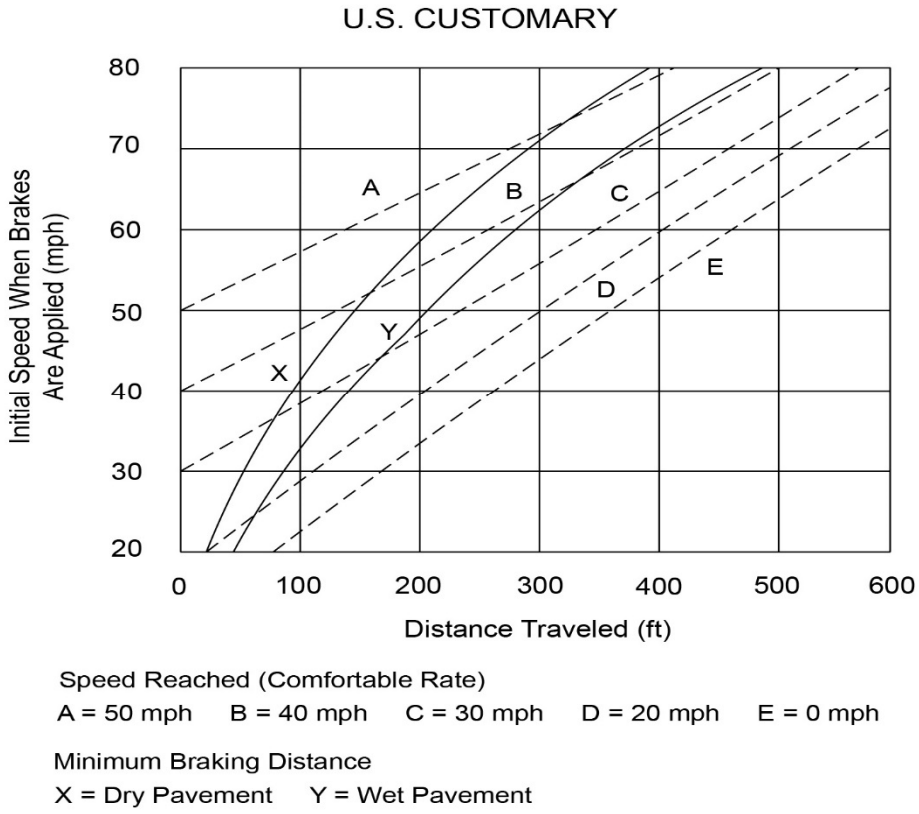


Figure 18. Deceleration Distances for Passenger Vehicles Approaching Intersections (AASHTO, 2018)

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Table 8. Deceleration Distances for Passenger Vehicles Approaching Intersections based on Steady Braking

Initial Speeds (mph)	Deceleration Distances for Passenger Vehicles Approaching Intersections (ft)				
	Final Speeds (mph)				
	0	20	30	40	50
20	75				
30	167	108			
40	268	204	115		
50	360	302	233	127	
60	460	404	346	264	127
70	567	512	458	383	275
80	675	620	567	500	413

Green Book Figure 2-34 can be used when referring to both initial and final design speeds and anticipated operating speeds. The speed distance relationships depicted in *Green Book* Figure 2-34 do not account for the effects of horizontal or vertical alignments. The lengths should be adjusted accordingly to account for the effects of horizontal curvature and/or grade.

When it comes to the ramp design at the end of the ramp proper near the crossroad ramp terminal, the operational characteristics of the crossroad ramp terminal control the design. The primary determinant in the design of the crossroad ramp terminal is the type of traffic control (i.e., whether operation onto the crossroad is stop-, yield-, or signal-controlled or free-flow). Where the crossroad ramp terminal is stop-, yield-, or signal-controlled, the ramp design should account for queue storage as depicted in Figure 19 which illustrates the functional area of a crossroad ramp terminal. The ramp should be designed to provide sufficient distance for a vehicle to come to a complete stop at the end of the queue storage length near the crossroad. The ramp should be designed so that vehicles enter the functional area of the crossroad ramp terminal at relatively low speeds (e.g., between 15 and 20 mph).

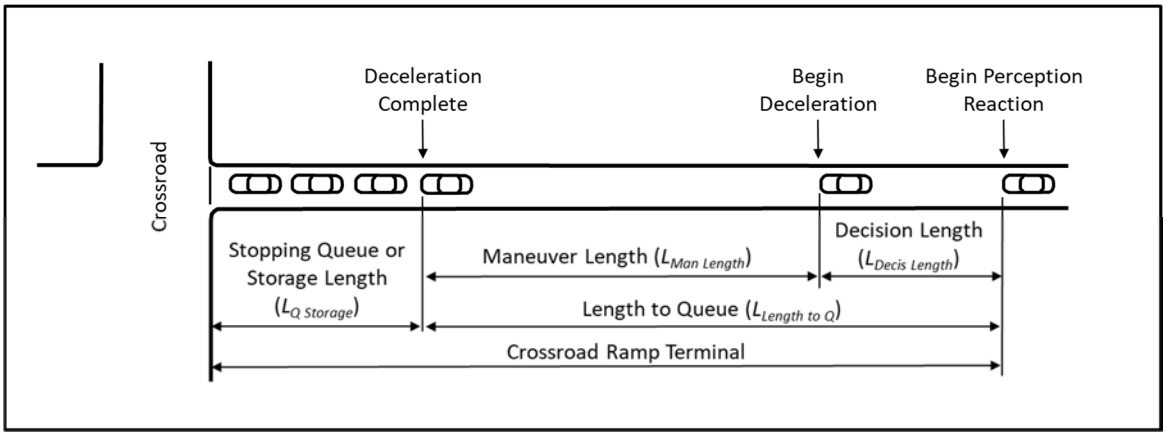


Figure 19. Elements of the Functional Area of a Crossroad Ramp Terminal (adapted from AASHTO, 2018)

The length from the beginning of the crossroad ramp terminal to the end of the storage length (i.e., $L_{Length\ to\ Q}$) consists of the decision length ($L_{Decis\ Length}$) and the maneuver length ($L_{Man\ Length}$).

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Similar to the brake reaction time used for stopping sight distance (SSD), the decision length is the distance covered by a vehicle from the instant that the driver recognizes the presence of a queue of vehicles stopped ahead until the instant that the driver actually applies the breaks. For design purposes, the decision length is based on an assumed perception reaction time of 2.5 seconds. The maneuver length approximates the distance that a vehicle travels after the brakes are applied. For design purposes, the maneuver length is based on deceleration distances for passenger vehicles approaching an intersection based on steady braking. *Green Book* Figure 2-34 (Figure 18) and Table 8 can again be used to determine the maneuver length to the rear of the queue. Table 9 provides the recommended minimum length from the beginning of the crossroad ramp terminal to the end of the storage length (i.e., $L_{Length\ to\ Q}$) based on the decision length ($L_{Decis\ Length}$) and the maneuver length ($L_{Man\ Length}$).

Table 9. Recommended Minimum Length to Queue at a Crosroad Ramp Terminal

Initial Speeds (mph)	Decision Length ($L_{Decis\ Length}$) (ft)	Maneuver Length ($L_{Man\ Length}$) (ft)	Length to Queue Calculated ($L_{Length\ to\ Q}$) (ft)	Length to Queue Design ($L_{Length\ to\ Q}$) (ft)
15	55	45	90	90
20	74	75	149	150
25	92	125	217	220
30	110	167	277	280
35	129	220	349	350

Where the crossroad ramp terminal provides for free-flow operations onto the crossroad, the speed at the downstream end of the final tangent or curve of the ramp proper should be equivalent to the initial speed entering the acceleration length connecting the ramp proper to the crossroad. *Green Book* Table 10-4 could be used to design the acceleration length for the crossroad ramp terminal.

Thus, *Green Book* Tables 10-6 and 10-5 serve to establish the minimum requirements for the deceleration length of a freeway mainline ramp terminal; *Green Book* Figure 2-34 provides guidance on the minimum lengths of the individual tangent and curve sections of the ramp proper; and the operational characteristics near the crossroad ramp terminal control the design of the terminal. The key element in coordinating the design of the individual components and features of an exit ramp is the ramp design speed. The ramp design speed is used to coordinate the design of adjacent sections of the ramp to accommodate sequential deceleration from one component or section to the next.

For each tangent section on the ramp proper, the speed at the downstream end is associated with the design speed of the tangent. The selected design speed of a tangent section should be consistent with the speeds that drivers are likely to be traveling at the end of the tangent. With horizontal curves on exit ramps, it is assumed that vehicle speeds will decrease slightly over the length of the curve, so the speeds of vehicles exiting a curve will be slightly lower than the speeds of vehicles entering the curve. Speeds associated with the midpoint of the curve should be used to design the curve rather than speeds at the beginning or end of the curve, and the design

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speed of the curve should be consistent with the speeds that drivers are likely to be traveling at the midpoint of the curve.

3.2.3 Guidelines for Freeway-to-Freeway Ramps

The same design controls and criteria for freeway mainline ramp terminals and the ramp proper for entrance and exit ramps are also applicable to freeway-to-freeway ramps. In other words, the same design controls and criteria used to coordinate the design of individual components and features of entrance and exit ramps based on the ramp design speed may also be used to design freeway-to-freeway ramps. This means that design criteria such as *Green Book* Tables 10-4, 10-5, and 10-6 may be used to determine the minimum acceleration and deceleration lengths for the freeway mainline ramp terminals of freeway-to-freeway ramps, and *Green Book* Figures 2-33 and 2-34 may be used to determine the minimum practical lengths for individual tangents and curves along the ramp proper of freeway-to-freeway ramps.

For freeway-to-freeway ramps, the sharpest curve on the ramp proper is considered the controlling curve, from which the ramp design speed for the entire ramp is based. The ramp design speed is used to coordinate the design of adjacent sections of the ramp to accommodate sequential acceleration or deceleration from one component or section to the next. For each tangent section on the ramp proper, the speed at the downstream end is associated with the design speed of the tangent. The selected design speed of a tangent section should be consistent with the speeds that drivers are likely to be traveling at the end of the tangent. With horizontal curves, speeds associated with the midpoint of the curve should be used to design the curve rather than speeds at the beginning or end of the curve, and the design speed of the curve should be consistent with the speeds that drivers are likely to be traveling at the midpoint of the curve. Although the freeway mainline ramp terminals do not have associated design speeds, the operational and design conditions of the freeway mainline ramp terminals are related to and coordinated with the operational and design conditions of the downstream and upstream components of the ramp.

For freeway-to-freeway ramps, rather than the design speeds of the contiguous segments along the ramp proper continuously increase or decrease in a stepwise manner; design speeds of the contiguous segments along the ramp proper should initially decrease and then increase. It is still desirable that the change in design speed between adjoining sections be limited to no more than 10 to 15 mph.

3.3 Measures for Reducing Speeds on Exit Ramps

On exit ramps, where anticipated operating speeds exceed the ramp design speed or design speed of an individual section, and changing the ramp design speed (or other design options) cannot resolve the issue, the following speed-control measures may be considered (Harwood and Mason, 1993):

- Provide signing with an appropriate advisory speed for the ramp and place the advisory speed sign so that drivers have sufficient length to reduce their speed prior to the most critical curve.
- Increase the deceleration length.

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- Realign the ramp to increase the distance from the gore point to the controlling curve.
- Supplement the standard advisory speed signs to make the signing more conspicuous, increase the distance from the signing to the controlling curve and draw attention of truck drivers to the signing. These objectives may be accomplished through:
  - Using more than one advisory speed sign.
  - Placing an advisory speed sign on the mainline highway in advance of the ramp.
  - Incorporating an exit speed panel in the guide signing for the exit ramp.
  - Using overhead signs.
  - Using a TRUCK SPEED advisory sign.
  - Using flashing beacons to call attention to the advisory speed.
- Avoid designs where the presence of the controlling curve is not obvious (e.g., where a tight horizontal curve follows a larger-radius curve).
- Consider use of C-D roads at the interchange to introduce an intermediate-speed roadway between the mainline freeway and the ramp.

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Section 4.
Design Tool to Evaluate Ramp Designs for Consistency with the Selected Ramp Design Speed

A tool to estimate vehicle speeds along interchange ramps could help designers and traffic engineers in conducting operational analyses and making design decisions. Speed estimation on ramps requires application of calibrated speed profile models. A database of vehicle speeds observed on ramps was assembled and statistical modeling conducted to develop speed profile models for ramps (Torbic et al., 2021). The models consist of equations that contain calibration coefficients and terms to account for the influence of variables such as initial speed, ramp segment length (curve or tangent segment), curve radius, and others. To apply the models, an Excel®-based spreadsheet tool was developed that allows designers and traffic engineers to enter key variables describing the ramp of interest to obtain a speed estimate for vehicles along the length of the ramp.

The spreadsheet tool is applicable to entrance and exit ramps at service interchanges including diagonal ramps, loop ramps, and outer connection ramps. Speed data were collected for ramps from several system interchanges, but modeling efforts did not yield reliable speed prediction models. Therefore, the ramp speed profile models incorporated into the spreadsheet tool are only applicable for the design and evaluation of entrance and exit ramps at service interchanges.

The spreadsheet-based Ramp Speed Profile Model (RSPM) is intended for several purposes:

1. It can be used to assess the adequacy of a new ramp design during the design process. After developing an initial design for a ramp, the engineer can input key variables into the RSPM to estimate vehicle speeds along the ramp and determine if the anticipated operating speeds along the ramp are as intended (e.g., predicted operating speeds along each sections of the ramp do not exceed the design speeds of the individual ramp sections). If the anticipated operating speeds are not consistent with the design of the individual ramp sections, then the ramp could be redesigned so the anticipated operating speeds along the ramp are more consistent and align with the design speeds of the individual ramp sections or speed-reduction measures could be implemented as appropriate to reduce speeds.
2. If a ramp is being considered for reconstruction, the RSPM could be used to assess the design consistency of different design alternatives such that the anticipated operating speeds along the ramp will be consistent and align with the design speeds of the individual ramp sections.

The RSPM is organized so the worksheets for entrance and exit ramps can each be printed on two pages, or one sheet of paper front and back. The first page for each type of ramp contains input data entry cells and the computed vehicle speeds in tabular format. The second page for each ramp type contains the computed vehicle speeds in graphical format, advisory notes, and acceleration rate calculations in tabular format.

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4.1 Input Data for the RSPM

The RSPM contains two analysis worksheets – one for entrance ramps and one for exit ramps. The input cells on these worksheets are color-coded blue or orange. Blue cells represent required input values, and orange cells represent optional input values. The analyst provides input values into these cells that describe key design characteristics of the freeway, freeway mainline ramp terminal, ramp proper, and crossroad ramp terminal. The RSPM also contains a Factors and Tables worksheet with various yellow-shaded cells. Yellow represents a calibration parameter. The analyst can change these values, but should do so only if the change is justified by an engineering evaluation of field data.

Some of the input data cells are configured with data validation features to prevent illogical values from being entered. For example, the speed-change lane type can only be specified as "parallel" or "taper". With other input data cells, warning messages are provided to indicate inputs that are permitted but may be outside of the range for which the models were developed. For example, curve radius values are preferred to be in the range of 0 to 2,000 ft based on the speed model calibration dataset. Curve radii greater than 2,000 ft may be entered, but a warning message will show that curves with radii greater than 2,000 ft will be treated as tangents.

Some cells or data boxes in the program have comment boxes that provide a description of the input data or interpretation of the output data. Red triangles indicate the presence of these comments. The comments can be viewed by placing the cursor on top of the red triangles.

4.1.1 RSPM Input Data – Entrance Ramps

Figure 20 provides a screen shot of the data input portion of the entrance ramp analysis worksheet. The following data are needed to describe an entrance ramp. For an entrance ramp, Milepost 0.000 is always located where the ramp connects to the crossroad ramp terminal. The milepost for all other locations along the entrance ramp are measured relative to Milepost 0.000.

Inputs for Entrance Ramps:

- Speeds
  - Freeway mainline design speed, $V_{DS\ Hwy}$ (mph). Values must be in the range of 50 to 85 mph and an increment of 5 mph.
  - Average operating speed on the freeway mainline, $V_{OS\ Hwy}$ (mph). If this speed is not known and the input cell is left blank, the freeway mainline regulatory speed limit value will be used as the average operating speed for the freeway.
  - Freeway mainline regulatory speed limit, $V_{SL\ Hwy}$ (mph). Values must be in the range of 45 to 85 mph and an increment of 5 mph. Values greater than 65 mph may be entered, but calculations will be based on a maximum value of 65 mph due to calibration of the models.
  - Estimated vehicle speed at the crossroad ramp terminal, V_{XRoad} (mph). This value may be based on field measurements. If field measurements are not available, a value of 15 mph is recommended where the traffic control is stop, yield, or signal controlled; otherwise, 30 mph is recommended. A minimum value of 5 mph must be entered.

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- General Characteristics
  - Ramp grade (percent). This grade describes the overall ramp and is computed from the elevations of the ramp's beginning and ending points.
- Ramp Proper
  - Input for individual tangents:
    - Design speed of tangent, $V_{DS\ Tan}$ (mph). Values must be in the range of 5 to 80 mph. This information is entered into the same data row as the information for the subsequent curve.
      - If the entire ramp is tangent, this information is entered into the first data row for curves and tangents on the ramp proper.
      - If no tangent design speeds are entered, the design speed for each tangent is assumed to be equal to that of the subsequent curve, or that of the freeway mainline if the entire ramp is tangent.
  - Input for individual curves:
    - Design speed of curve, $V_{DS\ Curve}$ (mph). Values must be in the range of 5 to 80 mph.
    - Milepost of the beginning of the curve [i.e., point of curvature (PC)], X_{PC} (mi).
    - Radius of the curve, R (ft). Values should be in the range of 0 to 2,000 ft. Curve radii greater than 2,000 ft may be entered, but a warning message will show that curves with radii greater than 2,000 ft will be treated as tangents.
    - Curve length, L_c (mi). Values should be in the range of 0.05 to 0.25 mi. Values outside of this range may be entered but a message box will prompt the analyst to adjust the value or continue if the input value is outside the range of the calibration dataset.
- Freeway Mainline Ramp Terminal
  - Milepost of the gore point, X_G (mi). The gore point is located where the pair of solid white pavement edge markings that separate the ramp from the intersecting roadway are 2 ft apart. If markings do not extend to a point where they are 2.0 ft apart, the gore point is found by extrapolating both markings until the extrapolated portion is 2.0 ft apart.
  - Gap acceptance length, $L_{Gap\ Acpt}$ (mi). This length begins at the gore point and ends at the beginning of the taper.
  - Taper length, L_{Taper} (mi). This length begins at the end of the gap acceptance length and ends at the end of the speed-change lane where there is no additional width for the speed-change lane.
  - Acceleration length on the ramp proper, $L_{Acc\ Length\ RP}$ (mi). This length is that portion of the acceleration length measured upstream from the gore. This length may coincide with the point of tangency (PT) of the last curve on the ramp or another location where it is believed acceleration to the freeway speed begins.

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- Proportion of gap acceptance length used by merging vehicles, $P_{Gap\ Acpt}$ [From a design perspective, $P_{Gap\ Acpt}$ = 1.0 meaning merging vehicles are expected to use the entire gap acceptance length. However, field data show that many drivers use only a portion of the gap acceptance length prior to merging onto the freeway. $P_{Gap\ Acpt}$ = 0.5 is provided as a default based on Torbic et al. (2012)]. This quantity is a calibration parameter and can be adjusted by the analyst.

Figure 21 illustrates the elements and dimensions in red text of the freeway mainline ramp terminal that are input into the RSPM for an entrance ramp. This figure is also included on the Factors and Tables tab within the spreadsheet tool.

NCHRP 15-56 Ramp Speed Calculations - Entrance Ramp

Calculate | Revise

Location: Sample Entrance Ramp

Component	Variable	Input Value	Unit	Notes
Speeds	$V_{DS\ Hwy}$	65	mph	
	$V_{OS\ Hwy}$	60	mph	
	$V_{SL\ Hwy}$	55	mph	
	V_{XRoad}	15	mph	
General Characteristics	Ramp grade	3	%	

Variables for Curves and Tangents on Ramp Proper

$V_{DS\ Tan}$ (mph)	$V_{DS\ Curve}$ (mph)	X_{PC} (mi)	R (ft)	L_c (mi)	
20	25	0.02	500	0.05	Curve 1
28	30	0.12	500	0.05	Curve 2
35	40	0.26	700	0.06	Curve 3

Freeway Mainline Ramp Terminal	X_G	0.400	mi	
	$L_{Gap\ Acpt}$	0.070	mi	
	L_{Taper}	0.050	mi	
	$L_{Acc\ Length\ RP}$	0.000	mi	
	$P_{Gap\ Acpt}$	0.50		

Figure 20. Input Data for RSPM for Entrance Ramps

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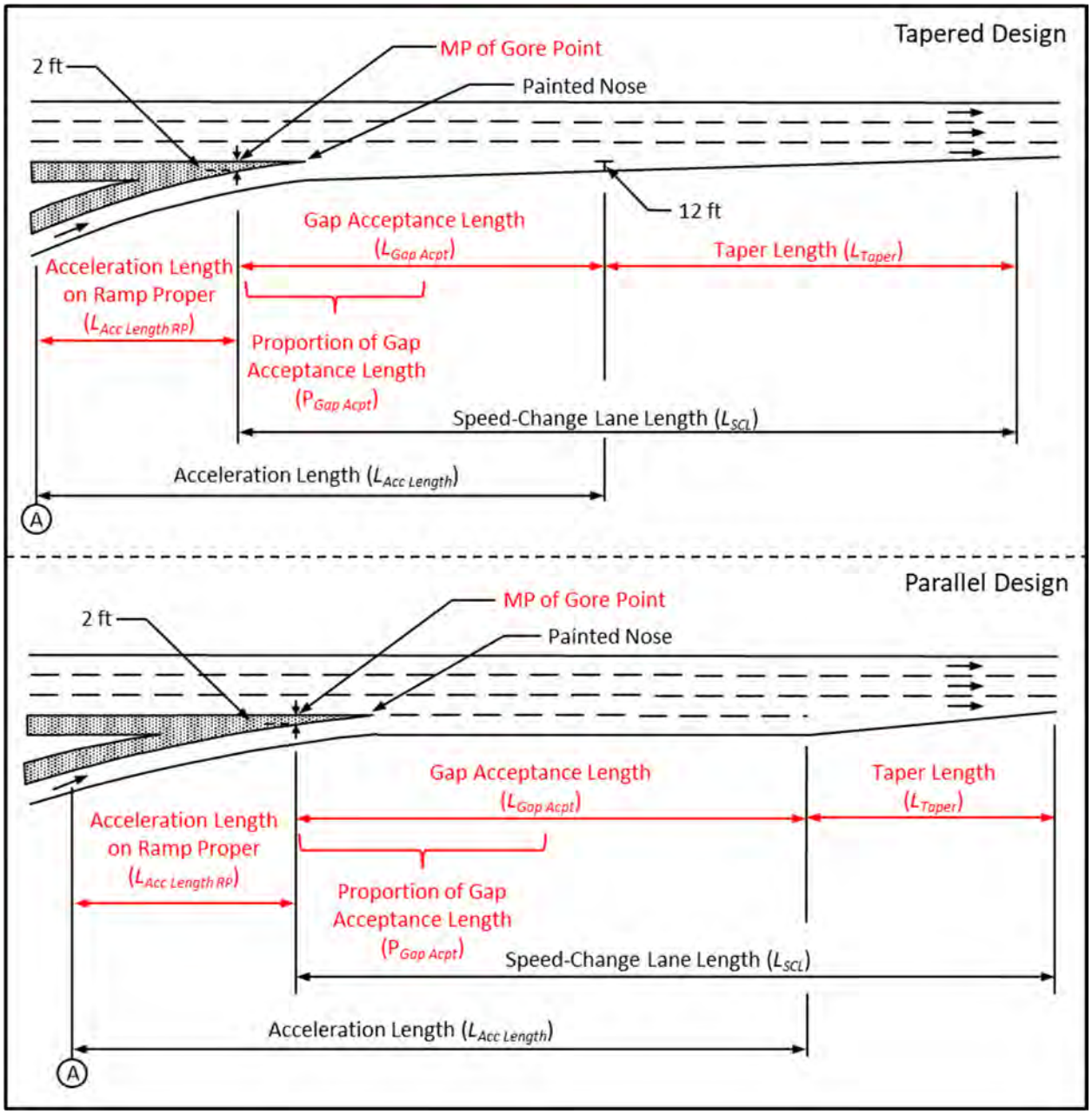


Figure 21. Elements and Dimensions of Freeway Mainline Ramp Terminals Input into RSPM for Entrance Ramps

4.1.2 RSPM Input Data – Exit Ramps

Figure 22 provides a screen shot of the data input portion of the exit ramp analysis worksheet. The following data are needed to describe an exit ramp. For an exit ramp, Milepost 0.000 is located at the gore point where the ramp diverges from the freeway mainline. Mileposts for all other locations along the exit ramp are measured relative to Milepost 0.000. Locations upstream of the gore point along the freeway mainline ramp terminal have negative milepost values, and locations downstream of the gore point have positive milepost values.

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Inputs for Exit Ramps:

- Speeds
  - Freeway mainline design speed, $V_{DS\ Hwy}$ (mph). Values must be in the range of 50 to 85 mph and an increment of 5 mph.
  - Average operating speed on the freeway mainline, $V_{OS\ Hwy}$ (mph). If this speed is not known and the input cell is left blank, the freeway mainline regulatory speed limit value will be used as the average operating speed for the freeway.
  - Freeway mainline regulatory speed limit, $V_{SL\ Hwy}$ (mph). Values must be in the range of 45 to 85 mph and an increment of 5 mph.
  - Estimated vehicle speed at the crossroad ramp terminal, V_{XRoad} (mph). This value may be based on field measurements. If field measurements are not available, a value of 15 mph is recommended where the traffic control is stop, yield, or signal controlled; otherwise, 30 mph is recommended. A minimum value of 5 mph must be entered.
- General Characteristics
  - Ramp grade (percent). This grade describes the overall ramp and is computed from the elevations of the ramp's beginning and ending points.
  - Ramp type. Permissible types are diagonal, loop, or outer connection.
- Freeway Mainline Ramp Terminal
  - Speed-change lane type. Permissible types are parallel or taper.
  - Taper length, L_{Taper} (mi). This length begins at the beginning of the speed-change lane where there is no additional width for the speed-change lane and ends at the beginning of the divergence zone (i.e., where the taper is at least 12 ft).
  - Divergence zone length, $L_{Div\ Zone}$ (mi). This length begins at the end of the taper and ends at the gore point. Values should be in the range of 0.0 to 0.14 mi. Values outside of this range may be entered but a message box will prompt the analyst to adjust the value or continue if the input value is outside the range of the calibration dataset.
  - Proportion of speed-change lane length located upstream of the diverge point, P_{SCL}. From a design perspective, vehicles are expected to diverge from the freeway at the end of the taper and the beginning of the divergence zone length. However, field data show that drivers diverge at different locations along the speed-change lane. P_{SCL} = 0.1 is provided as a default based on Torbic et al. (2012). This quantity is a calibration parameter and can be adjusted by the analyst. Note that the speed-change lane length equals the taper length plus the divergence zone length ($L_{SCL} = L_{Taper} + L_{Div\ Zone}$).
  - Deceleration length on the ramp proper, $L_{Dec\ Length\ RP}$ (mi). This length is that portion of the deceleration length measured downstream from the gore point. This length may coincide with the PC of the first curve on the ramp proper or another location where it is believed deceleration to the speed on the ramp proper transitions.

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- Ramp Proper
  - Input for individual tangents:
    - Design speed of tangent, $V_{DS\ Tan}$ (mph). This information is entered into the same data row as the information for the preceding curve.
      - If the entire ramp is tangent, this information is entered into the first data row for curves and tangents on the ramp proper. If no tangent design speeds are entered, the design speed for each tangent is assumed to be equal to that of the preceding curve, or that of the freeway mainline if the entire ramp is tangent.
  - Input for individual curves:
    - Design speed of curve, $V_{DS\ Curve}$ (mph). Values must be in the range of 5 to 80 mph.
    - Milepost of the beginning of the curve (i.e., PC), X_{PC} (mi).
    - Radius of the curve, R (ft). Values should be in the range of 0 to 2,000 ft. Curve radii greater than 2,000 ft may be entered, but a warning message will show that curves with radii greater than 2,000 ft will be treated as tangents.
    - Curve length, L_c (mi). Values should be in the range of 0.05 to 0.25 mi. Values outside of this range may be entered but a message box will prompt the analyst to adjust the value or continue if the input value is outside the range of the calibration dataset.
- Crossroad Ramp Terminal
  - Milepost of the crossroad ramp terminal, X_{XRoad} (mi).
  - Queue storage length, $L_{Q\ Storage}$ (mi). The queue storage length consists of the specified length (if any) at the end of the ramp where stopped vehicles are likely to queue because of traffic control at the ramp terminal intersection.

Figure 23 illustrates the elements and dimensions in red text of the freeway mainline ramp terminal that are input into the RSPM for an exit ramp. This figure is also included on the Factors and Tables tab within the spreadsheet tool.

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NCHRP 15-56 Ramp Speed Calculations - Exit Ramp

Location: Sample Exit Ramp

Calculate | Revise

Component	Variable	Input Value	Unit	Notes
Speeds	$V_{DS\ Hwy}$	65	mph	
	$V_{OS\ Hwy}$	60	mph	
	$V_{SL\ Hwy}$	60	mph	
	V_{XRoad}	15	mph	
General Characteristics	Ramp grade	3	%	
	Ramp type	Diagonal		
Freeway Mainline	L_{Taper}	0.050	mi	
Ramp Terminal	$L_{Div\ Zone}$	0.050	mi	
	$L_{Dec\ Length\ RP}$	0.000	mi	
	SCL type	Taper		

Variables for Curves and Tangents on Ramp Proper

$V_{DS\ Tan}$ (mph)	$V_{DS\ Curve}$ (mph)	X_{PC} (mi)	R (ft)	L_c (mi)	
40	45	0.05	1000	0.05	Curve 1
30	35	0.18	600	0.05	Curve 2
15	20	0.26	600	0.06	Curve 3

Crossroad	X_{XRoad}	0.400	mi	
	$L_{Q\ Storage}$	0.050	mi	
	P_{SCL}	0.10		

Figure 22. Input Data for RSPM for Exit Ramps

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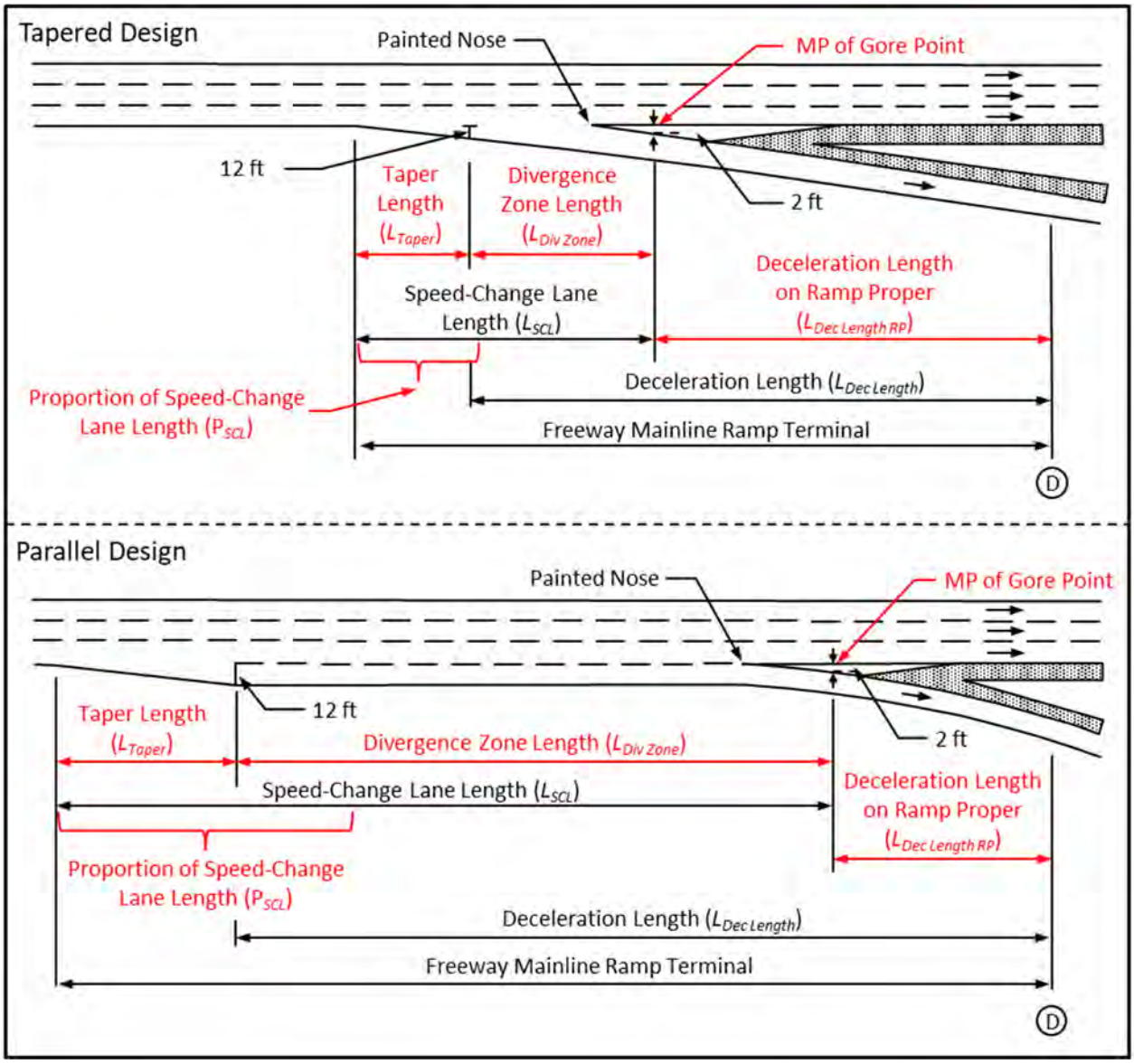


Figure 23. Elements and Dimensions of Freeway Mainline Ramp Terminals Input into RSPM for Exit Ramps

4.1.3 RSPM Input Data – Error Checks and Warning Messages

The RSPM performs several checks of the input data to determine if any errors were made or inconsistent values entered into the spreadsheets. On the entrance ramp worksheet, the following error checks and warning messages are provided:

- If any required data elements are not entered, a note will appear above the output graph to remind the analyst to "Provide all missing data elements and re-run the analysis."
- If the analyst specifies a beginning milepost for a curve that is less than the previous curve's ending milepost, the analyst will receive a message box that reads "Invalid data entry. The beginning milepost for curve *x* is located upstream of the end of curve *x*-1." A

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message will also appear to the right of the curve input data stating "Check milepost and length data," and a message will appear above the output graph stating "Check curve data."

- If the analyst specifies a milepost for the gore point that is less than the previous curve's ending milepost, the analyst will receive a message box that reads "Invalid data entry. The last curve cannot extend past the gore point." A message will also appear to the right of the input data stating "Check milepost and length data" and "Check gore location data", and a message will appear above the output graph stating "Check location data for gore and last curve."

On the exit ramp worksheet, the following error checks and warning messages are provided:

- If any required data elements are not entered, a note will appear above the output graph to remind the analyst to "Provide all missing data elements and re-run the analysis."
- If the analyst specifies a beginning milepost for a curve that is less than the previous curve's ending milepost, the analyst will receive a message box that reads "Invalid data entry. The beginning milepost for curve *x* is located upstream of the end of curve *x*-1." A message will also appear to the right of the curve input data stating "Check milepost and length data," and a message will appear above the output graph stating "Check curve data."
- If the analyst specifies a milepost for the ramp endpoint that is less than the last curve's ending milepost, the analyst will receive a message box that reads "Invalid data entry. The last curve extends past the ramp endpoint." A message will also appear to the right of the input data stating "Check milepost and length data" and "Check ramp endpoint location data," and a message will appear above the output graph stating "Check location data for ramp endpoint and last curve."
- The worksheet will display error messages if the specified queue storage length exceeds half the ramp length or extends upstream of the PT of the last curve.

In addition, various data entry cells on both worksheets are programmed with data validation checks to prevent entries that are erroneous or outside the range of the calibration dataset for the speed profile models. If the analyst enters an improper value, a message box will appear specifying the proper range of input values or prompt the analyst to adjust the value or continue if the input value is outside the range of the calibration dataset.

4.2 Output Data from the RSPM

The RSPM provides the capability to predict speeds for a given ramp design and visually compare predicted speeds for alternative designs. Section 4.2.1 describes the output for an initial ramp design. Section 4.2.2 describes the output to visually compare predicted speeds of alternative designs.

4.2.1 Output Data - Calculate

The analyst obtains output data by clicking the Calculate button after entering the input data (see Figures 20 and 22). Output data are provided in tabular and graphical form. The output includes estimated vehicle speeds and acceleration rates along the ramp and comparisons of estimated acceleration rates against design values from the *Green Book*.

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4.2.1.1 RSPM Output Data – Entrance Ramps

Figure 24 shows the graph containing estimated vehicle speeds along an entrance ramp based on the inputs in Figure 20. In this example, the ramp consists of four tangents and three curves in alternating series, followed by the speed-change lane. The speed profile is shown as a solid blue line, and the design speeds of individual sections are shown as a broken red line. Vehicle speeds are either input by the analyst or estimated for the following key points on the speed profile:

- The beginning point of the ramp at the crossroad ramp terminal.
- The beginning and ending mileposts for each tangent or curve segment.
- The midpoint of each curve segment.
- The gore point, labeled as Point G.
- The point on the speed-change lane where most vehicles are expected to merge into mainline traffic. By default, the location of this point is specified as 0.5 of the gap acceptance length in the direction of traffic flow but can be changed by adjusting the calibration factor in the worksheet.
- The endpoint (i.e., taper point) of the speed-change lane.

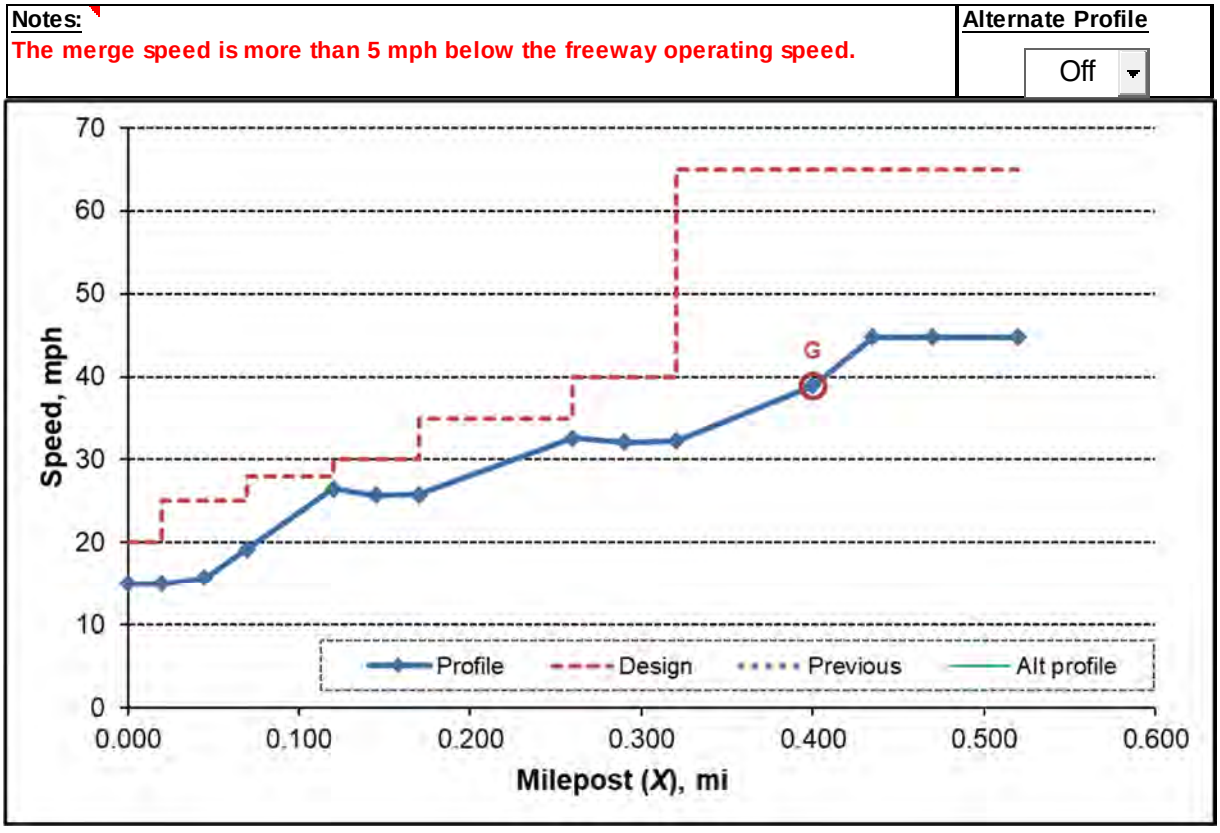


Figure 24. Graphical Illustration of Vehicle Speeds for Sample Entrance Ramp

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Design speeds for the ramp components are plotted as follows:

- The design speed for each curve is shown as the specified value for the curve.
- The design speed for each tangent is shown as the specified value for the tangent, or for the next downstream curve if no value is specified, or the freeway mainline if there is no downstream curve.
- Along the speed-change lane, the design speed of the freeway mainline is shown as an upper limit.

Two types of tabular outputs are provided with the RSPM. The predicted speeds used to generate the graphical output are provided in tabular form (Table 10), and acceleration calculations associated with the output are also generated (Table 11). The acceleration calculations are organized by segment, starting with the first tangent between the crossroad ramp terminal and the first curve, continuing through each successive curve and tangent, and ending with the last portion of the speed-change lane. Curves are treated as two segments because the speed prediction models provide speed estimates for the midpoint and end of each curve. The speed-change lane is treated as two segments because acceleration occurs only on the specified portion of its length prior to the merge (the first half, by default) but the specified portion can be adjusted. The following calculations are provided:

- Segment length, mi.
- Initial and final speed on the segment, mph.
- Average acceleration rate on the segment (computed from length and initial and final speeds), mph/s and ft/s^2. These values are provided in the purple-shaded cells.
- Design acceleration rate, computed using Equation 29, ft/s^2.
- Notes comparing the estimated average acceleration rates to design acceleration rates. If the estimated average acceleration rate exceeds the design rate, a message will indicate "accel > design". Otherwise, the message will indicate "OK".

Table 10. Plotted Speed Data for Sample Entrance Ramp

Point Type	Milepost	Profile, mph	Milepost	Design, mph
Xrd	0.000	15.00	0.000	20
Tan	0.020	15.00	0.020	20
Cmc	0.045	15.67	0.020	25
Cpt	0.070	19.27	0.070	25
Tan	0.120	26.46	0.070	28
Cmc	0.145	25.74	0.120	28
Cpt	0.170	25.79	0.120	30
Tan	0.260	32.58	0.170	30
Cmc	0.290	32.09	0.170	35
Cpt	0.320	32.27	0.260	35
Tan	0.400	38.92	0.260	40
SCm	0.435	44.77	0.320	40
SCe	0.470	44.77	0.320	65
Tpr	0.520	44.77	0.520	65

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Table 11. Acceleration Rate Calculations for Sample Entrance Ramp

Segment Data			Speed, mph		Average Acceleration		Design Acceleration	
Number	Type	Length, mi	Initial	Final	mph/s	ft/s^2	ft/s^2	Notes
1	Tangent	0.020	15.000	15.000	0.000	0.000	7.500	OK
2	Curve	0.025	15.000	15.667	0.114	0.167	7.500	OK
3	Curve	0.025	15.667	19.266	0.698	1.024	7.181	OK
4	Tangent	0.050	19.266	26.464	0.914	1.341	5.839	OK
5	Curve	0.025	26.464	25.735	-0.211	-0.310	4.251	OK
6	Curve	0.025	25.735	25.793	0.017	0.024	4.371	OK
7	Tangent	0.090	25.793	32.579	0.611	0.897	4.362	OK
8	Curve	0.030	32.579	32.087	-0.147	-0.216	3.453	OK
9	Curve	0.030	32.087	32.272	0.055	0.081	3.506	OK
10	Tangent	0.080	32.272	38.917	0.821	1.205	3.486	OK
11	Speed-change	0.035	38.917	44.772	1.944	2.852	2.891	OK
12	Speed-change	0.035	44.772	44.772	0.000	0.000	2.513	OK
13	Taper	0.050	44.772	44.772	0.000	0.000	2.513	OK

4.2.1.2 RSPM Output Data – Exit Ramps

Figure 25 shows the graph containing vehicle speed calculations for an exit ramp based on the inputs in Figure 22. In the example, the ramp consists of a speed-change lane followed by four tangents and three curves, with queue storage space at the end of the last tangent segment. Speeds are provided for the following key points:

- The beginning point of the speed-change lane, at the beginning of the taper from the freeway mainline.
- The point on the speed-change lane where most vehicles are expected to diverge from the mainline traffic. The location of this point is specified as 0.1 of the speed-change lane length in the direction of travel but can be changed by adjusting the calibration factor in the worksheet.
- The beginning and ending mileposts for each tangent or curve segment.
- The midpoint of each curve segment.
- The beginning point of the queue storage length.
- The endpoint of the queue storage length and the ramp.

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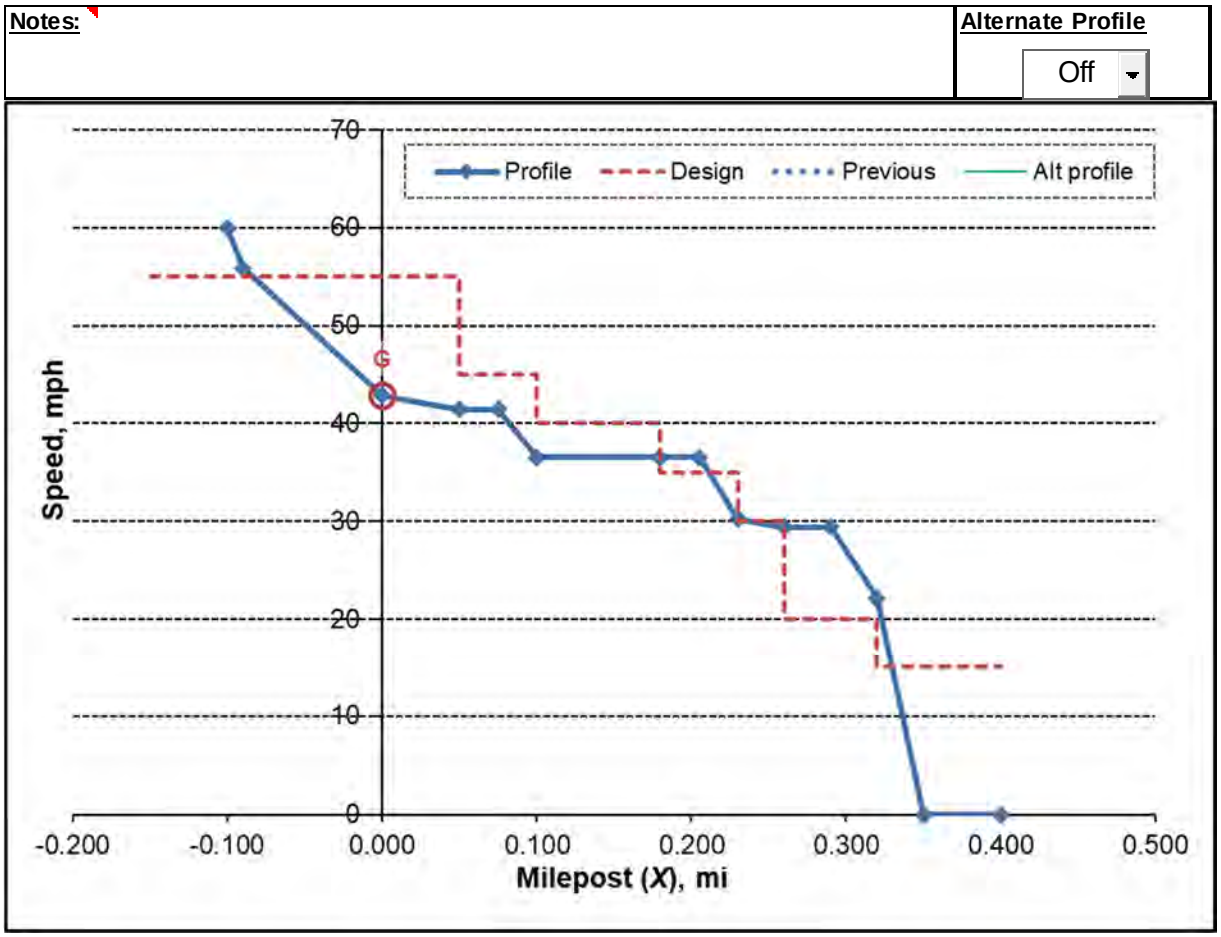


Figure 25. Graphical Illustration of Vehicle Speeds for Sample Exit Ramp

Design speeds for the ramp components are plotted as follows:

- Along the speed-change lane and the first tangent, the design speed of the freeway mainline is shown as an upper limit.
- The design speed for each curve is shown as the specified value for the curve.
- The design speed for each subsequent tangent is shown as the specified value for the tangent, or the specified value for the previous upstream curve if no value is specified, or the freeway mainline if there is no upstream curve.

Two types of tabular outputs are provided for exit ramps. The predicted speeds used to generate the graphical output are provided in tabular form (Table 12), and deceleration calculations associated with the output are also generated (Table 13). The deceleration calculations are organized by segment, starting with the speed-change lane, continuing through each successive curve and tangent, and ending with the queue storage space on the last tangent segment. Curves are treated as two segments because the speed prediction models provide speed estimates for the midpoint and end of each curve. The speed-change lane is treated as two segments. The first portion is where a vehicle begins the diverge maneuver from the freeway mainline onto the

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freeway mainline ramp terminal. The second portion begins where the diverge maneuver is complete and the vehicle is positioned within the speed-change lane. Similar calculations are provided as for entrance ramps, except design deceleration rates are computed using Equation 30. If the estimated average deceleration rate exceeds the design rate, a message will indicate "decel > design" for exit ramps. Otherwise, the message will indicate "OK".

Table 12. Plotted Speed Data for Sample Exit Ramp

Point Type	Milepost	Profile, mph	Milepost	Design, mph
Tpr	-0.100	60.00	-0.150	55
SCd	-0.090	55.90	0.050	55
Gor	0.000	42.86	0.050	45
Tan	0.050	41.43	0.100	45
Cmc	0.075	41.43	0.100	40
Cpt	0.100	36.55	0.180	40
Tan	0.180	36.55	0.180	35
Cmc	0.205	36.55	0.230	35
Cpt	0.230	30.24	0.230	30
Tan	0.260	29.38	0.260	30
Cmc	0.290	29.38	0.260	20
Cpt	0.320	22.14	0.320	20
Qst	0.350	0.00	0.320	15
Xrd	0.400	0.00	0.400	15

Table 13. Deceleration Rate Calculations for Sample Exit Ramp

Segment Data			Speed, mph		Average Acceleration		Design Acceleration	
Number	Type	Length, mi	Initial	Final	mph/s	ft/s^2	ft/s^2	Notes
1	Speed-change	0.010	60.000	55.900	-6.600	-9.680	-10.648	OK
2	Speed-change	0.090	55.900	42.864	-1.987	-2.914	-9.920	OK
3	Tangent	0.050	42.864	41.434	-0.335	-0.491	-7.607	OK
4	Curve	0.025	41.434	41.434	0.000	0.000	-7.353	OK
5	Curve	0.025	41.434	36.546	-2.117	-3.106	-7.353	OK
6	Tangent	0.080	36.546	36.546	0.000	0.000	-6.486	OK
7	Curve	0.025	36.546	36.546	0.000	0.000	-6.486	OK
8	Curve	0.025	36.546	30.242	-2.339	-3.431	-6.486	OK
9	Tangent	0.030	30.242	29.381	-0.237	-0.348	-5.367	OK
10	Curve	0.030	29.381	29.381	0.000	0.000	-5.214	OK
11	Curve	0.030	29.381	22.145	-1.726	-2.532	-5.214	OK
12	Tangent	0.030	22.145	0.000	-2.270	-3.330	-3.930	OK
13	Queue	0.050	0.000	0.000	0.000	0.000	0.000	OK

4.2.2 Output Data - Revise

In addition to the previously described calculations, the analysis worksheets provide the capability to easily compare results after the analyst changes some input values to evaluate an alternative design. This feature is invoked by clicking the revise button (see Figures 20 and 22).

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For example, the speed profile shown in Figure 25 is for an exit ramp with three curves (see input data in Figure 22). Figure 26 shows the revised speed profile if the radius of the first curve is decreased from 1,000 ft to 600 ft. The original speed profile is shown as a broken blue line and the revised speed profile is shown as a solid blue line. If the data inputs for design speed are adjusted, only the revised design speed plot is displayed.

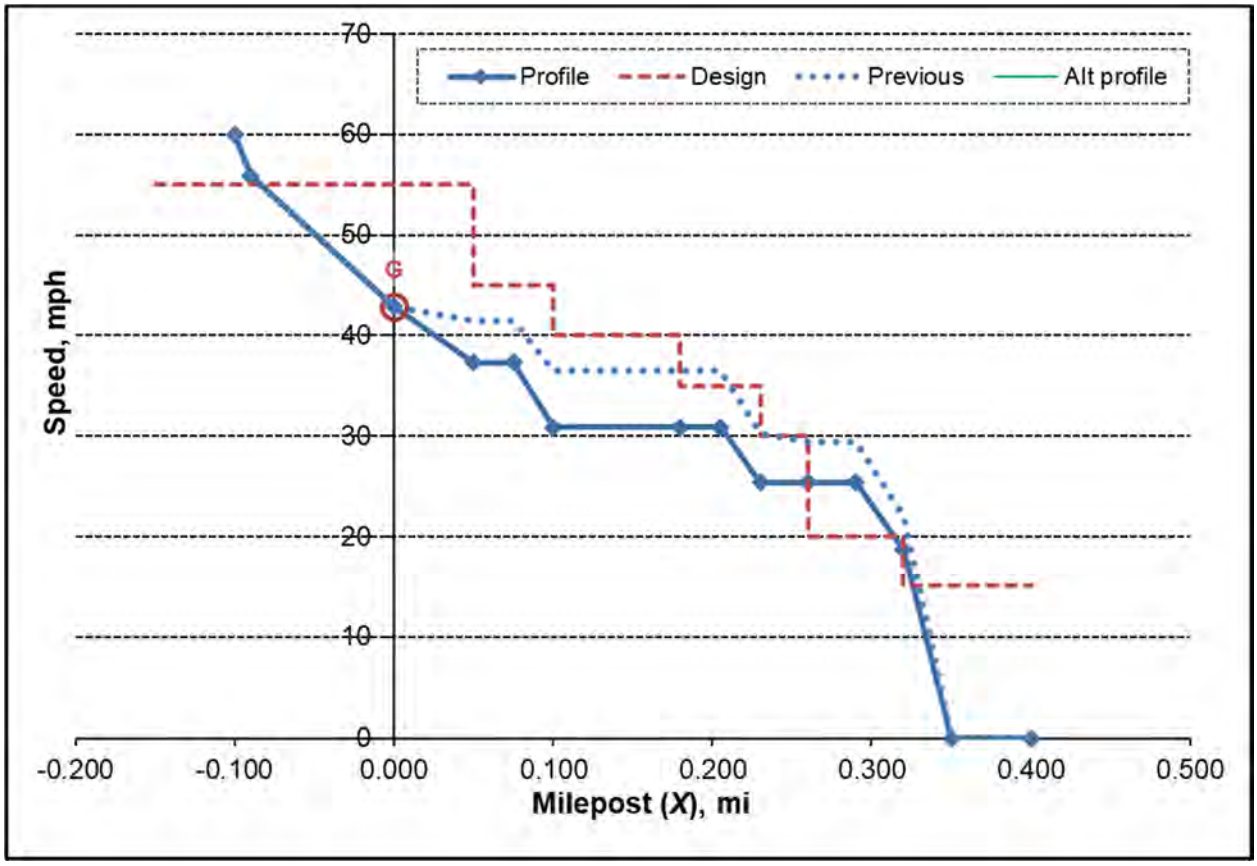


Figure 26. Revised Graphical Illustration of Vehicle Speeds for Sample Design Alternative (Exit Ramp)

4.3 RSPM Speed Calculations

This section describes the speed calculation framework used by the RSPM. The calculation framework is a direct implementation of curve and tangent speed profile models for ramps. The framework involves estimating the initial speed at the beginning of the ramp, and then estimating speeds on each successive ramp segment as a function of the speed at the end of the preceding segment.

The RSPM estimates vehicle speeds along the full length of a ramp, from the crossroad ramp terminal through the full length of the speed-change lane. A ramp is described linearly, with mileposts at the beginning and end of each ramp segment. For entrance ramps, Milepost 0.000 is located where the ramp proper connects to the edge of traveled way at the crossroad ramp terminal, and the ramp endpoint is located at the end of the speed-change lane of the freeway

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mainline ramp terminal. By definition, the speed-change lane includes the tapered area. For exit ramps, Milepost 0.000 is located at the gore point which is defined as the location where the pair of solid white pavement edge markings that separate the ramp from the intersecting roadway are 2.0 ft apart.

A ramp with *n* segments is generally analyzed as follows:

- Obtain an estimate of the initial speed at the beginning of the ramp. For entrance ramps, a default estimate of the initial speed is based on the type of traffic control at the crossroad ramp terminal. The analyst may also input an initial speed at the crossroad ramp terminal. For exit ramps, a default estimate is based on the regulatory speed limit on the freeway mainline, and the length and type of speed-change lane where the ramp diverges from the mainline. The analyst may also input the average operating speed of the freeway mainline in the vicinity of the speed-change lane. Then, the diverge speed is based on the average operating speed of the freeway mainline rather than the regulatory speed limit, taking into consideration the length and type of speed-change lane.
- If Segment 1 is a curve, apply a curve speed model to estimate vehicle speed at the curve midpoint (MC) and PT. If Segment 1 is a tangent, apply a tangent speed model to estimate vehicle speed at the end of the tangent.
- For each successive segment (Segment 2 through Segment *n*), use the vehicle speed at the end of the preceding segment and the appropriate speed model (curve or tangent) to estimate the vehicle speed at the end of the segment. If the segment is a curve, estimate vehicle speeds at the MC and the PT.
- Estimate the vehicle speed at the end of Segment *n* based on the appropriate speed model (curve or tangent) and the conditions at the end of the ramp. For entrance ramps, this speed is influenced by the length and type of speed-change lane of the freeway mainline ramp terminal. For exit ramps, this speed is influenced by the type of traffic control at the crossroad ramp terminal.
- Compute the average acceleration rate on every ramp segment based on the segment lengths and estimated vehicle speeds. Compare the average acceleration rate to design acceleration rates from the *Green Book* and note any cases where the estimated average acceleration rate exceeds the design value.

The vehicle speed calculations for entrance and exit ramps are described in greater detail in the following subsections, followed by a discussion of acceleration rate calculations used to assess the adequacy of the design and other checks of the design performed within the RSPM.

4.3.1 Speed Calculations for Entrance Ramps

Applicable speed calculations are presented in the order of the primary components of an entrance ramp based on the direction of travel:

- The crossroad ramp terminal.
- The ramp proper.
- The freeway mainline ramp terminal.

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4.3.1.1 Crossroad Ramp Terminal

The vehicle speed at the beginning point of an entrance ramp at the crossroad ramp terminal, V_{XRoad}, is estimated based on the traffic control at the crossroad ramp terminal or is input by the analyst. Default speeds are based on the traffic control type at the crossroad ramp terminal as follows:

- Stop, yield, or signal: 15 mph.
- Other traffic control types: 30 mph.

The default speed values are based on engineering judgement and recommendations by Bonneson et al. (2012).

4.3.1.2 Ramp Proper – Preliminary Speed Profile

A preliminary speed profile is computed for the ramp proper to determine if the ramp proper has a "controlling curve" that is sufficiently sharp to affect vehicle speeds. Preliminary speeds are computed for the PC, midpoint (MC), and PT for each curve assuming that the curves are sufficiently gradual that their geometry does not affect vehicle speeds. The following equations are used to compute the preliminary speed estimates at the PC, MC, and PT of each curve:

$$V_{i,j,NC} = 1.0118V_{XRoad} + 78.3087X_j \;\; ; \;\; \leq V_{OS\,Hwy} \quad \textbf{(5)}$$

$$V_{i,j,C} = 0.9667V_{XRoad} + 143.9664X_j - 5.3122I_{55} - 2.6028I_{60} + 7.9150I_{65} \; ; \; \leq V_{OS\,Hwy} \quad \textbf{(6)}$$

where:

$V_{i,j,NC}$ = Predicted vehicle speed for curve i at point j (j = PC, MC, or PT), by applying the model for tangents with no downstream curve (see Section 4.3.1.3), mph.

$V_{i,j,C}$ = Predicted vehicle speed for curve i at point j (j = PC, MC, or PT), by applying the model for tangents with a downstream curve (see Section 4.3.1.3), mph.

V_{XRoad} = Vehicle speed at the crossroad ramp terminal (provided by the analyst), mph.

$V_{OS\,Hwy}$ = Average operating speed on the highway (or freeway) mainline, mph.

X_j = Milepost for point j, mi.

I_{55} = Indicator variable for 55-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 55 mph, 0 otherwise).

I_{60} = Indicator variable for 60-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 60 mph, 0 otherwise).

I_{65} = Indicator variable for 65-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 65 mph or above, 0 otherwise).

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Equations 5 and 6 are the calibrated tangent speed models. By applying these models to the curve points, speeds are predicted along the ramp proper as if it is a tangent ramp. Once the speeds $V_{i,j,NC}$ and $V_{i,j,C}$ are computed for each curve point, they are compared to the design speed for the curve ($V_{DS\ Curve}$). A controlling curve is defined where either of the predicted vehicle speeds ($V_{i,j,NC}$ or $V_{i,j,C}$) exceeds the curve's design speed ($V_{DS\ Curve}$). If a controlling curve is identified, the preliminary speed profile is revised by applying calibrated curve and tangent speed models to all ramp proper points at or downstream of the midpoint of the controlling curve (see Section 4.3.1.3).

4.3.1.3 Ramp Proper Revised Speed Profile

If a controlling curve is identified, the speed profile is updated as follows. The model to estimate vehicle speed at the midpoint of a curve (MC) on an entrance ramp is described as follows:

$$V_{MC} = -8.7255 + 1.0125V_{PC} + 4.7053R - 2.0183R^2 + 0.1316V_{SL\ Hwy}\ ;\ V_{MC} = Max\ (V_{OS\ Hwy}) \quad \textbf{(7)}$$

where:

V_{MC} = Predicted vehicle speed at the MC, mph.
V_{PC} = Predicted vehicle speed at the beginning of a curve (i.e., initial point of horizontal curve – PC), mph.
R = Curve radius, mi.
$V_{SL\ Hwy}$ = Regulatory speed limit on the highway (or freeway) mainline, mph.

The model to estimate vehicle speed at the end of a curve (PT) on an entrance ramp is described as follows:

$$V_{PT} = -12.1179 + 1.0127V_{PC} + 8.3558R - 3.5183R^2 + 0.1882V_{SL\ Hwy}\ ;\ V_{PT} \geq V_{MC} \text{ and } V_{PT} = Max\ (V_{OS\ Hwy}) \quad \textbf{(8)}$$

where:

V_{PT} = Predicted vehicle speed at the end of a curve (PT), mph.

To obtain speed estimates at the midpoint (MC) and end (PT) of the curve, knowledge of vehicle speed at the beginning of the curve (PC) is required. If the curve is the first segment on the ramp, the speed at the beginning of the curve (V_{PC}) is estimated based on the traffic control type at the crossroad ramp terminal or a speed entered by the analyst. If the curve follows a tangent segment, the speed at the beginning of the curve (V_{PC}) is equal to the predicted tangent endpoint speed obtained from the tangent speed model.

For tangent segments followed by a curve on an entrance ramp, the tangent endpoint speed is estimated as follows:

$$V_{Tan-End,C} = Max\left(\begin{matrix} V_{Tan-Begin}, \\ 0.9667V_{Tan-Begin} + 143.9664L_T - 5.3122I_{55} - 2.6028I_{60} + 7.9150I_{65} \end{matrix}\right); \leq V_{OS\ Hwy} \quad \textbf{(9)}$$

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where:

$V_{Tan\text{-}End,C}$ = Predicted vehicle speed at the end of a tangent if the tangent is followed by a curve, mph.
$V_{Tan\text{-}Begin}$ = Predicted vehicle speed at the beginning of a tangent, mph.
L_T = Tangent length, mi.

For tangent segments not followed by a curve on an entrance ramp, the tangent endpoint speed is estimated as follows:

$$V_{Tan-End,NC} = Max\left(V_{Tan-Begin},\ 1.0118V_{Tan-Begin} + 78.3087L_T\right);\ \leq V_{OS\ Hwy} \quad \textbf{(10)}$$

where:

$V_{Tan\text{-}End,NC}$ = Predicted vehicle speed at the end of a tangent if the tangent is not followed by a curve, mph.

If the tangent is the first segment on the ramp, the speed at the beginning of the tangent is estimated based on the traffic control type at the crossroad ramp terminal or the analyst input value as previously described. If the tangent follows a curve segment, the speed at the beginning of the tangent is equal to the predicted curve PT speed obtained from the curve speed model (V_{PT}).

For each tangent, the analyst can input the design speed of the tangent. The design speed of the tangent section should be consistent with the speeds that drivers are likely to travel at the end of the tangent. The tangent design speed does not affect the calculation of vehicle speeds, but the RSPM plots the tangent design speed values so the analyst can check if vehicle speeds are consistent with the design speeds.

4.3.1.4 Freeway Mainline Ramp Terminal

On the freeway mainline ramp terminal, drivers accelerate until they merge into the highway (or freeway) mainline traffic. With the freeway mainline ramp terminal, there are three primary points of interest:

- The beginning point of the acceleration length (or Point A, with speed $V_{Acc\ Length(i)}$).
- The gore point (with speed V_G).
- The merge location (with speed $V_{Acc\ Length(f)}$).

Figure 27 illustrates the primary components of an entrance ramp near the freeway mainline ramp terminal and the associated speeds of vehicles at the primary points of interest.

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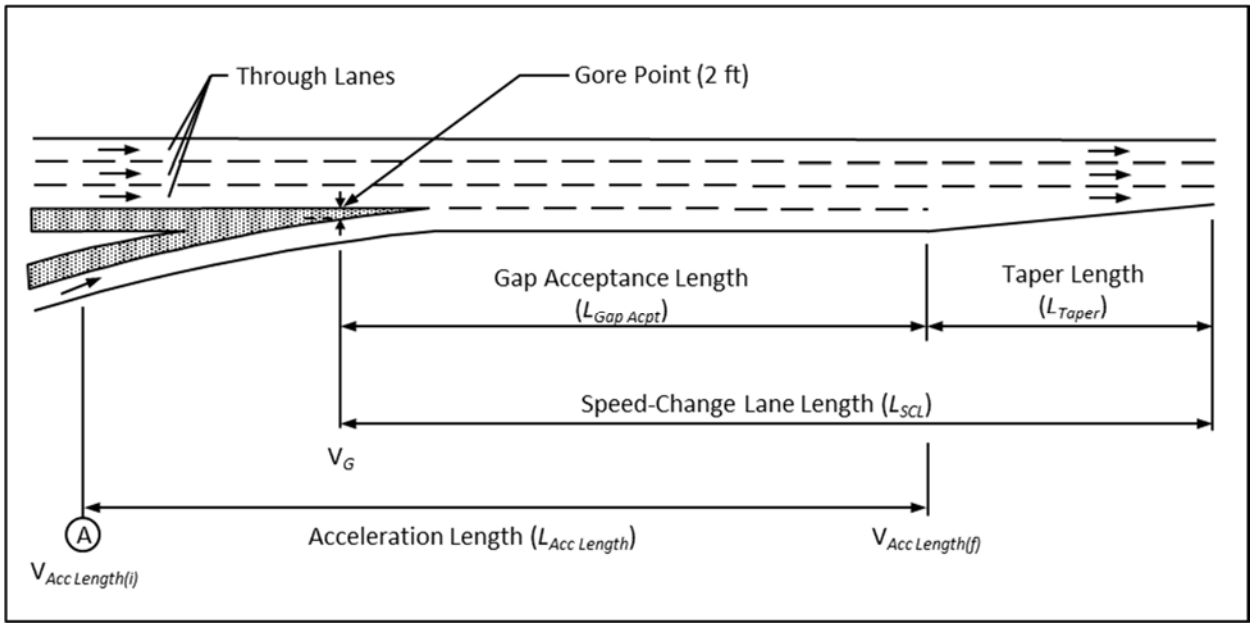


Figure 27. Primary Components of an Entrance Ramp near the Freeway Mainline Ramp Terminal and Associated Speed Points of Interest

The speed at the gore point (V_G) is computed using one of three methods based on the following scenarios:

1. The ramp proper has one or more curves and the final curve ends at the gore point $(n_c > 0 \text{ and } X_{PT,f} = X_G)$.
2. The ramp has one or more curves and the final curve ends upstream of the gore point $(n_c > 0 \text{ and } X_{PT,f} < X_G)$.
3. The ramp has no curves $(n_c = 0)$.
   V_G is computed in these cases as follows:

$$V_G = \begin{cases} Min\left(V_{OS\ Hwy}, V_{PT,f}\right);\ n_c > 0 \text{ and } X_{PT,f} = X_G \\ Min\left(V_{OS\ Hwy}, 1.0118V_{PT,f} + 78.3087\left[X_G - X_{PT,f}\right]\right);\ n_c > 0 \text{ and } X_{PT,f} < X_G \\ Min\left(V_{OS\ Hwy}, 1.0118V_{XRoad} + 78.3087X_G\right);\ n_c = 0 \end{cases} \quad \textbf{(11)}$$

where:

V_G = Predicted vehicle speed at the gore point, mph.
n_c = Number of curves on the ramp proper.
$X_{PT,f}$ = Milepost for the PT of final curve on the ramp proper, mi.
X_G = Milepost for the gore point, mi.
$V_{PT,f}$ = Predicted vehicle speed at the PT of final curve on the ramp proper, mph.

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The speed calculations for Scenarios 2 and 3 apply to the tangent speed model described by Equation 11. In all cases, the upper bound for the speed at the gore point is the average operating speed on the highway (or freeway) mainline ($V_{OS\ Hwy}$) as input by the analyst.

The gap acceptance length ($L_{Gap\ Acpt}$) begins at the gore point and ends at the beginning of the taper (i.e., the last location where the taper is at least 12 ft) and should be a minimum of 300 ft. The merge location is typically about half of the gap acceptance length (Torbic et al., 2012). V_{Merge} is estimated based on the gap acceptance length, the merge location, and field-measured acceleration rates (Torbic et al., 2012) as follows:

$$V_{Merge} = Min\left(V_{OS\ Hwy}\ ,\ \sqrt{V_G^2 + 2\left(\frac{3600^2}{5280}\right) a_{Gap\ Acpt} L_{Gap\ Acpt} P_{Gap\ Acpt}} \right) \quad \textbf{(12)}$$

where:

V_{Merge} = Predicted vehicle speed where the vehicle merges from the speed-change lane onto the highway (or freeway), mph.
$a_{Gap\ Acpt}$ = Acceleration rate along gap acceptance length, ft/s^2.
$L_{Gap\ Acpt}$ = Gap acceptance length, mi.
$P_{Gap\ Acpt}$ = Proportion of gap acceptance length used by merging vehicles [default = 0.50, based on Torbic et al., (2012)].

The acceleration rate along the gap acceptance length ($a_{Gap\ Acpt}$) is based on Table 27 from Torbic et al., (2012), but the table has been modified slightly for use with the spreadsheet tool and is presented here as Tables 14 and 15. Using V_G as the initial speed and the design speed of the highway (or freeway) mainline ($V_{DS\ Hwy}$), Tables 14 and 15 are used to determine the acceleration rate along the gap acceptance length ($a_{Gap\ Acpt}$) based on the following conditions:

1. Use the values from Table 14 if the absolute value of the ramp's overall grade G_r is less than 2.5 percent and Table 14 is populated for both initial speed values bounding the computed V_G value and the design speed of the highway (or freeway) mainline ($V_{DS\ Hwy}$). For example, if the ramp grade is 1 percent and the design speed of the highway (or freeway) mainline is 60 mph, Table 14 could be used for an initial speed of 30 mph (by interpolating) but not for an initial speed of 22 mph.
2. Use the values from the body of Table 15 if Condition 1 is not met.
3. Use the values from the footer of Table 15 in all other cases.

Table 14. Entrance Ramp Acceleration Rates along Gap Acceptance Length, $|G_r| < 2.5$ percent (adapted from Torbic et al., 2012)

Design Speed of Freeway ($V_{DS\ Hwy}$) mph	Acceleration Rate along Gap Acceptance Length ($a_{Gap\ Acpt}$), ft/s^2, by Initial Speed (V_G), mph								
	0	14	18	22	26	30	36	40	44
30									
35					3.39				
40					3.28	2.88			
45					3.22	2.97			
50					3.24	2.90	2.69		
55					3.40	2.72	2.68	2.47	
60						2.68	2.66	2.36	1.97

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65						3.18	2.57	2.36	1.96
70							2.62	2.49	2.05
75							2.76	2.60	2.12

Table 15. Entrance Ramp Acceleration Rates along Gap Acceptance Length, $|G_r| \geq 2.5$ percent (adapted from Torbic et al., 2012)

Design Speed of Freeway ($V_{DS\ Hwy}$) mph	Acceleration Rate along Gap Acceptance Length ($a_{Gap\ Acpt}$), ft/s^2, by Initial Speed (V_G), mph								
	0 *	**14 ***	**18**	**22**	**26**	**30**	**36**	**40**	**44**
30	2.46	2.46	2.83	2.94					
35	2.53	2.53	3.07	3.12	3.16				
40	2.48	2.48	3.07	3.18	3.16	3.17			
45	2.31	2.31	3.01	3.11	3.12	3.06			
50	2.41	2.41	2.97	3.08	3.04	3.00	2.78		
55	2.94	2.94	2.94	3.09	3.01	2.96	2.81	2.57	
60	3.00	3.00	3.00	3.10	3.02	2.99	2.89	2.83	2.48
65	3.11	3.11	3.11	3.18	3.15	3.11	2.91	2.83	2.67
70	3.10	3.10	3.10	3.27	3.28	3.21	3.07	2.91	2.80
75	3.29	3.29	3.29	3.37	3.38	3.36	3.37	3.21	3.00
Average	2.76	2.76	3.04	3.14	3.15	3.11	2.97	2.87	2.74

* Underlined numbers are extrapolated.

4.3.1.5 Alternate Speed Prediction Models for Entrance Ramps

In addition to the speed prediction models developed as part of this research, the RSPM spreadsheet tool includes speed prediction models developed for use with the HSM safety analysis procedures for interchange ramps (Bonneson et al., 2012). Bonneson et al. note that these speed models were not developed for the purpose of predicting vehicle speeds in the context of operational or design analyses, but the speed prediction models were incorporated into the RSPM spreadsheet tool as an alternative approach to estimating and evaluating the consistency of a ramp design as it relates to vehicle speeds. Within the RSPM spreadsheet tool, the analyst has the option to display or not display predicted speeds from the HSM safety analysis procedures.

Inputs into the speed profile models included in the HSM methodology for interchange ramps are shown in Table 16. When applied, the speed profile models included in the HSM methodology yield average entry and exit speeds for each curve on a ramp. A seven-step procedure described below provides details on how the models in the HSM methodology are applied to estimate vehicle speed along an entrance ramp. The nomenclature that Bonneson et al. used to describe the methodology of the speed profile models included in the HSM methodology is repeated below with a few adaptions.

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Table 16. Input Data for Ramp Curve Speed Prediction Procedures in ISATe (adapted from Bonneson et al., 2012)

Variable	Description	Default Value
X_i	Milepost of the point of change from tangent to curve (PC) for curve i [1], mi	None
R_i	Radius of curve i [2], ft	None
$L_{C,i}$	Length of horizontal curve i, mi	None
$V_{OS\ Hwy}$	Average traffic speed on freeway during off-peak periods of the typical day, mph	Estimate is equal to the speed limit
V_{XRoad}	Average speed at point where ramp connects to crossroad, mph	15—ramps with stop-, yield-, or signal-controlled crossroad ramp terminals 30—all other ramps at service interchanges

NOTES:

[1] If the curve is preceded by a spiral transition, then X_i *is* the average of the TS and SC mileposts, where TS is the milepost of the point of change from tangent to spiral and SC is the milepost of the point of change from spiral to curve.

[2] If the curve has spiral transitions, then R_i is equal to the radius of the central circular portion of the curve.

Step 1—Gather Input Data: The input data for this procedure are identified in Table 16.

Step 2—Compute Limiting Curve Speed: The limiting curve speed is computed for each curve on the ramp using Equation 13.

$$v_{max,i} = 3.24\ (32.2\ R_i)^{0.30} \quad \textbf{(13)}$$

where,

$v_{max,i}$ = Limiting speed for curve i, ft/s.

The analysis proceeds in the direction of travel with the first curve encountered on the ramp designated as Curve 1 (i=1). The value of v_{max} is computed for all curves prior to, and including, the curve of interest. The value obtained from Equation 13 is an upper limit on the curve speed. Vehicles may reach this speed if the distance between curves is long enough or the crossroad speed is high.

Step 3—Calculate Curve 1 Entry Speed: The average entry speed at Curve 1 is computed using Equation 14.

$$v_{ent,1} = ([1.47\ V_{XRoad}]^3 + 495 \times 5280\ X_1)^{1/3}\ ; \quad \leq 1.47\ V_{OS\ Hwy} \quad \textbf{(14)}$$

where,

$v_{ent,1}$ = Average entry speed for Curve 1, ft/s.

The boundary condition of Equation 14 indicates that the value computed ($v_{ent,1}$) cannot exceed the average highway (or freeway) speed ($V_{OS\ Hwy}$).

Step 4—Calculate Curve 1 Exit Speed: The average exit speed at Curve 1 is computed using Equation 15.

$$v_{ext,1} = (V^3_{ent,1} + 495 \times 5280\ L_{c,1})^{1/3}\ ; \quad \leq v_{max,1} \text{ and } \leq 1.47\ V_{OS\ Hwy} \quad \textbf{(15)}$$

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where,

$v_{ext,1}$ = Average exit speed for Curve 1, ft/s.

The boundary conditions of Equation 15 indicate that the value computed ($v_{ext,1}$) should not exceed the limiting curve speed ($v_{max,i}$) or the average highway (or freeway) speed ($V_{OS\ Hwy}$).

Step 5—Calculate Curve *i* Entry Speed: The average entry speed at Curve 2 (and all subsequent curves) is computed using Equation 16.

$$v_{ent,i} = \left(V^3_{ext,i-1} + 495 \times 5280 \left[X_i - X_{i-1} - L_{c,i-1}\right] \right)^{1/3} ; \quad \leq 1.47\ V_{OS\ Hwy} \qquad \textbf{(16)}$$

where,

$v_{ent,i}$ = Average entry speed for curve i (i = 2, 3, ...), ft/s.
$v_{ext,i}$ = Average exit speed for curve i (i = 2, 3, ...), ft/s.

Step 6—Calculate Curve *i* Exit Speed: The average exit speed at Curve 2 (and all subsequent curves) is computed using Equation 17.

$$v_{ext,i} = \left(V^3_{ent,i} + 495 \times 5280\ L_{c,i}\right)^{1/3} ; \quad \leq v_{max,i} \text{ and } \leq 1.47\ V_{OS\ Hwy} \qquad \textbf{(17)}$$

Step 7—Calculate Speed on Successive Curves: The entry and exit speeds for Curve 3 and successive curves are computed by applying Steps 5 and 6 for each curve.

Figure 28 illustrates a sample output from the RSPM worksheet for an entrance ramp that includes the alternate speed profile from the HSM procedures.

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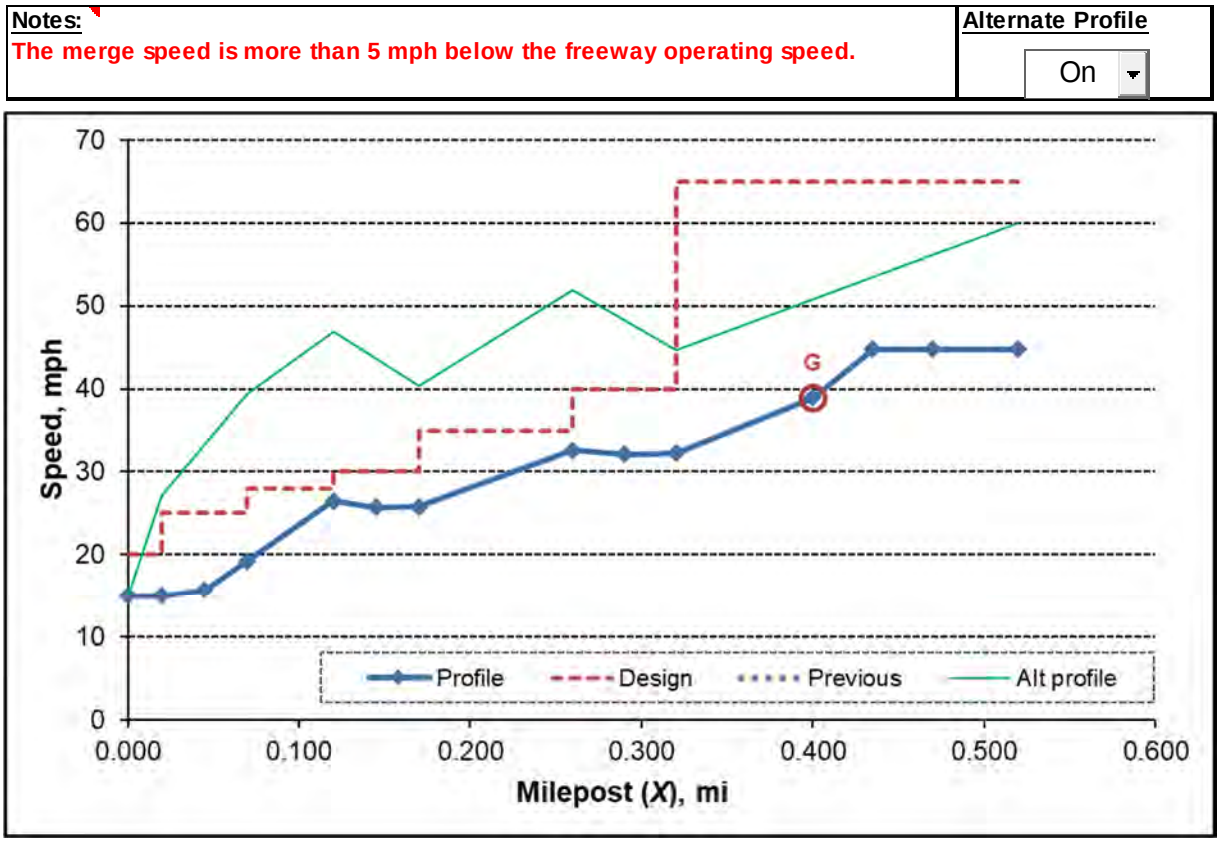


Figure 28. Graphical Illustration of Vehicle Speeds for Sample Entrance Ramp with Alternate Profile

4.3.2 Speed Calculations for Exit Ramps

Applicable speed calculations are presented in the order of the primary components of an exit ramp based on the direction of travel:

- The freeway mainline ramp terminal.
- The ramp proper.
- The crossroad ramp terminal.

4.3.2.1 Freeway Mainline Ramp Terminal

At the freeway mainline ramp terminal of an exit ramp, there are three points of interest associated with the speed calculations: the diverge location (with speed $V_{Diverge}$), the gore point (with speed V_G), and the end point of the deceleration length (or Point D, at the end of $L_{Dec\ Length}$). Figure 29 illustrates the primary components of an exit ramp near the freeway mainline ramp terminal and the associated speed points of interest.

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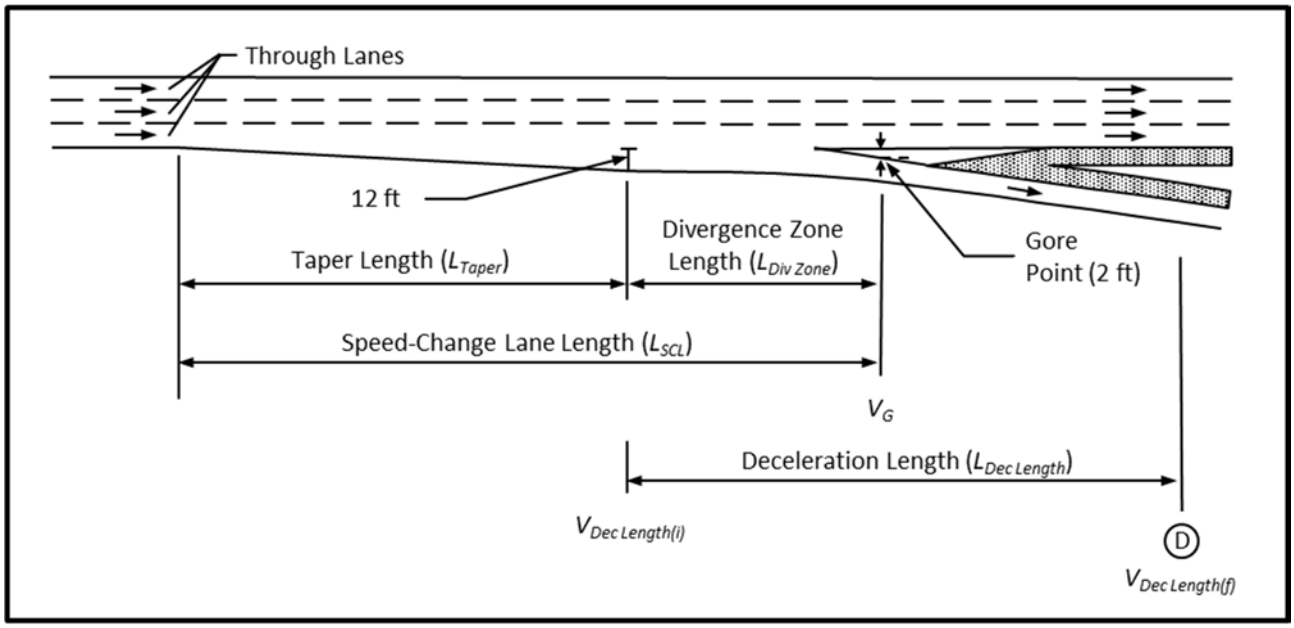


Figure 29. Primary Components of an Exit Ramp near the Freeway Mainline Ramp Terminal and Associated Speed Points of Interest

The diverge location where the vehicle enters the speed-change lane is typically about one-tenth the length of the speed-change lane (Torbic et al., 2012). The vehicle speed at the diverge location is estimated based on the average operating speed of the freeway mainline ($V_{OS\ Hwy}$) and default speed differential values (Table 17) as follows:

$$V_{Diverge} = V_{OS\ Hwy} + \Delta V \quad \textbf{(18)}$$

where:

$V_{Diverge}$ = Predicted vehicle speed at the diverge location, mph.
ΔV = Mean speed differential between freeway mainline speed and the initial speed of vehicles entering the speed-change lane (default values are provided in Table 17 based on the ramp type and type of speed-change lane), mph.

Table 17. Exit Ramp Speed Differentials (adapted from Torbic et al., 2012)

Ramp Type	Type of Speed-Change Lane	Mean Speed Differential ΔV, mph
Loop	Parallel	-4.4
Loop	Taper	-5.7
Diagonal	Parallel	-1.6
Diagonal	Taper	-4.1
Outer connection	Parallel	-1.6
Outer connection	Taper	-4.1

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Vehicle speed at the gore point is estimated based on vehicle speed at the diverge location and field-measured deceleration rates (Torbic et al., 2012) as follows:

$$V_G = \frac{3600}{5280}\sqrt{\left(\frac{5280}{3600}V_{Diverge}\right)^2 + 2a_{SCL}(1 - P_{SCL})L_{SCL}} \quad \textbf{(19)}$$

where:

a_{SCL} = Deceleration rate along the speed-change lane (= -2.914 ft/s^2).
L_{SCL} = Length of speed-change lane, ft.
P_{SCL} = Proportion of speed-change lane located upstream of the diverge point (default value is 0.10 based on Torbic et al. (2012)).

The deceleration rate (a_{SCL}) along the speed-change lane within the speed-change lane is derived from a weighted averaging of the deceleration rates and vehicle counts reported by Torbic et al. (2012).

4.3.2.2 Ramp Proper

The model to estimate vehicle speed at the MC on an exit ramp is described as follows:

$$V_{MC} = \begin{cases} -13.4726 + 0.5951V_{PC} + 208.5633R - 521.3073R^2 + 0.2361V_{SL\,Hwy} + 2.1981I_{DN} \\ \quad +0.7507I_{LV} - 2.9488I_{UP} \;; \quad \leq V_{PC}, R \leq 1{,}000\ ft \\ \\ V_{PC} + (V_{PC} - V_{PC-1})\dfrac{(X_{MC} - X_{PC})}{(X_{PC} - X_{PC-1})} \;; \quad R > 1{,}000\ ft \end{cases} \quad \textbf{(20)}$$

where:

I_{DN} = Indicator variable for downward slope (= 1 if the overall ramp slopes downward, 0 otherwise).
I_{LV} = Indicator variable for level slope [= 1 if the overall ramp is level (i.e., ± 2 percent), 0 otherwise].
I_{UP} = Indicator variable for upward slope (= 1 if the overall ramp slopes upward, 0 otherwise).
$V_{PC\text{-}1}$ = Predicted speed at the point preceding the curve PC, mph.
$X_{PC\text{-}1}$ = Milepost of the point preceding the curve PC, mi.

The model to estimate vehicle speed at the end of a curve (PT) on an exit ramp is described as follows:

$$V_{PT} = -0.6272 + 0.8637V_{PC} + 108.0929R - 265.9747R^2 - 190.8941L_c \;; \leq V_{MC} \quad \textbf{(21)}$$

where:

L_c = Curve segment length, mi.

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To obtain speed estimates at the midpoint and end of the curve, knowledge of vehicle speed at the beginning of the curve (PC) is required. If the curve is the first segment on the ramp, the speed at the beginning of the curve is set equal to V_G. If the curve follows a tangent segment, the speed at the beginning of the curve is equal to the estimated tangent endpoint speed obtained from the tangent speed model. If the radius of the curve exceeds 1,000 ft, the estimated speed at the MC is computed based on an extrapolation of the speed change between the curve PC and its preceding point. In this case, knowledge of the location and speed at the point preceding the curve PC is required.

For each curve, the analyst inputs the design speed of the curve, the milepost of the beginning of the curve (PC), the curve radius, and the curve length.

For tangent segments followed by a curve on an exit ramp, the tangent endpoint speed is estimated as follows:

$$V_{Tan-End,C} = 0.7980V_{Tan-Begin} + 228.6196R_{next} - 575.2145R_{next}^2 - 645.4397L_T + 6733.5203L_T^2 \quad \textbf{(22)}$$

with:

R_{next} = Min (R_{max}, radius of subsequent curve), mi.
R_{max} = 1,000 ft (0.189 mi).

For tangent segments not followed by a curve on an exit ramp, the tangent endpoint speed is estimated as follows:

$$V_{Tan-End,NC} = 0.4890V_{Tan-Begin} \quad \textbf{(23)}$$

If the tangent is the first segment on the ramp, the speed at the beginning of the tangent is set equal to V_G. If the tangent follows a curve segment, the speed at the beginning of the tangent is equal to the estimated speed at the end of the curve obtained from the curve speed model.

For each tangent, the analyst inputs the design speed of the tangent. The design speed of the tangent section should be consistent with the speeds that drivers are likely to be traveling at the end of the tangent.

4.3.2.3 Crossroad Ramp Terminals

Several factors are considered in the estimation of vehicle speed where the ramp meets the crossroad traveled way at the crossroad ramp terminal. First, the same default speeds used for the beginning point of entrance ramps (15 mph for stop, yield, or signal control; 30 mph otherwise) are provided as default speeds unless a field-measured value is available. Second, if queue storage space is needed at the end of the ramp, vehicle speed is estimated as 0 mph for the entirety of the needed queue storage length. The analyst enters the queue storage length.

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4.3.2.4 Alternate Speed Prediction Models for Exit Ramps

The RSPM spreadsheet tool also includes speed prediction models developed for use with the HSM safety analysis procedures for interchange ramps (Bonneson et al., 2012) that can be used to estimate vehicle speeds on exit ramps. Within the RSPM spreadsheet tool, the analyst has the option to display or not display predicted speeds from the HSM safety analysis procedures. Inputs into the speed profile models included in the HSM methodology for interchange ramps are shown in Table 16. The speed profile models yield average entry and exit speeds for each curve on a ramp. The seven-step procedure described below details how the models are applied to estimate vehicle speeds along an exit ramp.

Step 1—Gather Input Data: The input data for this procedure are identified in Table 16.

Step 2—Compute Limiting Curve Speed: This step is the same as Step 2 for the entrance ramp procedure. A limiting curve speed is computed for each curve on the ramp using Equation 13.

$$v_{max,i} = 3.24\ (32.2\ R_i)^{0.30} \quad \textbf{(13)}$$

The analysis proceeds in the direction of travel with the first curve encountered on the ramp designated as Curve 1 (i=1). The value of v_{max} is computed for all curves prior to, and including, the curve of interest. The value obtained from Equation 13 is an upper limit on the curve speed. A lower curve speed than that obtained from Equation 13 is possible as deceleration may occur along the ramp as the driver transitions from the freeway speed to the crossroad speed.

Step 3—Calculate Curve 1 Entry Speed: The average entry speed at Curve 1 is computed using Equation 24.

$$v_{ent,1} = 1.47\ V_{OS\ Hwy} - 0.034 \times 5280\ X_1 \quad ; \quad \geq 1.47\ V_{XRoad} \quad \textbf{(24)}$$

The boundary condition of Equation 24 indicates that the value computed ($v_{ent,1}$) cannot be less than the average speed at the point where the ramp connects to the crossroad (V_{XRoad}).

Step 4—Calculate Curve 1 Exit Speed: The average exit speed at Curve 1 is computed using Equation 25.

$$v_{ext,1} = v_{ent,1} - 0.034 \times 5280\ L_{c,1} \quad ; \quad \leq v_{max,1} \text{ and } \geq 1.47\ V_{XRoad} \quad \textbf{(25)}$$

The boundary conditions of Equation 25 indicate that the value computed ($v_{ext,1}$) should not exceed the limiting curve speed ($v_{max,i}$) and should not be less than the average speed at the point where the ramp connects to the crossroad (V_{XRoad}).

Step 5—Calculate Curve i Entry Speed: The average entry speed at Curve 2 (and all subsequent curves) is computed using Equation 26.

$$v_{ent,i} = v_{ext,i-1} - 0.034 \times 5280\ (X_1 - X_{i-1} - L_{c,i-1}) \quad ; \quad \geq 1.47\ V_{XRoad} \quad \textbf{(26)}$$

Step 6—Calculate Curve *i* Exit Speed: The average exit speed at Curve 2 (and all subsequent curves) is computed using Equation 27.

$$v_{ext,i} = v_{ent,i} - 0.034 \times 5280\ L_{c,i} \quad ; \quad \leq v_{max,i} \text{ and } \geq 1.47\ V_{XRoad} \quad \textbf{(27)}$$

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Step 7—Calculate Speed on Successive Curves: The entry and exit speeds for Curve 3 and successive curves are computed by applying Steps 5 and 6 for each curve. This step is the same as Step 7 for the entrance ramp procedure.

Figure 30 illustrates a sample output for an exit ramp that includes the alternate speed profile from the HSM procedures.

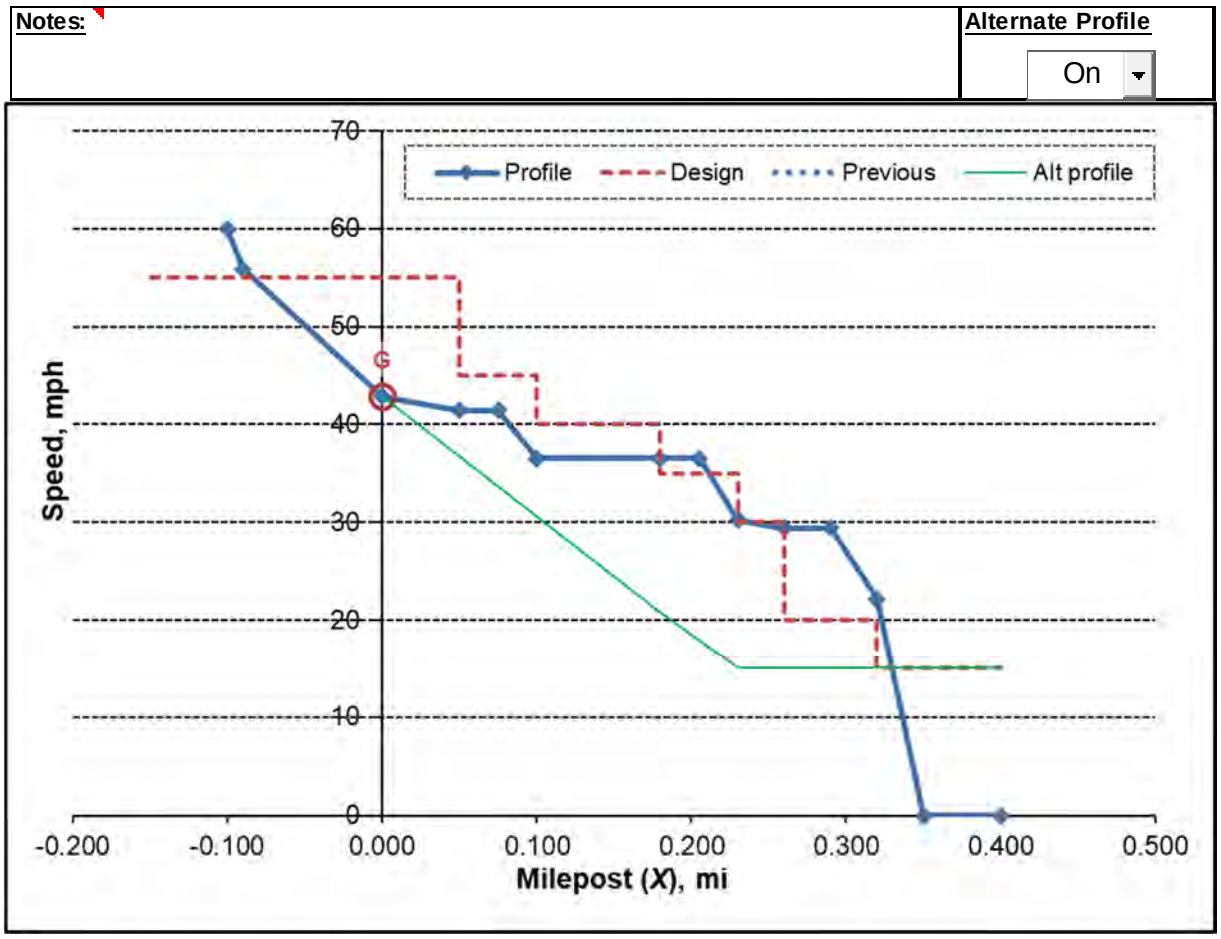


Figure 30. Graphical Illustration of Vehicle Speeds for Sample Exit Ramp with Alternate Profile

4.3.3 Quality Control Checks

The RSPM performs several quality control checks to assess whether the ramp design, as input by the analyst, meets or violates several general design controls and criteria for ramp design. The types of quality control checks are associated with:

- Acceptable acceleration/deceleration rates associated with individual portions of the ramp.
- Assessment of whether the merge speed (V_{Merge}) is within 5 mph of the average operating speed of the highway (or freeway) ($V_{OS\ Hwy}$).

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- Graphical illustration of estimated speeds to design speeds of individual ramp components.

4.3.3.1 Check of Acceleration/Deceleration Rates

After the speeds are estimated at the beginning and end of each ramp section, the average acceleration rate for the section is computed using basic kinematics as follows:

$$a_{avg} = \frac{V_f^2 - V_i^2}{2L} \quad \textbf{(28)}$$

where:

a_{avg} = Average acceleration rate, ft/s^2.
V_i = Predicted vehicle speed at the beginning of the section, ft/s.
V_f = Predicted vehicle speed at the end of the section, ft/s.
L = Segment length, ft.

By definition, positive acceleration (or increasing speed) is positive, and negative acceleration (or deceleration, or decreasing speed) is negative.

The computed average acceleration rates are compared to acceleration rates recommended by *Green Book* Figures 2-33 and 2-34 for design. *Green Book* Figures 2-33 and 2-34, reproduced above as Figures 15 and 18 respectively, illustrate acceleration and deceleration characteristics of passenger cars suitable for design applications for design features such as intersections and freeway ramps. As part of previous research on freeways and interchanges, Bonneson et al. (2012) developed regression models to estimate the acceleration and deceleration characteristics depicted in *Green Book* Figures 2-33 and 2-34. These acceleration and deceleration characteristics are interpreted as vehicle performance limits. For example, on an entrance ramp, vehicles can accelerate up to a maximum acceleration rate as depicted in *Green Book* Figure 2-33. Acceleration rates above those depicted in *Green Book* Figure 2-33 are not acceptable for design purposes. Based on the assessment by Bonneson et al. (2012), the maximum design acceleration rates along freeway entrance ramps may be calculated using Equation 29 and the maximum design deceleration rates along freeway exit ramps may be calculated using Equation 30:

$$\text{For entrance ramps: } a_{Design} = \frac{165}{V_i} \quad \textbf{(29)}$$

$$\text{For exit ramps: } a_{Design} = -0.121V_i \quad \textbf{(30)}$$

where:

a_{Design} = Maximum design acceleration/deceleration rate, ft/s^2.

The maximum design acceleration rates calculated using Equations 29 and 30 are compared to the average acceleration/deceleration rates for the individual ramp sections.

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4.3.3.2 Check of Merge Speed on Entrance Ramps

For the design of an entrance ramp, the *Green Book* states that motorists should attain a merge speed within 5 mph of the speed of the highway (or freeway) by the time they reach the end of the gap acceptance length ($L_{Gap\ Acpt}$). The RSPM performs a check to determine if the merge speed (V_{Merge}) is within 5 mph of the average operating speed of the highway (or freeway) ($V_{OS\ Hwy}$). If the difference between the average operating speed of the highway (or freeway) and the merge speed ($V_{OS\ Frwy}$ - V_{Merge}) is greater than 5 mph, a warning message is provided.

4.3.3.3 Check of Estimated Operating Speeds to Design Speeds on Individual Ramp Components

Graphs provide a visual indication of any locations on the ramp where the estimated operating speed of a vehicle is greater than the design speed of the individual ramp section.

4.4 Uses and Limitations of the RSPM

The RSPM is intended to be used to assess the adequacy of a ramp design or design alternatives during the design or reconstruction process. With the capability to adjust several input parameters, the RSPM can be used to predict vehicle speeds based upon anticipated driver behaviors or design conditions. The output from the RSPM should not be viewed as absolute truth but should be interpreted by the analyst or engineer to help design an interchange ramp in a consistent manner that meets driver expectations.

The RSPM has several known limitations. By listing these limitations, engineers can have a better sense of the strengths and weaknesses of the tool, know the conditions for which it can best be applied, and possibly figure out best approaches to modify the inputs to address design conditions or situations for which the tool was not specifically designed.

The RSPM was designed to predict vehicle speeds based upon anticipated driver behaviors but the inputs can also be adjusted to predict speeds based on assumed design conditions. The primary inputs to the tool for addressing both conditions are $P_{Gap\ Acpt}$ for entrance ramps and P_{SCL} for exit ramps. For entrance ramps, it is assumed for design purposes that vehicles use the entire acceleration length and gap acceptance length before merging onto the freeway. As such, the assumed acceleration of vehicles along the acceleration length and design values for minimum acceleration lengths are based on assumed merging speeds at the end of the deceleration length. However, field data show that many drivers use only a portion of the gap acceptance length prior to merging onto the freeway (Torbic et al., 2012). By adjusting the value of $P_{Gap\ Acpt}$, the analyst can use the RSPM to predict merge speeds based upon different merge locations. A default value of $P_{Gap\ Acpt} = 0.5$ is provided within the RSPM based on observed driver behaviors. A value of $P_{Gap\ Acpt} = 1.0$ more accurately represent assumed design conditions where vehicles merge onto the freeway at the end of the acceleration length and gap acceptance length.

For exit ramps, it is assumed that vehicles diverge from the freeway mainline at the end of taper and the beginning of both the divergence zone length and the deceleration length. It is also assumed that vehicles do not decelerate in the freeway lane prior to the diverge maneuver. However, field data show that drivers diverge at different locations along the speed-change lane (Torbic et al., 2012). Noting that $L_{SCL} = L_{Taper} + L_{Div\ Zone}$, a value of $P_{SCL} = 0.1$ is provided as a

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default based on field observation. However, this value could be adjusted by the analyst so that the diverge location corresponds with the beginning of the divergence zone length and deceleration length. For the diverge speed to be equivalent with the operating speed of the freeway, the input value for the operating speed of the freeway would have to be adjusted equivalent to the mean speed differentials used in the prediction methodology (see Table 17).

By understanding how the various inputs can be adjusted to more accurately model observed driver behaviors or design conditions, analyst can use the RSPM to evaluate and assess specific conditions. As with the development of most design tools, it was not possible to develop the RSPM to address all ramp design scenarios or conditions. Below is a list of several known limitations of the tool which are intended to help analyst more accurately apply the tool:

Key Limitations to the RSPM:

- The RSPM does not address the design of ramps for system interchanges. Speed data were collected along ramps at several system interchanges, but modeling efforts did not yield reliable speed prediction models for use with freeway-to-freeway ramps. Therefore, the RSPM was developed to address only entrance and exit ramps at service interchanges, not freeway-to-freeway ramps at system interchanges.
- The RSPM was designed to address the most common types of ramps at service interchanges including diagonal, loop, and outer connection ramps. The RSMP was not designed to address other types of ramps or ramps with more complex geometrics.
- The RSPM incorporates several regression models for tangents and curves that predict vehicle speeds at specific locations along the ramp including:
  - Entrance ramp: at the MC.
  - Entrance ramp: at the end of the curve (PT).
  - Entrance ramp: at the end of a tangent with a subsequent curve.
  - Entrance ramp: at the end of a tangent without a subsequent curve.
  - Exit ramp: at the MC.
  - Exit ramp: at the end of the curve (PT).
  - Exit ramp: at the end of a tangent with a subsequent curve.
  - Exit ramp: at the end of a tangent without a subsequent curve.

  Most of the models were developed based on observations of 500 or more vehicles across multiple sites. However, the model that predicts vehicle speeds at the end of a tangent without a subsequent curve for exit ramps is based on data for 77 vehicles at one location. Therefore, when modeling an exit ramp that includes a tangent without a subsequent curve, the analyst should be aware that predicted speeds for this design condition may be less reliable than for other design conditions.

  The regression models are also limited with respect to the range of site characteristics where actual speed data were collected. In most cases, the RSPM allows analysts to enter site characteristics outside of the limits from which the models were calibrated (e.g., curve length). The RSPM simply provides a message, showing the range of values for which the models were calibrated and prompts the analyst to continue or modify the input value, before estimating speeds for conditions outside of the range of site characteristics from which the models were calibrated (i.e., extrapolation of speed estimates).

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- The 85th percentile of the distribution of observed operating speeds is a frequently used measure associated with the design of geometric features. The speed prediction models incorporated in the RSPM yield estimates of average operating speeds, not the 85th percentile of operating speeds.

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Section 5.
Case Studies

This section demonstrates through several case studies how to evaluate the consistency of a ramp design based on the selected ramp design speed. It brings together information presented in three previous sections (Section 2.14, Section 3, and Section 4) to demonstrate how designers can assess the adequacy of a new ramp design during the design process or if an existing ramp is being considered for reconstruction how the guidelines herein can be used to evaluate the adequacy of design alternatives.

Several case studies are first presented for entrance ramps, followed by several case studies for exit ramps. The case studies are of actual ramps recently designed and/or built. When designing the ramps used in the case studies, agencies would have used design guidance from the 2011 *Green Book* or a manual with similar guidance. When gathering information for the case studies, the ramp design speed for each of the ramps was requested; however, none of the agencies provided the actual ramp design speeds from which the designs were based. Therefore, the ramp design speeds reported below in the case studies are inferred ramp design speeds based on the information available. As such, the case studies may illustrate potential differences that could arise when designing a ramp based on design guidance from the 2011 *Green Book* and design guidelines presented herein. In addition, other assumptions were made about the design of the ramp and/or characteristics of the connecting roads if the agencies did not provide the information.

5.1 Evaluating the Consistency of a Ramp Design for an Entrance Ramp at a Service Interchange Based on the Selected Ramp Design Speed

With entrance ramps, several key issues related to the selection of an appropriate ramp design speed should be considered during the design process and while evaluating the consistency of a ramp design:

- Because the controlling feature of an entrance ramp is towards the end of the ramp near the freeway mainline ramp terminal, ramp design speeds for entrance ramps are generally at least 35 mph or greater.
- Vehicle acceleration is a primary design control for entrance ramps. On the ramp proper, each tangent and curve section should be designed with sufficient length to accommodate the acceleration expected to occur over its length. *Green Book* Figure 2-33 may be used as a guide for determining minimum practical lengths of individual tangent and curve sections along the ramp proper.
- As vehicles transition from the ramp proper to the freeway mainline ramp terminal of an entrance ramp, the primary design criterion provided within AASHTO policy addresses minimum acceleration lengths along the freeway mainline ramp terminal (see *Green Book* Table 10-4). Vehicles should be able to attain a speed within 5 mph of the operating speed of the highway (or freeway) by the time they reach the end of the acceleration length.

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- Drivers tend to begin the merge, acceleration process only after gaining a clear view of the freeway right-traffic lane. If the driver's view from the ramp is obstructed, the acceleration length will not begin until near the ramp gore where the view of the freeway becomes unobstructed (Hunter and Machemehl, 1999).
- In general, the acceleration length begins at the end of the controlling curve or feature of the ramp proper. However, if the radius of the final curve of the ramp is greater than 1,000 ft or a tangent is present upstream of the gore point and drivers on the ramp have a clear view of traffic in the right lane of the freeway, the beginning of the acceleration length may be located upstream of the gore point (i.e., along the final curve of the ramp proper).

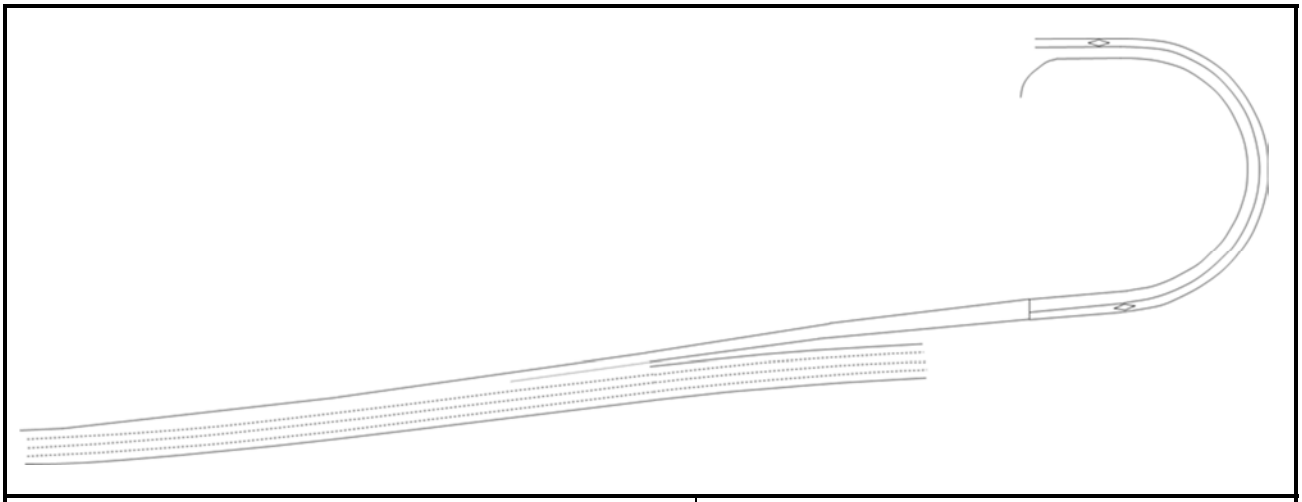
Below are two case studies of entrance ramps to demonstrate how to evaluate the consistency of a ramp design based on the selected ramp design speed.

5.1.1 Case Study No. 1: Loop Entrance Ramp in Suburban Area

5.1.1.1 Background Information

This case study features a loop ramp in a highly developed suburban area. Figure 31 provides a schematic and characteristics of the ramp. The loop ramp is part of a partial cloverleaf interchange. It begins at a signalized ramp terminal on a divided arterial. The speed limit on the crossroad is 45 mph. A ramp meter is present on the ramp, and a high-occupancy vehicle lane is adjacent to the travel lane leading up to the ramp meter. The ramp consists of a compound curve with radii of 182 ft and 224 ft, respectively, in the direction of travel. A third horizontal curve with a radius of 3,000 ft is located near the gore point. Based on the design speeds of Curves 1 and 2 and guidance in the 2011 and 2018 editions of the *Green Book*, the likely ramp design speed used for the design of this loop ramp is 25 mph.

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Interchange type:	Service		Conditions:	Constrained
Area type:	Urban/Suburban		Ramp configuration:	Loop
Freeway characteristics	Design speed:	70 mph	Operating speed:	65 mph
	Speed limit:	65 mph		
Crossroad characteristics	Type of roadway:	Primary/major rd	Speed limit:	45 mph
	Type of traffic control at terminal:		Signal	
Ramp Proper				
Ramp grade:	1 % downgrade		Max superelevation (e_{max}):	12 %
Curve 1	PC milepost:	0.044 mi	Curve length:	0.076 mi
	Curve radius	182 ft	Superelevation:	12 %
	Design speed	25 mph		
Curve 2	PC milepost:	0.120 mi	Curve length:	0.036 mi
	Curve radius	224 ft	Superelevation:	10 %
	Design speed	25 mph		
Curve 3	PC milepost:	0.223 mi	Curve length:	0.032 mi
	Curve radius	3000 ft	Superelevation:	4 %
	Design speed	50 mph		
Freeway Mainline Ramp Terminal				
Milepost of gore point			0.338 mi	
Acceleration length (upstream of the gore point)			0.083 mi	
Gap acceptance length			0.057 mi	
Taper length			0.110 mi	
Speed-change lane type			Parallel	

Figure 31. Case Study No. 1 (Entrance Ramp: General Ramp Characteristics and Dimensions)

5.1.1.2 Design Consistency Assessment

Based on the design guidelines presented herein, given the freeway design speed is 70 mph, the crossroad is a divided arterial, and the ramp is located in a highly developed suburban area, Table 4 indicates the ramp design speed for a loop ramp should be between 25 and 35 mph. Since space is limited, a lower ramp design speed of 25 mph seems logical for this ramp. Thus, the design speed of the last curve encountered along the ramp proper that significantly affects vehicle speed should be 25 mph. The design speed of Curve 2 is 25 mph; so as designed, the ramp design speed of this ramp is consistent with the design guidelines presented herein. All

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other sections and elements of the ramp should be designed consistent with this ramp design speed to provide appropriate speed transitions along the ramp.

To evaluate the adequacy and consistency of the existing ramp design, the ramp proper should be divided into individual tangents and curves. Design speeds and anticipated operating speeds should be evaluated for these tangents and curves on the ramp proper, along with the acceleration length and anticipated operating speeds for the freeway mainline ramp terminal. In Figure 31, the design speeds of the curves are presented, but the design speeds for the tangents were not provided for the ramp so this information is assumed as follows.

With entrance ramps, the first check of the adequacy of the ramp is to assess whether the design speeds increase in a stepwise manner between the contiguous segments on the ramp proper. For this case study, the first tangent begins at the crossroad ramp terminal and extends to the PC of Curve 1 at MP 0.044. Since the design speed of Curve 1 is 25 mph, it is logical for the design speed of the first tangent to be 20 mph. From MP 0.044 to MP 0.156, the design speeds of the individual curves for the compound curve are both 25 mph. The second tangent begins at the end of Curve 2 (i.e., MP 0.156) and extends to the PC of Curve 3 at MP 0.223. Since the design speed of Curve 3 is 50 mph, it is logical for the design speed of the second tangent to be 45 mph. From MP 0.233 to MP 0.255, the design speed of Curve 3 is 50 mph. The third tangent begins at the end of Curve 3 (i.e., MP 0.255) and extends to the gore point at MP 0.338. Since the design speed of Curve 3 is 50 mph, it is logical for the design speed of the third tangent to be 55 mph. Based on the assumed design speeds for the individual tangents and the design speeds of the curves as specified, the existing design of the ramp proper appears adequate as the design speeds of contiguous segments either increase or are the same in the direction of travel, and the change in design speed between adjoining sections is at most 5 mph.

The next step to evaluate the adequacy of the ramp proper is to confirm that sufficient distance is provided for a vehicle to accelerate as intended along the length of the ramp. A quick assessment that could be made is to simply compare the overall length of the ramp proper (i.e., from the crossroad ramp terminal to the gore point) with the length necessary to accelerate from the assumed entry speed of a vehicle at the crossroad ramp terminal and the design speed of the final segment of the ramp proper. The length from the crossroad ramp terminal to the gore point is 0.338 mi (1785 ft). The assumed entry speed of a vehicle onto the ramp proper from the crossroad ramp terminal is 15 mph, and the design speed of the tangent immediately upstream of the gore point is 55 mph. From Table 5, the minimum length required to accelerate from a speed of 15 mph to 55 mph is 0.179 mi (945 ft). This initial check suggests that sufficient distance is provided along the ramp proper for a vehicle to accelerate from a speed of 15 mph to 55 mph prior to entering the speed-change lane at the gore point. However, this assessment does not account for the individual tangent and curve sections that comprise the ramp proper. Therefore, further assessments should be made in reference to both the design speeds of the individual components and the anticipated operating speeds along the individual components.

In reference to design speeds and evaluating the ramp proper to confirm that sufficient distance is provided for a vehicle to accelerate as intended along the length of the ramp, Table 18 shows the beginning and ending mileposts and actual length of the individual components, the beginning and ending design speeds of the individual segments for this assessment, and the minimum acceleration lengths necessary to accelerate from the beginning and ending design speeds based upon acceleration capabilities of lower performance vehicles (see *Green Book*

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Figure 2-33). For this assessment, the design speed of the upstream segment is used as the beginning design speed for the adjacent downstream segment.

Table 18. Case Study No. 1: Assessment of Design Speeds for Individiual Segment of the Ramp Proper and Minimum Acceleration Lengths

Section	Mileposts (MP)		Actual Length (mi)	Design Speeds (mph)		Minimum Acceleration Length (mi)
	Begin	End		Begin	End	
Tangent 1	0.000	0.044	0.044	15	20	0.009
Curve 1	0.044	0.120	0.076	20	25	0.013
Curve 2	0.120	0.156	0.036	25	25	0.000
Tangent 2	0.156	0.223	0.067	25	45	0.084
Curve 3	0.223	0.255	0.032	45	50	0.038
Tangent 3	0.255	0.338	0.083	50	55	0.040

For this ramp, the lengths of the individual tangents and curve sections all exceed the minimum accelerate lengths required based on the initial and ending design speeds, except for Tangent 2 and Curve 3. As designed, the length of Tangent 2 is 0.067 mi, and the minimum length for a vehicle to accelerate from 25 to 45 mph is 0.084 mi. Similarly, the length of Curve 3 is 0.032 mi, and the minimum length for a vehicle to accelerate from 45 to 50 mph is 0.038 mi.

Before deciding that this ramp design is inadequate and should be redesigned, it should be recognized that the alignment of Curve 3 is fairly gentle with a radius of 3,000 ft so this curve is not expected to significantly affect vehicle speeds. With this in mind, the actual length from the beginning of Tangent 2 (MP 0.156) to the end of Tangent 3 (MP 0.338) is 0.182 mi. For a beginning design speed of 25 mph at the beginning of Tangent 2 and an ending design speed of 55 mph at the end of Tangent 3, the minimum length for a vehicle to accelerate from 25 to 55 mph is 0.162 mi. Thus, as designed, it appears that sufficient length is provided for vehicles to accelerate exiting Curve 2 up to the gore point at the end of Tangent 3, before entering the speed-change lane. However, before such a decision is made, further assessment should be made to evaluate minimum acceleration lengths based on the anticipated operating speeds along the individual components. To do so requires the use of speed prediction models to estimate vehicle speeds along the ramp.

Figure 32 illustrates the inputs that would be specified to estimate speeds along the ramp using the RSPM. For this ramp, since the radius of Curve 3 is greater than 2,000 ft, from the end of Curve 2 (MP 0.156) to the gore point (MP 0.338), the predicted speed along the ramp is calculated assuming a tangent section. Figure 33 shows the predicted speed profile along the ramp, including additional information in tabular form. Plotted speed data are provided as output from the RSMP as well as acceleration data as follows:

- Segment length, mi.
- Initial and final speed on the segment, mph.
- Average acceleration rate on the segment (computed from length and initial and final speeds), mph/s and ft/s^2.

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- Design acceleration rate, computed using Equation 29, ft/s^2.
- Notes comparing the estimated average acceleration rates to design acceleration rates. If the estimated average acceleration rate exceeds the design rate, the message will indicate "accel > design". Otherwise, the message will indicate "OK".

NCHRP 15-56 Ramp Speed Calculations - Entrance Ramp

Location: Case Study No 1 (Loop Ramp)

Calculate | Revise

Component	Variable	Input Value	Unit	Notes
Speeds	$V_{DS\ Hwy}$	70	mph	
	$V_{OS\ Hwy}$	65	mph	
	$V_{SL\ Hwy}$	65	mph	
	V_{XRoad}	15	mph	
General Characteristics	Ramp grade	-1	%	

Variables for Curves and Tangents on Ramp Proper

$V_{DS\ Tan}$ (mph)	$V_{DS\ Curve}$ (mph)	X_{PC} (mi)	R (ft)	L_c (mi)	
20	25	0.044	182	0.076	Curve 1
	25	0.12	224	0.036	Curve 2
55					

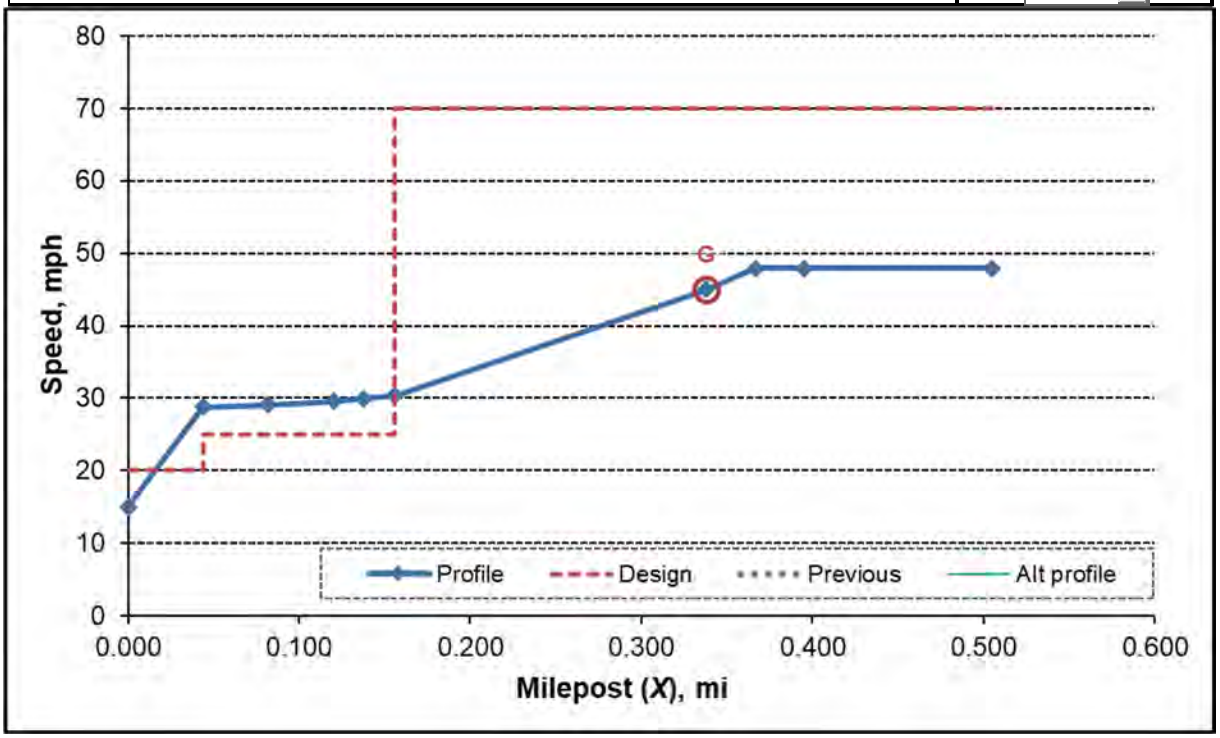
Freeway Mainline	X_G	0.338	mi	
Ramp Terminal	$L_{Gap\ Acpt}$	0.057	mi	
	L_{Taper}	0.110	mi	
	$L_{Acc\ Length}$	0.054	mi	
	$P_{Gap\ Acpt}$	0.50		

Figure 32. Case Study No. 1 (Entrance Ramp: Inputs to RSPM)

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Notes:
The merge speed is more than 5 mph below the freeway operating speed.

Alternate Profile
Off



Plotted Speed Data

Point Type	Milepost	Profile, mph	Milepost	Design, mph
Xrd	0.000	15.00	0.000	20
Tan	0.044	28.75	0.044	20
Cmc	0.082	29.10	0.044	25
Cpt	0.120	29.51	0.120	25
Cmc	0.138	29.91	0.120	25
Cpt	0.156	30.35	0.156	25
Tan	0.338	44.96	0.156	70
SCm	0.366	47.89	0.505	70
SCe	0.395	47.89		
Tpr	0.505	47.89		

Acceleration Data

Segment Data			Speed, mph		Average Acceleration		Design Acceleration	
Number	Type	Length, mi	Initial	Final	mph/s	ft/s^2	ft/s^2	Notes
1	Tangent	0.044	15.000	28.750	1.899	2.785	7.500	OK
2	Curve	0.038	28.750	29.098	0.074	0.108	3.913	OK
3	Curve	0.038	29.098	29.514	0.089	0.131	3.866	OK
4	Curve	0.018	29.514	29.908	0.180	0.265	3.812	OK
5	Curve	0.018	29.908	30.352	0.207	0.303	3.762	OK
6	Tangent	0.182	30.352	44.963	0.840	1.232	3.706	OK
7	Speed-change	0.028	44.963	47.892	1.326	1.944	2.502	OK
8	Speed-change	0.029	47.892	47.892	0.000	0.000	2.349	OK
9	Taper	0.110	47.892	47.892	0.000	0.000	2.349	OK

Figure 33. Case Study No. 1 (Entrance Ramp: Outputs from RSPM)

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A couple of points are worth noting from the speed profile illustrated in Figure 33:

- The predicted operating speed at the end of the first tangent (28.75 mph) is greater than the design speed of Curves 1 and 2 (25 mph). Because this is an entrance ramp and speeds are intended to increase along the ramp, having the predicted operating speeds entering and along Curves 1 and 2 being greater than the design speeds for the individual sections may not be much of a concern. However, if it is a concern, several options are to either decrease the length of tangent entering Curve 1 or install signing (e.g., an advisory speed sign in advance of Curve 1 and/or chevrons along the curve) to warn drivers of the sharpness of the curves.
- The predicted merge speed (47.89 mph) is approximately 17 mph below the operating speed of the freeway (65 mph). The goal is for vehicles to enter the highway at a speed within 5 mph of the speed of traffic on the highway. That is not expected to happen here, at least based on the current assumptions (see note above the speed profile). As input, it is assumed that vehicles merge on the highway at MP 0.366, using only half of the gap acceptance length ($P_{Gap\ Acpt}$ = 0.50). This assumption is based on merging behavior observed in the field (Torbic et al., 2012). According to the design guidelines, vehicles should attain a merge speed within 5 mph of the speed of the highway (or freeway) by the time they reach the end of the gap acceptance length ($L_{Gap\ Acpt}$), not the middle of the gap acceptance length, so the predicted speed at the end of the gap acceptance length should be looked into further.
- The actual acceleration length ($L_{Acc\ Length}$) is 0.111mi (0.057 mi + 0.054 mi). From Table 6, given the design speed of the final tangent is 55 mph and the design speed of the highway is 70 mph, the minimum acceleration length is 0.066 mi (350 ft). Thus, as designed, the actual acceleration length meets the design criteria for minimum acceleration length, but several inconsistencies exist between the predicted speed profile and design criteria as specified in Table 6. The design speed for the final section of the ramp proper is 55 mph, and from Table 6 the assumed operating speed for a design speed of 55 mph is 48 mph. However, the predicted operating speed at the end of the tangent is 45 mph, slightly less than the assumed operating speed corresponding to a design speed of 55 mph. Similarly, in Table 6, for a freeway design speed of 70 mph, the assumed merge speed is 53 mph. In this example, it is given that operating speed and speed limit of the highway are both 65 mph. Thus, the desirable merge speed is 60 mph (i.e., 5 mph below the operating speed of the highway). Assuming an initial entry speed of 45 mph as predicted from the RSPM and a final merge speed of 60 mph, the minimum acceleration length is approximately 0.227 mi (1,200 ft) (i.e., more than double the actual acceleration length).

  Considering these inconsistencies between the predicted speed profile and the assumed initial entry and merge speeds from Table 6, this ramp should potentially be redesigned to allow for a longer acceleration length between 0.189 and 0.227 mi (1,000 and 1,200 ft). Assuming an acceleration length of 0.220 mi (1,160 ft), the RSPM predicts an operating speed of 60 mph at the end of the acceleration length and gap acceptance length. Figure 34 shows the revised inputs to the RSPM. The gap acceptance length was increased to 0.166 mi (which by default increases the acceleration length) and the P_{Gap}

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Acpt was changed to 1.0. Figure 35 shows the revised speed profile with a merge speed of 60 mph at the end of the acceleration length and gap acceptance length (MP 0.614).

NCHRP 15-56 Ramp Speed Calculations - Entrance Ramp

Location: Case Study No 1 (Loop Ramp)

Calculate | Revise

Component	Variable	Input Value	Unit	Notes
Speeds	$V_{DS\ Hwy}$	70	mph	
	$V_{OS\ Hwy}$	65	mph	
	$V_{SL\ Hwy}$	65	mph	
	V_{XRoad}	15	mph	
General Characteristics	Ramp grade	-1	%	

Variables for Curves and Tangents on Ramp Proper

$V_{DS\ Tan}$ (mph)	$V_{DS\ Curve}$ (mph)	X_{PC} (mi)	R (ft)	L_c (mi)	
20	25	0.044	182	0.076	Curve 1
	25	0.12	224	0.036	Curve 2
55					

Freeway Mainline	X_G	0.338	mi	
Ramp Terminal	$L_{Gap\ Acpt}$	0.166	mi	
	L_{Taper}	0.110	mi	
	$L_{Acc\ Length\ RP}$	0.054	mi	
	$P_{Gap\ Acpt}$	1.00		

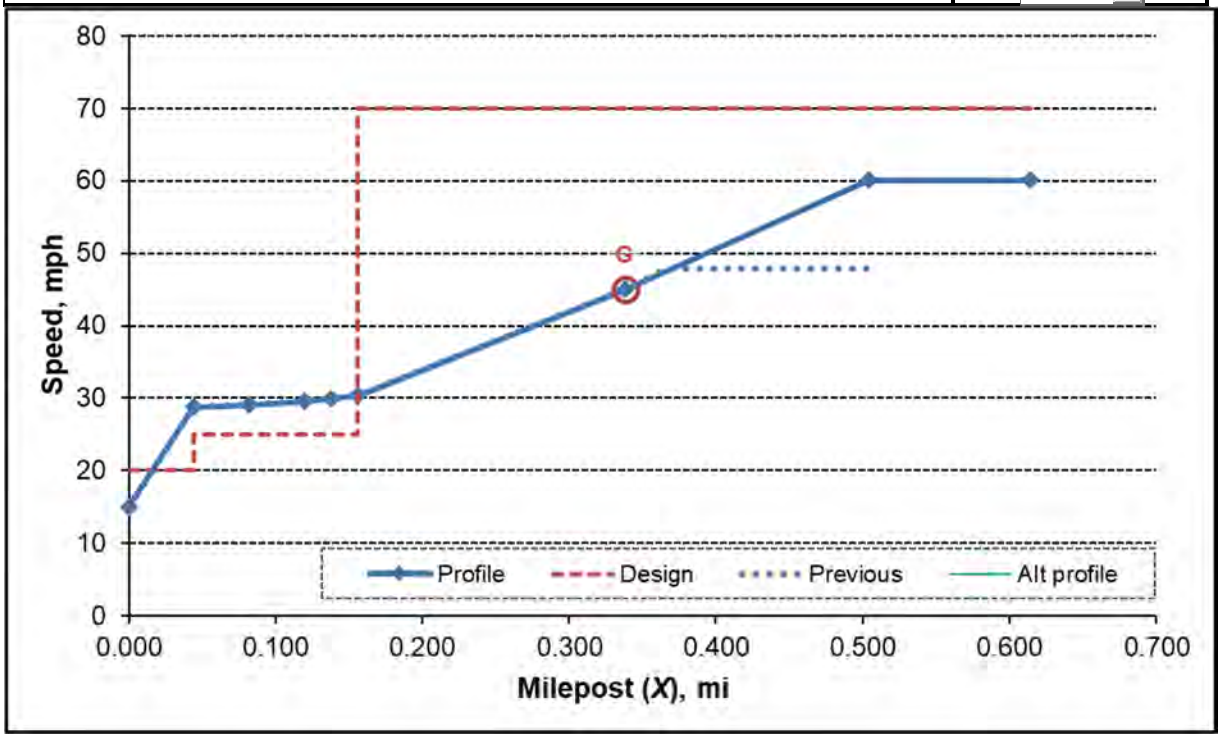
Figure 34. Case Study No. 1 (Entrance Ramp: Revised Inputs to RSPM)

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Notes:

Alternate Profile

Off



Plotted Speed Data

Point Type	Milepost	Profile, mph	Milepost	Design, mph
Xrd	0.000	15.00	0.000	20
Tan	0.044	28.75	0.044	20
Cmc	0.082	29.10	0.044	25
Cpt	0.120	29.51	0.120	25
Cmc	0.138	29.91	0.120	25
Cpt	0.156	30.35	0.156	25
Tan	0.338	44.96	0.156	70
SCe	0.504	60.05	0.614	70
Tpr	0.614	60.05		

Acceleration Data

Segment Data			Speed, mph		Average Acceleration		Design Acceleration	
Number	Type	Length, mi	Initial	Final	mph/s	ft/s^2	ft/s^2	Notes
1	Tangent	0.044	15.000	28.750	1.899	2.785	7.500	OK
2	Curve	0.038	28.750	29.098	0.074	0.108	3.913	OK
3	Curve	0.038	29.098	29.514	0.089	0.131	3.866	OK
4	Curve	0.018	29.514	29.908	0.180	0.265	3.812	OK
5	Curve	0.018	29.908	30.352	0.207	0.303	3.762	OK
6	Tangent	0.182	30.352	44.963	0.840	1.232	3.706	OK
7	Speed-change	0.166	44.963	60.049	1.326	1.944	2.502	OK
8	Taper	0.110	60.049	60.049	0.000	0.000	1.873	OK

Figure 35. Case Study No. 1 (Entrance Ramp: Revised Outputs from RSPM)

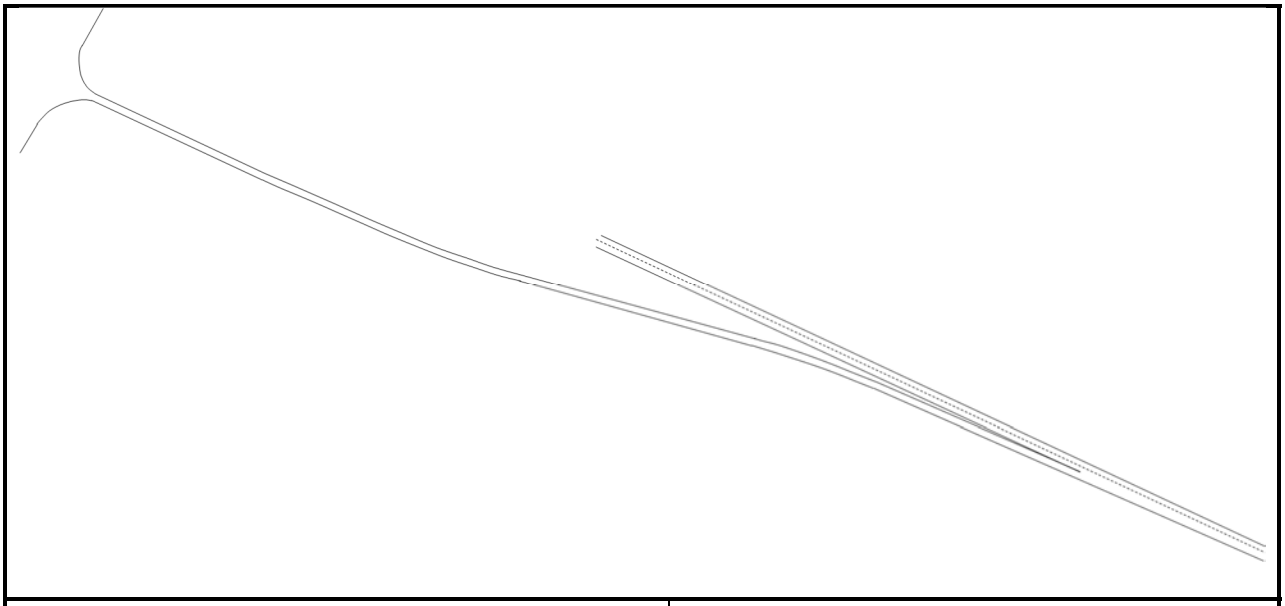
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5.1.2 Case Study No. 2: Diagonal Entrance Ramp in Rural Area

5.1.2.1 Background Information

This case study features a diagonal entrance ramp to a high-speed rural freeway. Figure 36 provides a schematic and characteristics of the ramp. The ramp begins at a minor-road stop-controlled intersection and consists of three curves with radii of 8,056 ft, 2,497 ft, and 2,856 ft. The posted speed limit on the freeway is 85 mph, and the average operating speed on the freeway is 72 mph. Based on the design speeds of the three curves on this ramp and guidance in the 2011 and 2018 editions of the *Green Book*, it is difficult to assess what ramp design speed would have been selected for the design of this diagonal ramp.

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Interchange type:	Service		Conditions:	Unconstrained
Area type:	Rural		Ramp configuration:	Diagonal
Freeway characteristics	Design speed:	85 mph	Operating speed:	72 mph
	Speed limit:	85 mph		
Crossroad characteristics	Type of roadway:	Local/minor rd	Speed limit:	45 mph
	Type of traffic control at terminal:		Stop control	

Ramp Proper				
Ramp grade: 0.17 %			Maximum superelevation (e_{max}): 6 %	
Curve 1	PC milepost:	0.123 mi	Curve length:	0.055 mi
	Curve radius	8056 ft	Superelevation:	3.51 %
	Design speed	80 mph		
Curve 2	PC milepost:	0.179 mi	Curve length:	0.062 mi
	Curve radius	2497 ft	Superelevation:	3.51 %
	Design speed	40 mph		
Curve 3	PC milepost:	0.272 mi	Curve length:	0.078 mi
	Curve radius	2856 ft	Superelevation:	2.50 %
	Design speed	35 mph		

Freeway Mainline Ramp Terminal	
Milepost of gore point	0.398 mi
Acceleration length (upstream of the gore point)	0.047 mi
Gap acceptance length	0.042 mi
Taper length	0.094 mi
Speed-change lane type	Taper

Figure 36. Case Study No. 2 (General Ramp Characteristics and Dimensions)

5.1.2.2 Design Consistency Assessment

The alignment of this ramp is relatively straight or flat, with the sharpest curve having a radius of 2,497 ft, so it is unlikely that the horizontal alignment will affect vehicle speeds. Based on the

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design guidelines presented herein, the footnote at the bottom of Table 4 reads where the horizontal alignment of a diagonal entrance ramp is relatively straight and has little impact on vehicle speeds, the ramp design speed is dependent on the operational characteristics of the freeway mainline ramp terminal. For this situation, it is recommended that the ramp design speed be within 15 to 20 mph of the highway design speed. Thus, a logical ramp design for this diagonal ramp would be either 65 or 70 mph. For purposes of this case study, 65 mph is selected as the ramp design speed. All other sections and elements of the ramp should be designed consistent with this ramp design speed to provide appropriate speed transitions along the ramp.

To evaluate the adequacy and consistency of the existing ramp design, one of the first checks for an entrance ramp is to assess whether the design speeds of contiguous segments along the ramp proper increase sequentially. Without even considering possible design speeds for the tangent sections between curves, it is apparent from Figure 36 that the design speeds of contiguous segments along this ramp do not increase sequentially in a stepwise manner. Curve 1 has the highest design speed (80 mph). Curve 2 has the second highest design speed (40 mph), and Curve 3 has the lowest design speed at 35 mph. Thus, rather than the design speeds increasing sequentially along the ramp proper, the design speeds (at least for the curves) decrease along the ramp proper as currently designed. This check does not necessarily mean that the current ramp design is inadequate and should be redesigned, but further evaluation is necessary. As indicated in the design guidelines presented herein, it is desirable but not essential that the design speeds of the contiguous segments along the ramp proper of an entrance ramp increase in a stepwise manner. It is more important that the contiguous segments along the ramp provide for appropriate speed transitions from segment to segment.

To further evaluate the adequacy and consistency of the existing ramp design, the ramp proper should be divided into individual tangent and curve sections to assess if the contiguous segments along the ramp provide for appropriate speed transitions from segment to segment. In this case, because all of the curve radii along the ramp proper are over 2,000 ft, it is unlikely that the curves will significantly affect vehicle speeds so the entire ramp proper can be evaluated as a single tangential section. Thus, a quick assessment can be made to simply compare the overall length of the ramp proper (i.e., from the crossroad ramp terminal to the gore point) with the length necessary to accelerate from the assumed entry speed of a vehicle at the crossroad ramp terminal and the design speed at the gore point. For this ramp, the length from the crossroad ramp terminal to the gore point is 0.398 mi (2,101 ft). The assumed entry speed of a vehicle entering the ramp proper from the crossroad ramp terminal is 15 mph, and the design speed of the overall tangential alignment is 65 mph (i.e., the assumed ramp design speed as indicated above). From Figure 15, the minimum length required to accelerate from a speed of 15 mph to 65 mph is approximately 0.317 mi. This check suggests that sufficient distance is provided along the ramp proper for a vehicle to accelerate from a speed of 15 mph to 65 mph prior to entering the speed-change lane at the gore point. However, before such a decision is made, further assessment should be made to evaluate the anticipated operating speeds along the individual components of the ramp.

To evaluate the anticipated operating speeds along the ramp, the RSPM can be used. Figure 37 shows the inputs into the RSPM for this ramp, and Figure 38 shows the predicted speed profile along the ramp and other outputs in tabular form.

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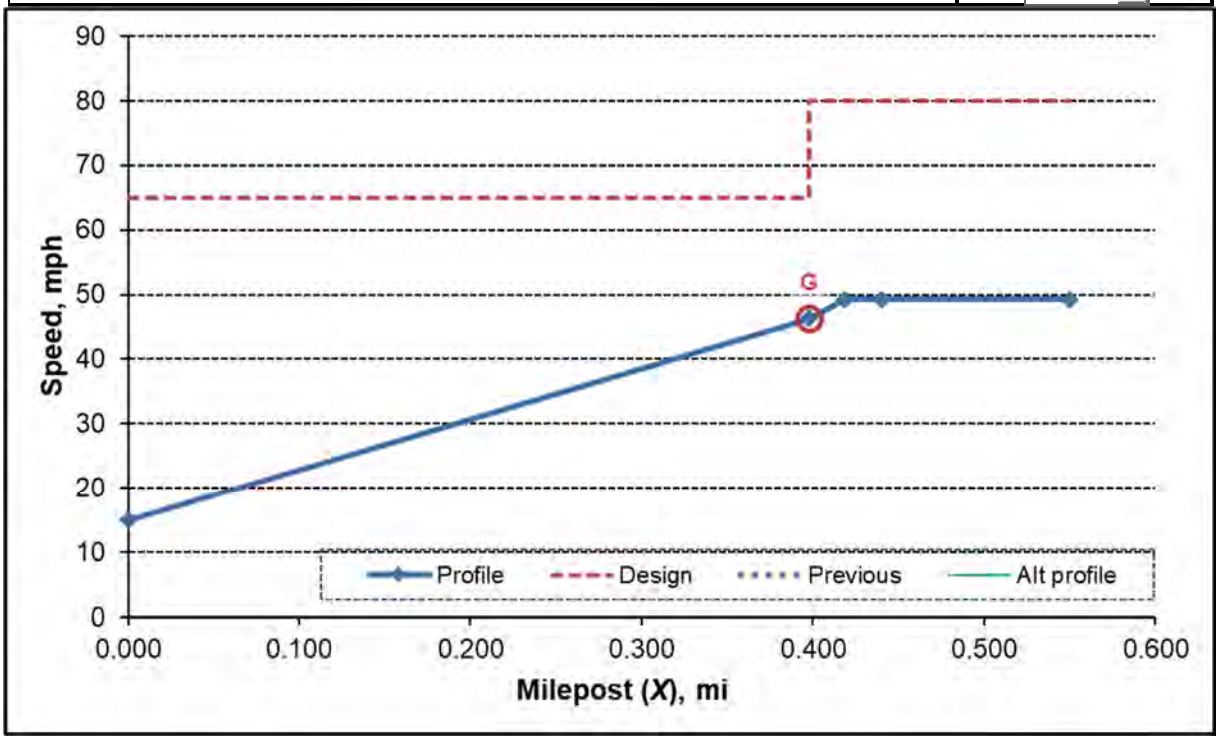
NCHRP 15-56 Ramp Speed Calculations - Entrance Ramp						Calculate	Revise
Location:	Case Study No 2 (Diagonal Ramp)						
Component	Variable	Input Value	Unit			Notes	
Speeds	$V_{DS\ Hwy}$	85	mph				
	$V_{OS\ Hwy}$	72	mph				
	$V_{SL\ Hwy}$	85	mph				
	V_{XRoad}	15	mph				
General Characteristics	Ramp grade	0.17	%				
Variables for Curves and Tangents on Ramp Proper							
$V_{DS\ Tan}$ (mph)	$V_{DS\ Curve}$ (mph)	X_{PC} (mi)	R (ft)	L_c (mi)			
65							
Freeway Mainline	X_G	0.398	mi				
Ramp Terminal	$L_{Gap\ Acpt}$	0.042	mi				
	L_{Taper}	0.110	mi				
	$L_{Acc\ Length\ RP}$	0.047	mi				
	$P_{Gap\ Acpt}$	0.50					

Figure 37. Case Study No. 2 (Entrance Ramp: Inputs to RSPM)

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Notes:
The merge speed is more than 5 mph below the freeway operating speed.

Alternate Profile
Off



Plotted Speed Data

Point Type	Milepost	Profile, mph	Milepost	Design, mph
Xrd	0.000	15.00	0.000	65
Tan	0.398	46.34	0.398	65
SCm	0.419	49.29	0.398	80
SCe	0.440	49.29	0.550	80
Tpr	0.550	49.29		

Acceleration Data

Segment Data			Speed, mph		Average Acceleration		Design Acceleration	
Number	Type	Length, mi	Initial	Final	mph/s	ft/s^2	ft/s^2	Notes
1	Tangent	0.398	15.000	46.344	0.671	0.984	7.500	OK
2	Speed-change	0.021	46.344	49.295	1.866	2.738	2.428	accel > design
3	Speed-change	0.021	49.295	49.295	0.000	0.000	2.282	OK
4	Taper	0.110	49.295	49.295	0.000	0.000	2.282	OK

Figure 38. Case Study No. 2 (Entrance Ramp: Outputs from RSPM)

A couple of points of interest are worth noting here comparing the output from the RSPM and design guidelines presented herein:

- Although the alignment of the ramp proper shows three curves (see Figure 36), because all of the curve radii are greater than 2,000 ft, the inputs into the RSPM essentially treat the ramp proper as a single tangent section. The predicted speed at the end of this ramp

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proper (i.e., the gore point) is approximately 46 mph, which is less than the ramp design speed of 65 mph.

- Although the alignment of the ramp proper is treated as a single tangent section, it is important to check that the anticipated operating speeds along each of the curves are consistent with the design speeds of the curves. From the graph in Figure 38, it can be seen that for Curves 1 and 2, the predicted operating speeds are below the design speeds of the curves; but for Curve 3 from MP 0.272 to MP 0.350, predicted speeds increase from approximately 35 to 42 mph. Considering that it is recommended that the design speed of a curve be based on the speed at the MC rather than the speeds entering or exiting the curve, the fact that vehicles are predicted to exit the curve approximately 7 mph above the design speed may be a concern. If so, several options for addressing this concern would be to potentially increase the curve radius or provide additional superelevation to increase the design speed of Curve 3.
- Based on previous research (Torbic et al., 2012), drivers tend to merge onto the freeway using only about half of the gap acceptance length rather than the full gap acceptance length and acceleration length. Assuming vehicles merge onto the freeway using half of the gap acceptance length as input in Figure 37 (i.e., $P_{Gap\ Acpt}$ = 0.5), vehicles are expected to merge onto the freeway at MP 0.419 at a speed of approximately 49 mph, which is 23 mph below the average operating speed of the freeway. If $P_{Gap\ Acpt}$ = 1.0, vehicles are expected to merge onto the freeway at MP 0.440 at a speed of approximately 52 mph, which is still 20 mph below the average operating speed of the freeway. With either scenario (i.e., $P_{Gap\ Acpt}$ = 0.5 or $P_{Gap\ Acpt}$ = 1.0), vehicles are not predicted to merge onto the freeway at a speed within 5 mph of the freeway operating speed.
- The actual acceleration length ($L_{Acc\ Length}$) is 0.089 mi (0.047 mi + 0.042) mi). From Table 6, given the design speed of the final tangent is 65 mph and the design speed of the highway is 85 mph, the minimum acceleration length is 0.095 mi (500 ft). Thus, as designed, the actual acceleration length meets the design criteria for minimum acceleration length, but several inconsistencies exist between the predicted speed profile and design criteria as specified in Table 6.

  The overall ramp design speed assuming a simple tangential ramp proper is 65 mph, and from Table 6 for a design speed of 65 mph the assumed operating speed entering acceleration length is 56 mph. However, the predicted operating speed at the gore point is 46 mph, which is less than the assumed operating speed corresponding to a design speed of 65 mph. Similarly, from Table 6, for a freeway design speed of 85 mph, the assumed merge speed is 59 mph. However, in this example, it is given that operating speed is 72 mph. Thus, the desirable merge speed is 67 mph (i.e., 5 mph below the operating speed of the highway). Table 6 (even as adapted) does not provide sufficient information to estimate the minimum acceleration length assuming a merge speed of 67 mph.

  In the absence of sufficient information from Table 6, Figure 15 could be used to estimate the minimum acceleration length based on an initial speed of 46 mph and a final speed of 67 mph (Note: 46 mph is the predicted speed at the gore point MP 0.398. To be more accurate, the predicted speed at MP 0.351 where the actual acceleration lane begins should be used for determining the minimum acceleration length). From Figure 15, the minimum acceleration length is approximately 0.227 mi (1,200 ft), compared to the

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actual acceleration length of 0.089 mi. Considering these inconsistencies between the predicted speed profile and the assumed initial and merge speeds from Table 6, this ramp should potentially be redesigned to allow for a longer acceleration length, gap acceptance length, and/or speed-change lane length.

Based on speed predictions from the RSPM, the acceleration length and gap acceptance length should extend to MP 0.573 for vehicles to merge onto the freeway at a speed of 67 mph (i.e., within 5 mph of the freeway operating speed). Figure 39 illustrates the revised inputs into the RSPM, and Figure 40 shows the revised output for the design alternative that predicts a merge speed of 67 mph by extending the acceleration length and gap acceptance length. In Figure 39, the gap acceptance length was increased from 0.042 mi to 0.175 mi, and $P_{Gap\ Acpt}$ was changed from 0.5 to 1.0.

NCHRP 15-56 Ramp Speed Calculations - Entrance Ramp

Location: Case Study No 2 (Diagonal Ramp)

Calculate | Revise

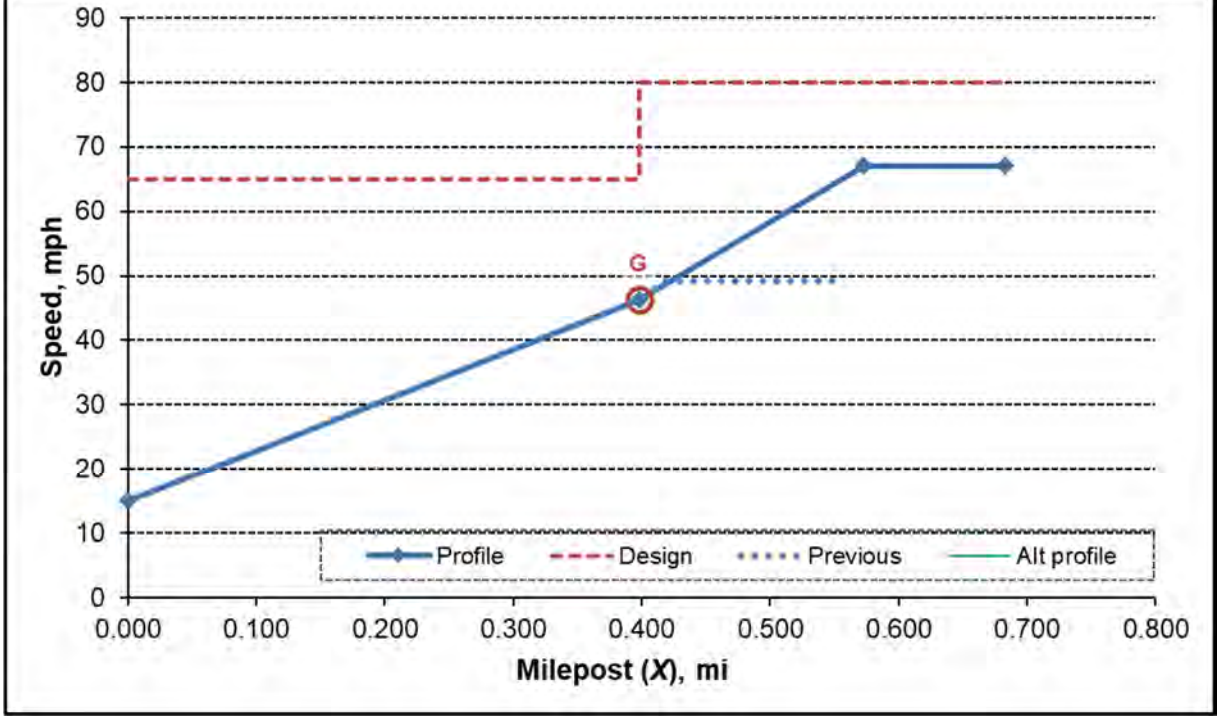
Component	Variable	Input Value	Unit	Notes
Speeds	$V_{DS\ Hwy}$	85	mph	
	$V_{OS\ Hwy}$	72	mph	
	$V_{SL\ Hwy}$	85	mph	
	V_{XRoad}	15	mph	
General Characteristics	Ramp grade	0.17	%	
Variables for Curves and Tangents on Ramp Proper				

$V_{DS\ Tan}$ (mph)	$V_{DS\ Curve}$ (mph)	X_{PC} (mi)	R (ft)	L_c (mi)
65				

Freeway Mainline	X_G	0.398	mi	
Ramp Terminal	$L_{Gap\ Acpt}$	0.175	mi	
	L_{Taper}	0.110	mi	
	$L_{Acc\ Length\ RP}$	0.047	mi	
	$P_{Gap\ Acpt}$	1.00		

Figure 39. Case Study No. 2 (Entrance Ramp: Revised Inputs to RSPM)

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Plotted Speed Data

Point Type	Milepost	Profile, mph	Milepost	Design, mph
Xrd	0.000	15.00	0.000	65
Tan	0.398	46.34	0.398	65
SCe	0.573	67.08	0.398	80
Tpr	0.683	67.08	0.683	80

Acceleration Data

Segment Data			Speed, mph		Average Acceleration		Design Acceleration	
Number	Type	Length, mi	Initial	Final	mph/s	ft/s^2	ft/s^2	Notes
1	Tangent	0.398	15.000	46.344	0.671	0.984	7.500	OK
2	Speed-change	0.175	46.344	67.078	1.866	2.737	2.428	accel > design
3	Taper	0.110	67.078	67.078	0.000	0.000	1.677	OK

Figure 40. Case Study No. 2 (Entrance Ramp: Revised Outputs from RSPM)

5.2 Evaluating the Consistency of a Ramp Design for an Exit Ramp at a Service Interchange Based on the Selected Ramp Design Speed

With exit ramps, several key issues related to the selection of an appropriate ramp design speed should be considered during the design process and while evaluating the consistency of a ramp design:

- A wide range of ramp design speeds are applicable to exit ramps because the controlling feature may be towards the upstream end of the ramp (i.e., near the freeway mainline

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ramp terminal) or the downstream end of the ramp (i.e., associated with the crossroad ramp terminal). For example, where the horizontal alignment of the ramp is curvilinear and the first curve encountered on the exit ramp is the controlling feature, ramp design speeds will likely be higher, but if the horizontal alignment of the ramp is relatively straight and the operational characteristics of the crossroad ramp terminal are the controlling feature, ramp design speeds will be lower. Ramp design speeds for exit ramps may range from 15 mph to 65 mph or higher.

- As vehicles transition from the freeway to the freeway mainline ramp terminal to the ramp proper of an exit ramp, the primary design criterion provided within AASHTO policy addresses minimum deceleration lengths along the freeway mainline ramp terminal (see *Green Book* Table 10-6). Vehicles should be able to maintain their speed on the freeway, diverge from the freeway onto the freeway mainline ramp terminal, decelerate at a comfortable rate within the freeway mainline ramp terminal, and enter the ramp proper at a speed consistent with the design of the ramp proper.
- The deceleration length begins where the width of the auxiliary lane increases to 12 ft or greater (i.e., the end of taper) and extends at least to the gore point. If the first curve that significantly affects vehicle speed begins at the gore point, then the boundary between the freeway mainline ramp terminal and the ramp proper is at the gore point. However, if the beginning of the first curve that significantly affects vehicle speed is downstream of the gore point, then the boundary between the freeway mainline ramp terminal and the ramp proper may be downstream of the gore point. It may be assumed that any curve radius less than or equal to 1,000 ft will significantly affect vehicle speed.
- Vehicle deceleration is a primary design control for exit ramps. On the ramp proper, each tangent and curve section should be designed with sufficient length to accommodate the deceleration expected to occur over its length. *Green Book* Figure 2-34 may be used as guide for determining minimum practical lengths of individual tangent and curve sections along the ramp proper.
- At the downstream end of the ramp, the ramp should be designed so that vehicles enter the functional area of the crossroad ramp terminal at relatively low speeds (e.g., between 15 and 20 mph); and as appropriate, sufficient distance should be provided for a vehicle to come to a complete stop at the end of the queue storage length near the crossroad.

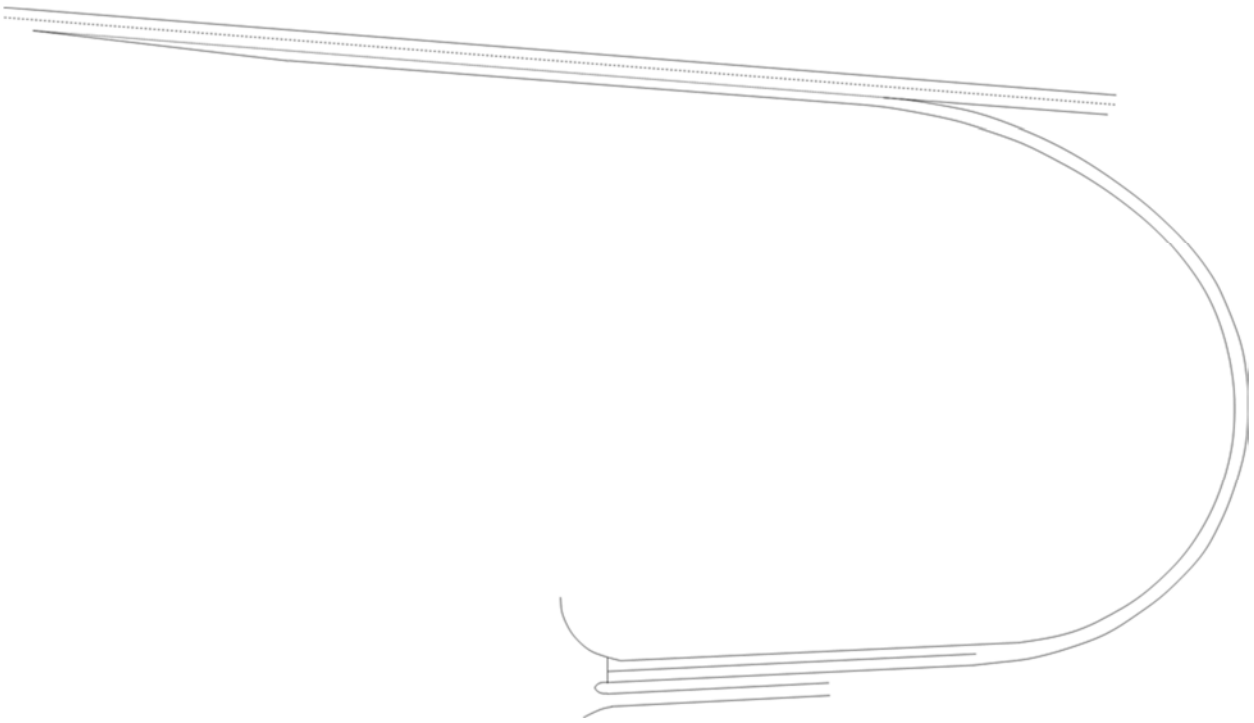
Below are two case studies of exit ramps to demonstrate how to evaluate the consistency of a ramp design based on the selected ramp design speed.

5.2.1 Case Study No. 3: Loop Exit Ramp in Suburban Area

5.2.1.1 Background Information

This case study features a loop ramp in a suburban area. Figure 41 provides a schematic and characteristics of the ramp. The ramp ends at a signalized intersection. The ramp consists of a compound curve, with radii in the following order: 525 ft and 265 ft. Based on the design speeds of Curves 1 and 2 and guidance in the 2011 and 2018 editions of the *Green Book*, the likely ramp design speed used to design this loop ramp was either 25 or 30 mph.

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Interchange type:	Service		Conditions:	Unconstrained
Area type:	Urban/Suburban		Ramp configuration:	Loop
Freeway characteristics	Design speed:	75 mph	Operating speed:	Unknown
	Speed limit:	70 mph		
Crossroad characteristics	Type of roadway:	Local/minor rd	Speed limit:	35 mph
	Type of traffic control at terminal:		Signal	
	Queue storage length:		0.050 mi	
	Milepost of crossroad ramp terminal		0.256 mi	

Ramp Proper				
Ramp grade:	1 % downgrade		Max superelevation (e_{max}):	8 %
Curve 1	PC milepost:	0.000 mi	Curve length:	0.068 mi
	Curve radius	525 ft	Superelevation:	7 %
	Design speed	30 mph		
Curve 2	PC milepost:	0.068 mi	Curve length:	0.116 mi
	Curve radius	265 ft	Superelevation:	6.6 %
	Design speed	25 mph		

Freeway Mainline Ramp Terminal	
Taper length	0.057 mi
Divergence zone length	0.127 mi
Deceleration length (downstream of the gore point)	0.000 mi
Speed-change lane type	Parallel

Figure 41. Case Study No. 3 (Exit Ramp: General Ramp Characteristics and Dimensions)

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5.2.1.2 Design Consistency Assessment

Based on the design guidelines presented herein, given the freeway design speed is 75 mph and the ramp is located in a suburban area, Table 4 indicates the ramp design speed for a loop ramp should be between 25 and 40 mph. Given the first curve encountered on the ramp proper that significantly affects vehicle speeds has a design speed of 30 mph, the ramp design speed would be 30 mph based on the design guidelines presented herein. All other sections and elements of the ramp should be designed consistent with this ramp design speed to provide appropriate speed transitions along the ramp.

To evaluate the adequacy and consistency of the existing ramp design, one of the first checks for an exit ramp is to assess whether the design speeds of contiguous segments along the ramp proper decrease sequentially in a stepwise manner. For this case study, the design speed of the first curve from MP 0.000 to MP 0.068 is 30 mph, and the design speed of the second curve from MP 0.068 to MP 0.184 is 25 mph. A tangent extends from the end of Curve 2 to the functional area of the crossroad ramp terminal. Since the crossroad ramp terminal is signalized and vehicles would be expected to come to a stop at the end of the queue storage length, a logical design speed for this tangent is 20 mph. Based on the design speeds of the two curves and the assumed design speed for the tangent upstream of the crossroad ramp terminal, the existing design of the ramp proper appears adequate as the design speeds of contiguous segments decrease in the direction of travel, and the change in design speed between adjoining sections is at most 5 mph.

The next step to evaluate the adequacy of an exit ramp design is to confirm that sufficient distance is provided for a vehicle to decelerate as intended along the length of the ramp. This involves evaluating the individual components of the freeway mainline ramp terminal, the ramp proper, and to some level the crossroad ramp terminal. First, an assessment should be made to confirm that the deceleration length is sufficiently long to allow for a vehicle to decelerate as intended along the freeway mainline ramp terminal. Using Table 8, for a highway design speed of 75 mph and a design speed of 30 mph for the first controlling feature on the ramp proper, a minimum deceleration length of 0.109 mi (575 ft) is recommended. The actual deceleration length is equivalent to the divergence zone length which is 0.127 mi in length. Thus, the existing design of the freeway mainline ramp terminal appears adequate as the actual deceleration length is greater than the minimum deceleration from Table 8.

A second type of assessment that should be made is to confirm that sufficient distance is provided for a vehicle to decelerate as intended along each individual tangent and curve of the ramp proper. This assessment is best accomplished by assuming or predicting an entering speed and an exiting speed for each individual section of the ramp proper and determining if sufficient distance is provided to decelerate as intended. The RSPM provides the tools to perform such an assessment.

Figure 42 shows the inputs into the RSPM for this ramp, and Figure 43 shows the predicted speed profile along the ramp and other outputs in tabular form.

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NCHRP 15-56 Ramp Speed Calculations - Exit Ramp

Location: Case Study No. 3 (Suburban Exit Loop Ramp)

Calculate | Revise

Component	Variable	Input Value	Unit	Notes
Speeds	$V_{DS\ Hwy}$	75	mph	
	$V_{OS\ Hwy}$		mph	
	$V_{SL\ Hwy}$	70	mph	
	V_{XRoad}	15	mph	
General Characteristics	Ramp grade	-1	%	
	Ramp type	Loop		
Freeway Mainline	L_{Taper}	0.057	mi	
Ramp Terminal	$L_{Div\ Zone}$	0.127	mi	
	$L_{Dec\ Length\ RP}$	0.000	mi	
	SCL type	Parallel		

Variables for Curves and Tangents on Ramp Proper

$V_{DS\ Tan}$ (mph)	$V_{DS\ Curve}$ (mph)	X_{PC} (mi)	R (ft)	L_c (mi)	
	30	0	525	0.068	Curve 1
20	25	0.068	265	0.116	Curve 2

Component	Variable	Input Value	Unit	Notes
Crossroad	X_{XRoad}	0.256	mi	
	$L_{Q\ Storage}$	0.050	mi	
	P_{SCL}	0.10		

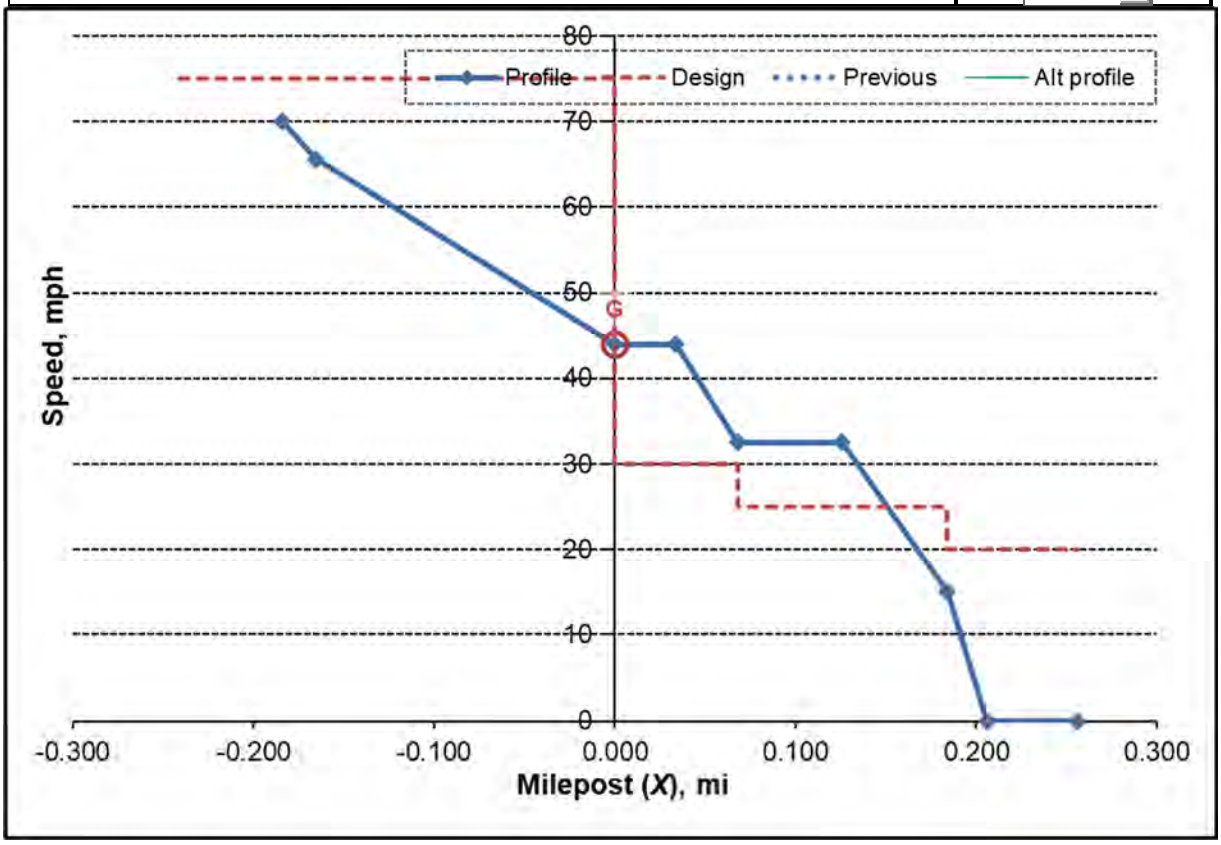
Figure 42. Case Study No. 3 (Exit Ramp: Inputs to RSPM)

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Notes:

Alternate Profile

Off



Plotted Speed Data

Point Type	Milepost	Profile, mph	Milepost	Design, mph
Tpr	-0.184	70.00	-0.241	75
SCd	-0.166	65.60	0.000	75
Gor	0.000	43.98	0.000	30
Cmc	0.034	43.98	0.068	30
Cpt	0.068	32.50	0.068	25
Cmc	0.126	32.50	0.184	25
Cpt	0.184	15.00	0.184	20
Qst	0.206	0.00	0.256	20
Xrd	0.256	0.00		

Acceleration Data

Segment Data			Speed, mph		Average Acceleration		Design Acceleration	
Number	Type	Length, mi	Initial	Final	mph/s	ft/s^2	ft/s^2	Notes
1	Speed-change	0.018	70.000	65.600	-4.504	-6.605	-12.423	OK
2	Speed-change	0.166	65.600	43.982	-1.987	-2.914	-11.642	OK
3	Curve	0.034	43.982	43.982	0.000	0.000	-7.805	OK
4	Curve	0.034	43.982	32.498	-3.588	-5.262	-7.805	OK
5	Curve	0.058	32.498	32.498	0.000	0.000	-5.767	OK
6	Curve	0.058	32.498	15.000	-1.990	-2.919	-5.767	OK
7	Tangent	0.022	15.000	0.000	-1.420	-2.083	-2.662	OK
8	Queue	0.050	0.000	0.000	0.000	0.000	0.000	OK

Figure 43. Case Study No. 3 (Exit Ramp: Outputs from RSPM)

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Figure 43 shows that vehicles are predicted to exit the freeway mainline ramp terminal at the gore point (i.e., MP 0.000) and enter Curve 1 at a speed of 44 mph. Vehicles will continue at this speed until the midpoint of Curve 1 (MP 0.034) and then decelerate to a speed of 32 mph at the end of Curve 1 (MP 0.068). Vehicle speeds are predicted to remain constant over the first half of Curve 2 (MP 0.0126) and then decelerate to a speed of 15 mph at the end of Curve 2 (MP 0.184). As vehicles exit Curve 2 and enter into a tangent portion of the ramp and functional area of the crossroad ramp terminal, vehicle speeds are predicted to decrease from 15 mph to 0 mph at the end of the queue storage length (MP 0.206).

In summary, vehicle speeds are predicted to decrease from 44 mph to 32 mph over the length of Curve 1 (L_C = 0.068 mi), decrease from 32 mph to 15 mph over the length of Curve 2 (L_C = 0.116 mi), and decrease from 15 mph to a stop condition over the length of the first and only tangent (L = 0.022 mi). From *Green Book* Figure 2-34 (see Figure 18 and associated Table 8), a minimum of 0.029 mi (154 ft) is required to decelerate from 44 mph to 32 mph, a minimum of 0.012 mi (61 ft) is required to decelerate from 32 mph to 15 mph, and a minimum of 0.011 mi (56 ft) is required to decelerate from 15 mph to 0 mph. The actual length of Curve 1 (i.e., 0.068 mi) is greater than 0.012 mi (61 ft), the actual length of Curve 2 (0.116 mi) is greater than 0.012 mi (61 ft), and the actual length from the end of Curve 2 to the end of the queue (0.022 mi) is greater than 0.011 mi (56 ft). Thus, the existing design of the individual segments of the ramp proper appears adequate for a vehicle to decelerate as intended along each section. The output from the RSPM also confirms similar findings as the final column under the acceleration data table shows that the average deceleration rates along the individual sections or subsections do no exceed the design decelerations.

Although the alignment of the ramp proper appears adequate in terms of providing sufficient distance for a vehicle to decelerate as intended over the length of individual sections, the predicted speeds over the entire length of Curve 1 are above the design speed of Curve 1, and the predicted speeds over a majority of the length of Curve 2 are above the design speed of Curve 2. For both Curves 1 and 2, the design superelevations are less than the e_{max} (i.e., 8 percent) used by the agency. Without changing the curve radius of either curve, it would be possible to increase the design superelevation of each curve such that the design speed of Curve 1 would be 40 mph and the design speed of Curve 2 would be 30 mph. By doing so, the design speeds for each curve would be more aligned with the predicted operating speeds (see Figure 44).

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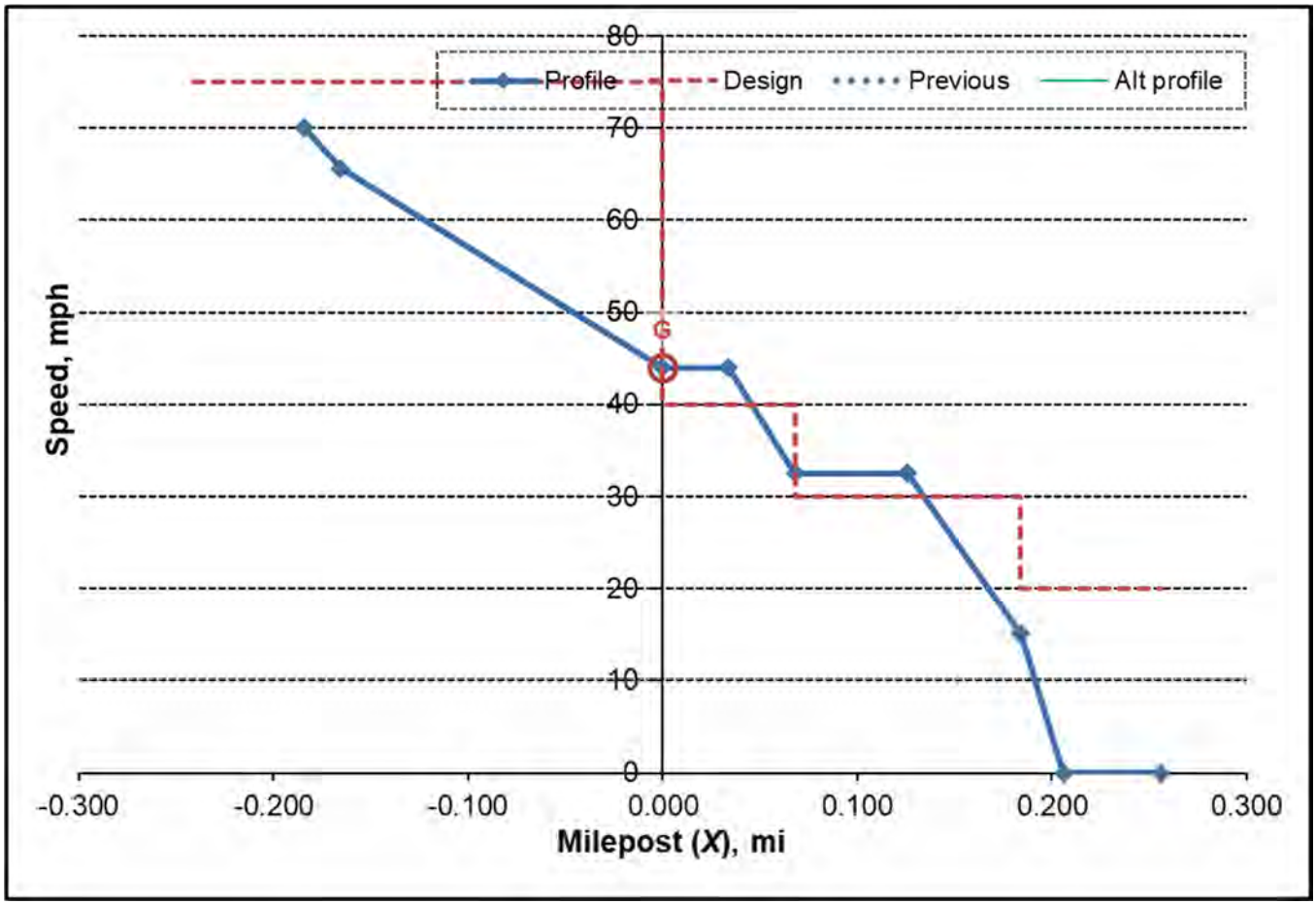


Figure 44. Case Study No. 3 (Exit Ramp: Revised Outputs from RSPM)

A couple of points of interest are worth noting here comparing the output from the RSPM, design guidelines presented herein, and current AASHTO policy:

- AASHTO policy assumes that vehicles do not decelerate in the freeway lanes before diverging from the freeway onto the freeway mainline ramp terminal. However, field observations suggest otherwise (i.e., Torbic et al., 2012). Therefore, rather than the diverge speed being equal to the operating speed of the freeway, the predicted speed at the diverge location is less than the operating speed of the freeway. In this example, the operating speed of the freeway is unknown (see Figure 42), so within the RSPM the operating speed of the freeway is assumed equal to the posted speed limit of the freeway (i.e., 70 mph). In this case study, based on characteristics of the freeway mainline ramp terminal, the diverge speed is predicted to be 65.60 mph.
- Based on AASHTO design policy, vehicles diverge from the freeway at the end of the taper length and beginning of the deceleration length and divergence zone length. Within the RSPM, the diverge location is based on a percentage or portion of the speed-change lane length (i.e., P_{SCL}) as input by the analyst. The default value within the RSPM is $P_{SCL} = 0.1$. In this case study, the end of the taper length and beginning of the deceleration length and divergence zone length corresponds to a value of $P_{SCL} = 0.31$.

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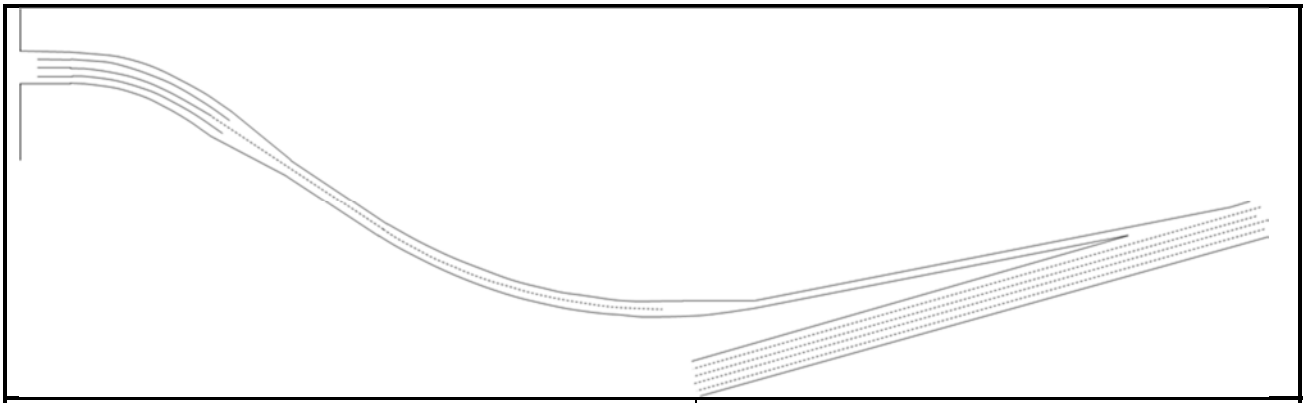
- Based on a highway design speed of 75 mph and a design speed of 30 mph for the controlling feature on the ramp proper, AASHTO policy assumes an average running speed of 26 mph at the end of the deceleration length entering the ramp proper. Based on the given inputs describing this ramp, the RSPM predicts a speed of 44 mph at the gore point which corresponds to the end of the deceleration length. Thus, there is an inconsistency in the predicted speed at the gore point output from the RSPM and the assumed value of running speed at the end of deceleration length in the design table from AASHTO policy (i.e., *Green Book* Table 10-6).

5.2.2 Case Study No. 4: Diagonal Exit Ramp in Suburban Area

5.2.2.1 Background Information

This case study features a diagonal ramp in a suburban area. Figure 45 provides a schematic and characteristics of the ramp. The ramp consists of two horizontal curves in opposite directions with a tangent section in between. The ramp ends at a signalized intersection, and the queue storage is within the second horizontal curve. Based on the design speed of Curve 1 and guidance in the 2011 and 2018 editions of the *Green Book*, the likely ramp design speed used for the design of this diagonal ramp is 45 mph.

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Interchange type:	Service		Conditions:	Constrained	
Area type:	Urban/Suburban		Ramp configuration:	Diagonal	
Freeway characteristics	Design speed:	70 mph	Operating speed:	62 mph	
	Speed limit:	65 mph			
Crossroad characteristics	Type of roadway:	Primary/major rd	Speed limit:	35 mph	
	Type of traffic control at terminal:		Signal		
	Queue storage length:		0.060 mi		
	Milepost of crossroad ramp terminal		0.368 mi		
Ramp Proper					
Ramp grade:	1 % downgrade		Max superelevation (e_{max}):	12 %	
Curve 1	PC milepost:	0.114 mi	Curve length:	0.124 mi	
	Curve radius	850 ft	Superelevation:	10 %	
	Design speed	45 mph			
Curve 2	PC milepost:	0.302 mi	Curve length:	0.035 mi	
	Curve radius	335 ft	Superelevation:	10 %	
	Design speed	30 mph			
Freeway Mainline Ramp Terminal					
Taper length			0.028 mi		
Divergence zone length			0.006 mi		
Deceleration length (downstream of the gore point)			0.114 mi		
Speed-change lane type			Taper		

Figure 45. Case Study No. 3 (Exit Ramp: General Ramp Characteristics and Dimensions)

5.2.2.2 Design Consistency Assessment

Based on the design guidelines presented herein, given the freeway design speed is 70 mph, the crossroad is a primary/major arterial, and the ramp is located in a suburban area, Table 4 indicates the ramp design speed for a diagonal ramp should be between 35 and 50 mph. The design speed of the first curve encountered along the ramp proper that significantly affects vehicle speed is 45 mph. Thus, the ramp design speed for this ramp is 45 mph, so as designed, the ramp design speed of this ramp is consistent with the design guidelines presented herein. All other sections and elements of the ramp should be designed consistent with this ramp design speed to provide appropriate speed transitions along the ramp.

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To evaluate the adequacy and consistency of the existing ramp design, one of the first checks for an exit ramp is to assess whether the design speeds of contiguous segments along the ramp proper decrease sequentially in a stepwise manner. For this case study, the design speed of Curve 1 from MP 0.114 to MP 0.238 is 45 mph. A tangent begins at the end of Curve 1 (i.e., MP 0.238) and extends to the PC of Curve 2 at MP 0.302. Since the design speed of Curve 2 is 30 mph, it is logical for the design speed of the tangent between Curves 1 and 2 to be 40 mph. Thus, the existing design of the ramp proper appears adequate as the design speeds of contiguous segments decrease in the direction of travel, and the change in design speed between adjoining sections is at most 10 mph.

The next step to evaluate the adequacy of an exit ramp design is to confirm that sufficient distance is provided for a vehicle to decelerate as intended along the length of the ramp. This involves evaluating the individual components of the freeway mainline ramp terminal, the ramp proper, and to some level the crossroad ramp terminal. First, an assessment should be made to confirm that the deceleration length is sufficiently long to allow for a vehicle to decelerate as intended along the freeway mainline ramp terminal. For this ramp the actual deceleration length is $L_{Dec\ Length}$ = 0.120 mi (i.e., 0.006 mi + 0.114 mi) which extends beyond the gore point and ends at the beginning of Curve 1. Using Table 8, for a highway design speed of 70 mph and a design speed of 45 mph for the first controlling feature on the ramp proper, a minimum deceleration length of 0.074 mi (390 ft) is recommended. Thus, the existing design of the freeway mainline ramp terminal appears adequate as the actual deceleration length is greater than the minimum deceleration from Table 8.

A second type of assessment that should be made is to confirm that sufficient distance is provided for a vehicle to decelerate as intended along each individual tangent and curve of the ramp proper. Of particular interest is the distance between the end of Curve 1 (MP 0.238) and the end of the queue storage length (MP 0.308), which measures 0.070 mi (370 ft). Assuming vehicles exit Curve 1 at its design speed of 45 mph and need to come to a complete stop at the end of the queue storage length, Table 9 indicates that 0.060 mi (314 ft) is required to come to a complete stop from an initial speed of 45 mph. Thus, the existing design of the ramp proper appears adequate as the actual length from the end of Curve 1 to the end of the queue storage length is greater than the minimum length specified from Table 9 for this design condition. However, note that the functional area of the crossroad ramp terminal extends into Curve 2, so the operational characteristics near Curve 2 and the crossroad ramp terminal are of particular interest. An assessment of the operational conditions along the entire ramp and near the crossroad ramp terminal can be performed using the RSPM.

Figure 46 shows the inputs into the RSPM for this ramp, and Figure 47 shows a graphical representation of the predicted speed profile along the ramp and other outputs in tabular form. The predicted speed entering Curve 2 is 25 mph. However, the functional area of the crossroad ramp terminal extends into Curve 2 and to decelerate from 25 mph at MP 0.302 to a stop condition at MP 0.308 (i.e., the end of the queue storage length) requires a deceleration rate that exceeds the maximum design deceleration rate [in Figure 45, under Acceleration Data see the Note (i.e., last column) for Segment 7]. This raises some concerns about the consistency of the current design near the crossroad ramp terminal, but as indicated the existing design of the ramp proper provides sufficient length for a vehicle exiting Curve 1 at its design speed (i.e., 45 mph) to come to a complete stop at the end of the queue storage length. Several options could be

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considered to redesign the ramp by either changing the horizontal alignment or lengthening the ramp to reduce speeds in proximity to the crossroad ramp terminal. Another option to consider is providing signing (e.g., Signal Ahead) as vehicles exit Curve 1 to reduce speeds in advance of the functional area of the crossroad ramp terminal.

NCHRP 15-56 Ramp Speed Calculations - Exit Ramp

Location: Case Study No. 4 (Suburban Exit Diagonal Ramp)

Calculate | Revise

Component	Variable	Input Value	Unit	Notes
Speeds	$V_{DS\ Hwy}$	70	mph	
	$V_{OS\ Hwy}$	62	mph	
	$V_{SL\ Hwy}$	65	mph	
	V_{XRoad}	15	mph	
General Characteristics	Ramp grade	-1	%	
	Ramp type	Diagonal		
Freeway Mainline	L_{Taper}	0.028	mi	
Ramp Terminal	$L_{Div\ Zone}$	0.006	mi	
	$L_{Dec\ Length\ RP}$	0.114	mi	
	SCL type	Taper		

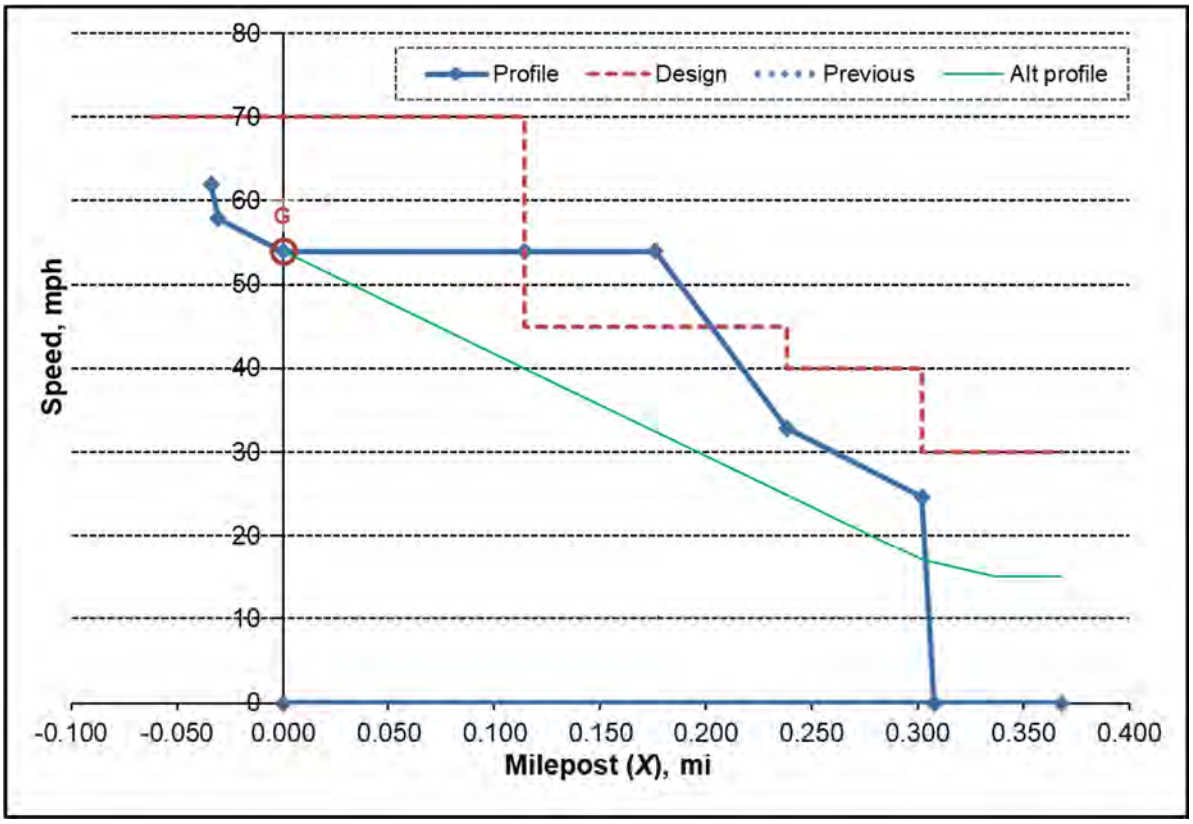
Variables for Curves and Tangents on Ramp Proper

$V_{DS\ Tan}$ (mph)	$V_{DS\ Curve}$ (mph)	X_{PC} (mi)	R (ft)	L_c (mi)	
40	45	0.114	850	0.124	Curve 1
	30	0.302	335	0.035	Curve 2

Crossroad	X_{XRoad}	0.368	mi	
	$L_{Q\ Storage}$	0.060	mi	
	P_{SCL}	0.10		

Figure 46. Case Study No. 4 (Exit Ramp: Inputs to RSPM)

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Plotted Speed Data

Point Type	Milepost	Profile, mph	Milepost	Design, mph	Milepost	Alt profile, mph
Tpr	-0.034	62.00	-0.062	70	0.000	53.99
SCd	-0.031	57.90	0.114	70	0.114	40.03
Gor	0.000	53.99	0.114	45	0.238	24.86
Tan	0.114	53.99	0.238	45	0.302	17.02
Cmc	0.176	53.99	0.238	40	0.337	15.00
Cpt	0.238	32.84	0.302	40	0.368	15.00
Tan	0.302	24.67	0.302	30		
Qst	0.308	0.00	0.368	30		
Xrd	0.368	0.00				
	0.000	0.00				

Acceleration Data

Segment Data			Speed, mph		Average Acceleration		Design Acceleration	
Number	Type	Length, mi	Initial	Final	mph/s	ft/s^2	ft/s^2	Notes
1	Speed-change	0.003	62.000	57.900	-20.081	-29.453	-11.003	**decel > design**
2	Speed-change	0.031	57.900	53.988	-1.987	-2.914	-10.275	**OK**
3	Tangent	0.114	53.988	53.988	0.000	0.000	-9.581	**OK**
4	Curve	0.062	53.988	53.988	0.000	0.000	-9.581	**OK**
5	Curve	0.062	53.988	32.839	-4.113	-6.033	-9.581	**OK**
6	Tangent	0.064	32.839	24.668	-1.020	-1.496	-5.828	**OK**
7	Tangent	0.006	24.668	0.000	-14.086	-20.659	-4.378	**decel > design**
8	Queue	0.060	0.000	0.000	0.000	0.000	0.000	**OK**
9		-0.368	0.000	0.000	0.000	0.000	0.000	**OK**

Figure 47. Case Study No. 4 (Exit Ramp: Outputs from RSPM)

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Section 6.
Summary of Design Guide and Future Research Needs

This section summarizes guidelines for selecting appropriate ramp design speeds. The guidelines facilitate the selection of an appropriate ramp design speed based on a combination of contextual considerations and quantitative information. The guidelines are expected to result in ramp designs consistent with driver expectations and behaviors over a range of traffic conditions and the functional classification of the two intersecting roads.

Ramp design speed is a selected speed used to determine the various geometric design features of a ramp. The selected ramp design speed should be a logical one with respect to the type of intersecting highways, area type (i.e., urban or rural), ramp configuration, and site constraints (including physical, environmental, and social). Table 4 provides guide values for ramp design speeds for system and service interchanges. The guide values are presented as a range of values given the context, ramp configuration, and type of interchange. The lower guide values are generally associated with constrained conditions, and the higher guide values are associated with unconstrained conditions. The guide values in Table 4 were developed to be consistent with guidance presented in the 2018 edition of the *Green Book*.

Where the horizontal alignment of the ramp is curvilinear, the ramp design speed applies to the controlling curve on the ramp proper. For an entrance ramp, the last curve encountered along the ramp proper that significantly affects vehicle speed is considered the controlling curve. For an exit ramp, the controlling curve is the first curve encountered along the ramp proper that significantly affects vehicle speed. For direct connection and outer connection ramps, the sharpest curve on the ramp proper is considered the controlling curve. The freeway mainline ramp terminal, the crossroad ramp terminal, and the adjoining sections of the ramp proper (i.e., adjoining tangents and horizontal curves in the direction of travel) that connect to the controlling curve should be designed to allow for appropriate speed transitions, consistent with driver behavior and expectations and vehicle performance capabilities, for the selected ramp design speed.

Where the horizontal alignment of the ramp is relatively straight and has little or no influence on vehicle speeds, the ramp design speed is based on the operational characteristics of the freeway mainline ramp terminal for an entrance ramp and the operational characteristics of the crossroad ramp terminal for an exit ramp and applies to the tangential section of the ramp proper.

The ramp design speed is typically associated with an individual segment of a ramp near the upstream or downstream end of the ramp. For an entrance ramp, the ramp design speed is normally associated with a segment on the downstream portion of the ramp. With an exit ramp, the ramp design speed is usually associated with a segment on the upstream portion of the ramp.

Each individual section of the ramp proper (i.e., tangent and curve) has its own design speed. The design speed of the controlling feature of the ramp (i.e., the ramp design speed) is selected first, and then the design speeds of the other elements or segments of the ramp proper should be determined accordingly. It is desirable (but not essential) that the design speeds of the contiguous segments along the ramp proper increase or decrease in a stepwise manner. The change in design speed between adjoining sections should be limited to no more than 10 to 15 mph. For entrance ramps, this means that the design speeds of the contiguous segments along the ramp proper

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should typically increase in a stepwise manner; and for exit ramps the design speeds of the contiguous segments along the ramp proper should decrease in a stepwise manner. Decreasing design speeds sequentially along a ramp in a stepwise manner is more critical for exit ramps than increasing design speeds along a ramp for entrance ramps, as unexpected and abrupt changes in design speeds along an exit ramp are more likely to be associated with potential lane keeping and/or loss of control issues.

The design speed of each individual curve and tangent along the ramp proper should be consistent with the speeds that drivers are likely to be traveling along the respective curve and tangent. If the design speed of an individual curve or tangent along the ramp proper is not consistent with the speeds that drivers are likely to be traveling along the respective section, consider potential options so that the design speed would be more consistent with operating speeds. For example, if the anticipated operating speed exceeds the design speed, consider changing the design speed for the respective section. If changing the design speed of an individual section means that the design speeds of the contiguous segments along the ramp proper no longer increase or decrease in a stepwise manner, consider changing the ramp design speed. If the design speed or ramp design speed cannot be changed, consider implementing speed-control measures so that the design speed or ramp design speed would be more consistent with operating speeds.

The freeway mainline ramp terminal portion of a ramp does not have an associated design speed. However, the operational and design conditions of the freeway mainline ramp terminal are related to and coordinated with the operational and design conditions of the adjoining sections of the ramp proper and freeway.

Each component of the ramp should be designed such that a vehicle has sufficient distance to accelerate or decelerate within the performance capabilities of the vehicle and driver expectations given the design speeds and features of the individual components of the ramp. Several design controls and criteria may be used in the design of ramps so that the individual components and features of the ramps are designed in a consistent manner, accounting for sequential speed transitions from one component or section to the next. *Green Book* Figures 2-33 and 2-34 may be used as a guide for determining minimum practical lengths of individual tangent and curve sections along the ramp proper, and *Green Book* Tables 10-4, 10-5, and 10-6 may be used to determine minimum acceleration and deceleration lengths for the freeway mainline ramp terminal portion of the ramp.

Green Book Tables 10-4 and 10-6 are presented herein as Tables 6 and 7, respectively, with a few adaptions. The headings were adapted to be more consistent with the design guidelines presented herein. In addition, the guide values were expanded to include additional design conditions for higher ramp design speeds and higher highway (or freeway) design speeds.

A spreadsheet-based tool was developed to estimate vehicle speeds along interchange ramps to help designers and traffic engineers conduct operational analyses and make design decisions. The spreadsheet tool is applicable to entrance and exit ramps at service interchanges including diagonal ramps, loop ramps, and outer connection ramps. The spreadsheet-based tool incorporates several speed prediction models based on vehicle speeds observed on ramps. The models consist of equations that contain calibration coefficients and terms to account for the influence of variables such as initial speed, ramp segment length (curve or tangent segment), curve radius, and others. With the capability to adjust several input parameters, the spreadsheet-

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based tool may be used to predict vehicle speeds based upon anticipated driver behaviors or design conditions. The output from the tool may be interpreted to help design an interchange ramp in a consistent manner that meets driver expectations.

Because the range of guide values for ramp design speeds in Table 4 are relatively consistent with guidance presented in the *Green Book*, the enhanced guidelines are not expected to introduce unforeseen safety or operational issues, create the need for more design exceptions, nor increase or decrease construction costs. The enhanced guidelines should result in the design of ramps that would be similar to ramps designed using guidance from the 2018 edition of the *Green Book* as intended. The enhanced guidelines simply provide more clarity on the primary elements that factor into selecting an appropriate ramp design speed and how the ramp design speed is used to coordinate the design of adjacent sections of the ramp to accommodate sequential speed transition from one component or section on the ramp to next.

The enhanced guidelines may help to improve safety and operational issues related to ramp design by reducing the number of ramps designed and built in an inconsistent manner due to confusion or inconsistent interpretation of existing AASHTO policy. Ramps designed in an inconsistent manner may not sufficiently accommodate sequential speed transitions from one component to the next. As such, driver expectations may be violated, resulting in abrupt changes in speed and/or creating safety and operational concerns. The enhanced guidelines may also benefit an agency's overall program budget by reducing the number of ramps that need to be reconstructed at a later date because they were originally designed and built in an inconsistent manner.

The enhanced guidelines are not expected to create any specific tort liability concerns related to ramp design. The enhanced guidelines developed through this research are consistent with guidance presented in the *Green Book* and simply provide more clarity on selecting an appropriate ramp design speed and how the ramp design speed is used to coordinate the design of adjacent sections of the ramp. There is no indication that guidance presented in the *Green Book* has created any tort liability concerns related to the selection and application of ramp design speed, so there is no apparent need for the enhanced guidelines to alleviate tort liability concerns based on use of current design guidance. (Note, adherence to standards or design guidance does not in itself provide tort immunity.) Terms such as "must", "shall", or "required" that could create obvious tort liability concerns are not used in the enhanced guidelines; and in general, if engineering decisions are made sensibly and are well documented, no tort liability concerns should arise. Thus, the enhanced guidelines are not expected to create any specific tort liability concerns related to ramp design.

Based on the findings from this research, several research needs were identified:

- *Green Book* Table 10-4, which provides guide values for minimum acceleration lengths, was adapted (see Table 6). In addition to modifying the caption headings, the design values were expanded to include additional design conditions for higher ramp design speeds and higher highway (or freeway) design speeds. These new design values were estimated through numeric extrapolation of the minimum acceleration lengths for the lower speed design conditions and to be consistent with the trends and acceleration rates of the design values in the original table. They were not developed based on field data or vehicle dynamics models that estimate vehicle performance. Similarly, *Green Book* Table

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10-6 which provides guide values for minimum deceleration lengths was adapted (see Table 7) for use with higher design speed facilities. Additional research should be conducted to verify and/or modify these design values for higher speed facilities.

- A first version of a spreadsheet tool was developed as part of this research to estimate vehicle speeds along interchange ramps to help designers and traffic engineers conduct operational analyses and make design decisions. Section 4.4 documents several known limitations of this tool. Additional research should be conducted to address these limitations and expand the capabilities of this tool.

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Section 7. References

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Section 8.
Abbreviations, Acronyms, Initialisms, and Symbols

Term	Description	Equation No.
General Terms		
AASHTO	American Association of State Highway and Transportation Officials	
NCHRP	National Cooperative Highway Research Program	
ITE	Institute of Transportation Engineers	
HSM	*Highway Safety Manual*	
HCM	*Highway Capacity Manual*	
VPI	Vertical point of intersection	
SSD	Stopping sight distance	
DSD	Decision sight distance	
RSPM	Ramp speed profile model	
FMRT	Freeway mainline ramp terminal	
CRT	Crossroad ramp terminal	
RP	Ramp proper	
C-D	Collector-distributor road	
PC	Point of curvature (i.e., point at which the curve begins)	
MC	Midpoint of a curve	
PT	Point of tangent (i.e., point at which the curve ends)	
G_r	Overall ramp grade	
Terms Associated with Speeds / Accelerations		
$V_{DS\ Hwy}$	Highway (or freeway) design speed	
$V_{OS\ Hwy}$	Average operating speed on the highway (or freeway) mainline	5,6,7,8,9,10,11, 12,14,15,16,17,24
$V_{SL\ Hwy}$	Regulator speed limit on the highway (or freeway) mainline	7,9,20
$V_{Acc\ Length(i)}$	Speed at the beginning of the acceleration length	2
$V_{Acc\ Length(f)}$	Speed at the end of the acceleration length	2
V_{Merge}	Predicted vehicle speed where the vehicle merges from the freeway speed-change lane onto the highway (or freeway)	12
$V_{Dec\ Length(i)}$	Speed at the beginning of the deceleration length	3,4
$V_{Diverge}$	Predicted vehicle speed at the diverge location	18,19
$V_{Dec\ Length(f)}$	Speed at the end of the deceleration length	3
$V_{i,j,NC}$	Predicted vehicle speed for curve *i* at point *j* (*j* = PC, MC, or PT), by applying the model for tangents with no downstream curve	5
$V_{i,j,C}$	Predicted vehicle speed for curve *i* at point *j* (*j* = PC, MC, or PT), by applying the model for tangents with a downstream curve	6
V_{XRoad}	Vehicle speed at the crossroad ramp terminal	5,6,11,14,24,25,26,27
$V_{DS\ Curve}$	Design speed of a curve	1
$V_{DS\ Tan}$	Design speed of tangent	
V_a	Speed after t_n sec of deceleration without brakes	3,4
V_{MC}	Predicted vehicle speed at the MC	7,8,20
V_{PC}	Predicted vehicle speed at the beginning of a curve (i.e., initial point of horizontal curve – PC)	7,8,20,21
V_{PT}	Predicted vehicle speed at the end of a curve (PT)	8,21
$V_{Tan\text{-}End,C}$	Predicted vehicle speed at the end of a tangent if the tangent is followed by a curve	9,22

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Term	Description	Equation No.
$V_{Tan\text{-}Begin}$	Predicted vehicle speed at the beginning of a tangent.	9,10,22,23
$V_{Tan\text{-}End,NC}$	Predicted vehicle speed at the end of a tangent if the tangent is not followed by a curve.	10,23
V_G	Predicted vehicle speed at the gore point.	11,12,19
$V_{PT,f}$	Predicted vehicle speed at the PT of final curve on the ramp proper.	11
a	Acceleration rate.	2
t_n	Deceleration time without brakes (assumed to be 3 sec).	3,4
d_n	Deceleration rate without brakes.	3,4
d_{wb}	Deceleration rate with brakes.	3
$a_{Gap\ Acpt}$	Acceleration rate along gap acceptance length.	12
$P_{Gap\ Acpt}$	Proportion of gap acceptance length used by merging vehicles [default = 0.50, based on Torbic et al., (2012)].	12
$vmax,i$	Limiting speed for curve *i*.	13,15,17,25,27
$v_{ent,1}$	Average entry speed for curve 1.	14,15,24,25
$vext,1$	Average exit speed for curve 1.	15,25
$vent,i$	Average entry speed for curve *i* (*i* = 2, 3, ...).	16,17,26,27
$vext,i$	Average exit speed for curve *i* (*i* = 2, 3, ...).	16,17,26,27
ΔV	Mean speed differential between freeway mainline speed and the initial speed of vehicles entering the speed-change lane.	18
a_{SCL}	Deceleration rate along the speed-change lane.	19
a_{avg}	Average acceleration rate.	28
V_i	Vehicle speed at the beginning of the section.	28,29,30
V_f	Vehicle speed at the end of the section.	28
a_{Design}	Maximum design acceleration rate.	29,30
Terms Associated with Lengths, Mileposts, and Locations		
$L_{Acc\ Length}$	Acceleration length.	2
$L_{Acc\ Length\ RP}$	This length is that portion of the acceleration length measured upstream from the gore.	
$L_{Dec\ Length}$	Deceleration length.	3
$L_{Dec\ Length\ RP}$	That portion of the deceleration length measured downstream from the gore point.	
$L_{Gap\ Acpt}$	Gap acceptance length.	12
$L_{Div\ Zone}$	Divergence zone length.	
L_{SCL}	Length of speed-change lane.	19
P_{SCL}	Proportion of speed-change lane located upstream of the diverge point (default value is 0.10 based on Torbic et al. (2012)).	19
L_{Taper}	Taper length.	
X_j	Milepost for point *j*.	5,6
L_T	Tangent length.	9,10,22
L_C	Curve length.	21
$L_{C,i}$	Length of horizontal curve *i*.	15,16,17,25,26,27
L	Segment length.	28
$X_{PT,f}$	Milepost for the PT of final curve on the ramp proper.	11
X_G	Milepost for the gore point.	11
X_i	Milepost of the point of change from tangent to curve (PC) for curve *i*.	14,16,24,26
X_{PC}	Milepost of the beginning of the curve.	20
X_{MC}	Milepost of the MC.	20
X_{XRoad}	Milepost at the crossroad ramp terminal.	

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Term	Description	Equation No.
A	Location of the beginning of the acceleration length (i.e., the boundary between the ramp proper and freeway mainline ramp terminal of an entrance ramp).	
D	Location of the end of the deceleration length (i.e., the boundary between the freeway mainline ramp terminal and ramp proper of an exit ramp).	
$L_{Length\ to\ Q}$	Length from the beginning of the crossroad ramp terminal to the end of the storage length.	
$L_{Decis\ Length}$	Distance covered by a vehicle from the instant that the driver recognizes the presence of a queue of vehicles stopped ahead until the instant that the driver actually applies the breaks (see Figure 19).	
$L_{Man\ Length}$	Distance that a vehicle travels after the breaks are applied (see Figure 19).	
$L_{Q\ Storage}$	Stopping queue or storage length (see Figure 19).	
Terms Associated with Curve Elements and Parameters		
R_{min}	Minimum radius of curvature.	1
e_{max}	Maximum superelevation rate.	1
f_{max}	Maximum side friction factor.	1
R	Curve radius.	7,8,20,21
n_c	Number of curves on the ramp proper.	11
R_i	Radius of curve *i*	13
I_{DN}	Indicator variable for downward slope (= 1 if the overall ramp slopes downward, 0 otherwise).	20
I_{LV}	Indicator variable for level slope (= 1 if the overall ramp is level, 0 otherwise).	20
I_{UP}	Indicator variable for upward slope (= 1 if the overall ramp slopes upward, 0 otherwise).	20
R_{next}	Radius of subsequent curve.	22
R_{max}	Maximum radius of curve.	22
Terms Associated with Tangent Elements and Parameters		
I_{55}	Indicator variable for 55-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 55 mph, 0 otherwise).	6,9
I_{60}	Indicator variable for 60-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 60 mph, 0 otherwise).	6,9
I_{65}	Indicator variable for 65-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 65 mph or above, 0 otherwise).	6,9

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